

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250277746

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

LaChapelle; Joseph G. et al.

RAMAN SPECTROSCOPY SYSTEM WITH PHOTONIC INTEGRATED CIRCUIT

Abstract

In one embodiment, a system includes a pump light source configured to produce a pump beam of light at a pump frequency and a Stokes light source configured to produce a Stokes beam of light at a Stokes frequency, where the pump and Stokes frequencies are offset by a frequency offset Ω . The system further includes a photonic integrated circuit (PIC) that includes multiple optical waveguides configured to: direct the pump and Stokes beams of light to a sample; receive a Raman signal produced by coherent Raman scattering of the pump and Stokes beams of light at the sample; and direct the Raman signal to an optical receiver. The optical receiver is configured to detect the Raman signal. The optical receiver includes a probe light source configured to produce a probe beam of light at a probe frequency, an optical detector, and an electronic circuit.

Inventors: LaChapelle; Joseph G. (Austin, TX), Cannon; Roger S. (Austin, TX), Stat; David C. (Austin, TX)

Applicant: Haemanthus, Inc. (Austin, TX)

Family ID: 96881094

Appl. No.: 19/187630

Filed: April 23, 2025

Related U.S. Application Data

parent US continuation-in-part 18762420 20240702 PENDING child US 19187630
us-provisional-application US 63560521 20240301

Publication Classification

Int. Cl.: G01N21/65 (20060101); A61B5/00 (20060101); A61B5/145 (20060101); G01J3/44 (20060101); H01S5/026 (20060101); H01S5/125 (20060101); H01S5/50 (20060101)

U.S. Cl.:

CPC **G01N21/65** (20130101); **A61B5/0075** (20130101); **A61B5/14532** (20130101); **G01J3/4412** (20130101); **H01S5/026** (20130101); **H01S5/125** (20130101); **H01S5/50** (20130101); G01N2021/653 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part of U.S. patent application Ser. No. 18/762,420, filed 2 Jul. 2024, which claims the benefit, under 35 U.S.C. § 119 (e), of U.S. Provisional Patent Application No. 63/560,521, filed 1 Mar. 2024, the entireties of which are incorporated by reference herein.

TECHNICAL FIELD

[0002] This disclosure generally relates to Raman spectroscopy systems and methods that use Raman scattering.

BACKGROUND

[0003] Raman spectroscopy is an optical measurement technique that can be applied to the study of molecular dynamics (e.g., to investigate vibrational and rotational states of molecules). Molecules typically exhibit molecular vibrations with frequencies ranging from less than 10 terahertz (THz) to approximately 100 THz, which corresponds to wavenumbers of approximately 300 cm.^{sup.}−1 to 3000 cm.^{sup.}−1 and wavelengths of approximately 30 to 3 micrometers (μm). Raman spectroscopy is based on the inelastic scattering of photons (referred to as Raman scattering) that occurs when light interacts with molecular vibrations or phonons in a sample. Raman scattering causes the energy (or equivalently, the frequency) of scattered light to be shifted, and this shift in energy can provide information about the vibrational modes of molecules in the sample.

[0004] Raman spectroscopy can be used in various chemical sensing applications to identify molecular components in a sample. Since many molecules exhibit a unique Raman scattering spectrum, the spectrum of Raman-scattered light produced when light interacts with a sample can serve as a fingerprint to sense or identify various molecular species within the sample. A sample illuminated with light may produce Raman scattered light at different wavelengths from the illumination light, and measurement of the spectrum of the Raman scattered light is typically performed in the optical domain. For example, the spectrum of Raman scattered light can be measured using an optical spectrometer which separates the Raman scattered light into its optical frequency components using a diffractive element, such as a diffraction grating.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0005] FIGS. 1-2 each illustrate an example Raman spectroscopy system.

[0006] FIGS. 3-4 each illustrate an example Raman signal produced by coherent Raman scattering.

[0007] FIG. 5 illustrates the example Raman signal of FIG. 4 along with a probe beam.

[0008] FIG. 6 illustrates an expanded view of a portion of the Raman signal of FIG. 5.

[0009] FIG. 7 illustrates an expanded view of a portion of the Raman signal of FIG. 6.

[0010] FIGS. 8-10 illustrate time-domain and frequency-domain plots of an example electronic signal resulting from coherent mixing of the Raman signal and probe beam of FIG. 7.

[0011] FIG. 11 illustrates an example Raman signal that is measured at multiple probe frequencies.

[0012] FIG. 12 illustrates an example Raman spectrum corresponding to the Raman signal of FIG. 11.

[0013] FIG. **13** illustrates another example Raman signal that is measured at multiple probe frequencies.

[0014] FIGS. **14-15** each illustrate a second example Raman signal obtained by changing the frequency offset between a pump beam and a Stokes beam.

[0015] FIG. **16** illustrates an example Raman signal along with two probe beams of light.

[0016] FIG. **17** illustrates an example optical receiver for measuring the Raman signal and pump beam of light from FIG. **16**.

[0017] FIG. **18** illustrates an example Raman spectroscopy system for measuring a Raman signal produced by spontaneous Raman scattering.

[0018] FIG. **19** illustrates an example Raman signal produced by the Raman spectroscopy system of FIG. **18**.

[0019] FIG. **20** illustrates an example laser diode that produces a free-space beam of light.

[0020] FIG. **21** illustrates an example laser diode that produces seed light that is amplified by a semiconductor optical amplifier (SOA).

[0021] FIG. **22** illustrates an example laser diode that produces seed light that is amplified by a fiber-optic amplifier.

[0022] FIG. **23** illustrates an example sampled-grating distributed Bragg reflector (SG-DBR) laser.

[0023] FIG. **24** illustrates an example light source with multiple laser diodes and an optical multiplexer that combines light produced by the laser diodes into a single output beam of light.

[0024] FIG. **25** illustrates an example pump laser and Stokes laser with a fiber-optic combiner that produces a combined pump-Stokes beam coupled into an optical fiber.

[0025] FIG. **26** illustrates an example laser diode coupled to a waveguide of a photonic integrated circuit (PIC).

[0026] FIG. **27** illustrates an example pump laser and Stokes laser with a photonic integrated circuit (PIC) that produces a combined pump-Stokes beam coupled into an optical waveguide of the PIC.

[0027] FIG. **28** illustrates an example fiber-optic combiner that combines a Raman signal with a probe beam.

[0028] FIG. **29** illustrates an example photonic integrated circuit (PIC) with a waveguide combiner that combines a Raman signal with a probe beam.

[0029] FIGS. **30-35** each illustrate example frequency ranges of a pump beam and a Stokes beam.

[0030] FIG. **36** illustrates an example optical receiver with two detectors.

[0031] FIG. **37** illustrates an example optical receiver configured for polarization-sensitive detection of a Raman signal.

[0032] FIG. **38** illustrates an example optical receiver configured to detect in-phase and quadrature components of a Raman signal.

[0033] FIG. **39** illustrates an example optical receiver configured to detect polarization as well as in-phase and quadrature components of a Raman signal.

[0034] FIGS. **40-43** each illustrate an example Raman spectroscopy system that includes a photonic integrated circuit (PIC).

[0035] FIGS. **44-45** illustrate an example Raman spectroscopy system that includes a photonic integrated circuit (PIC) with an evanescent waveguide.

[0036] FIG. **46** illustrates an example Raman spectroscopy system that includes a photonic integrated circuit (PIC) and a hollow waveguide.

[0037] FIGS. **47-49** each illustrate an example Raman spectroscopy system that includes a photonic integrated circuit (PIC) with a grating.

[0038] FIG. **50** illustrates an example Raman spectroscopy system that includes a photonic integrated circuit (PIC) and a mirror.

[0039] FIG. **51** illustrates an example Raman spectroscopy system that is affixed to a patient.

[0040] FIGS. **52-54** each illustrate an example Raman spectroscopy system that includes a

photonic integrated circuit (PIC) and one or more optical fibers.

[0041] FIG. 55 illustrates an example Raman spectroscopy system that includes a multi-chip PIC with two secondary PICs.

[0042] FIG. 56 illustrates an example Raman spectroscopy system that includes multiple optical fibers.

[0043] FIG. 57 illustrates an example method for measuring a Raman signal.

[0044] FIG. 58 illustrates an example computer system.

DETAILED DESCRIPTION

[0045] FIGS. 1-2 each illustrate an example Raman spectroscopy system **100**. The Raman spectroscopy system **100** in each of FIGS. 1-2 may detect a Raman signal **160** by coherently mixing the Raman signal **160** with a probe beam of light **120_{pr}** produced by a probe light source **111_{pr}**. The Raman spectroscopy system **100** in each of FIGS. 1-2 may be referred to as a coherent Raman spectroscopy system, a coherent Raman spectroscopy system with heterodyne detection, or a high-resolution coherent Raman spectroscopy system. One or more of the systems or methods described herein may be applied to any suitable form of coherent Raman spectroscopy or coherent Raman scattering (CRS), such as for example, coherent anti-Stokes Raman scattering (CARS), stimulated Raman scattering (SRS), or Raman-induced Kerr effect (RIKE).

[0046] The Raman spectroscopy system **100** in FIG. 1 includes a pump light source **111_{pu}** that produces a pump beam of light **120_{pu}** and a Stokes light source **111_S** that produces a Stokes beam of light **120_S**. The pump light source **111_{pu}** produces the pump beam of light **120_{pu}** at a pump frequency, which may be referred to as a first frequency and may be represented by $\nu_{\text{sub.1}}$, $\nu_{\text{sub.pu}}$, $\omega_{\text{sub.1}}$, or $\omega_{\text{sub.pu}}$. The pump light source **111_{pu}** may be referred to as a first light source, and the pump beam of light **120_{pu}** may be referred to as a first beam of light. The Stokes light source **111_S** produces the Stokes beam of light **120_S** at a Stokes frequency, which may be referred to as a second frequency and may be represented by $\nu_{\text{sub.2}}$, $\nu_{\text{sub.s}}$, $\omega_{\text{sub.2}}$, or $\omega_{\text{sub.s}}$. The Stokes light source **111_S** may be referred to as a second light source, and the Stokes beam of light **120_S** may be referred to as a second beam of light. The pump and Stokes frequencies may be offset by a frequency offset Ω , where Ω equals $\nu_{\text{sub.pu}} - \nu_{\text{sub.s}}$ (or equivalently, $\Omega = \nu_{\text{sub.1}} - \nu_{\text{sub.2}}$). Generally, the pump frequency $\nu_{\text{sub.pu}}$ is greater than the Stokes frequency $\nu_{\text{sub.s}}$, and the frequency offset Ω is a positive value. The pump and Stokes light sources in FIG. 1 may each include one or more lasers. For example, the Raman spectroscopy system **100** in FIG. 2 includes a pump laser **110_{pu}** that produces a pump beam of light **120_{pu}** and a Stokes laser **110_S** that produces a Stokes beam of light **120_S**.

[0047] In FIGS. 1-2, the pump beam **120_{pu}** and the Stokes beam **120_S** are directed to a sample **150**, and the sample **150** produces a Raman signal **160** in response to the pump and Stokes beams. For example, the Raman signal **160** may be produced by coherent Raman scattering of the pump and Stokes beams within the sample **150**. A Raman spectroscopy system may include one or more optical elements that direct the pump beam **120_{pu}** and the Stokes beam **120_S** to a sample **150**. Additionally, the optical elements may collect the Raman signal **160** produced by the sample **150** in response to the pump beam **120_{pu}** and Stokes beam **120_S** and may direct the Raman signal **160** to an optical receiver **200**. The optical elements may include free-space optics (which may be referred to as bulk optics), fiber-optic components, waveguide-based optics, metamaterials, or any suitable combination thereof. For example, the optical elements may include a mirror, lens, optical combiner (e.g., beamsplitter), optical fiber, photonic integrated circuit (PIC), optical waveguide, or metamaterial-based optic. As another example, the optical combiner **130_a** in each of FIGS. 1-2 may be a free-space dichroic beamsplitter that transmits light at the pump-beam wavelength and reflects light at the Stokes-beam wavelength. The combiner **130_a** may combine the pump beam **120_{pu}** and the Stokes beam **120_S** to produce a combined pump-Stokes beam **140** that is directed to the sample **150**. The pump and Stokes beams may be combined so that they are substantially overlapped with one another and propagate in the same direction along approximately the same optical axis.

Alternatively, a Raman spectroscopy system **100** may not include a pump-Stokes optical combiner **130a**, and the pump beam **120pu** and the Stokes beam **120S** may be directed to a sample **150** as separate beams (e.g., the pump and Stokes beams may be overlapped or combined at the sample rather than being combined earlier). In this embodiment, the pump and Stokes beams may enter the sample **150** from opposite sides (e.g., the pump and Stokes beams may propagate to the sample in opposite directions along approximately the same optical axis). As another example, the optical elements may include a lens (not illustrated in FIGS. **1-2**) that focuses the pump-Stokes beam **140** onto the sample **150**. Additionally, the optical elements may include a lens (not illustrated in FIGS. **1-2**) that collects the Raman signal **160** to produce a Raman-signal beam that is directed to the optical receiver **200**.

[0048] In FIGS. **1-2** the combined pump-Stokes beam **140** is directed to one side of the sample **150**, and the Raman signal **160** is emitted from the opposite side of the sample. A Raman signal **160** that is collected and directed to an optical receiver **200** may be emitted from a sample **150** in any suitable direction. For example, a Raman signal **160** that is sent to an optical receiver **200** may be emitted from a sample **150** in a forward-scattered direction (e.g., as illustrated in FIGS. **1-2**), in a backward-scattered direction (e.g., back toward the pump or Stokes beam), or in a sideways-scattered direction (e.g., in a direction approximately orthogonal to the combined pump-Stokes beam **140** in FIGS. **1-2**).

[0049] The Raman spectroscopy system **100** in each of FIGS. **1-2** includes an optical receiver **200** that detects the Raman signal **160** produced by the sample **150**. The optical receiver **200** (which may be referred to as a heterodyne optical receiver or a high-resolution optical receiver) may detect the Raman signal **160** using an optical heterodyne technique in which the Raman signal **160** is coherently mixed with a probe beam of light **120pr**. The optical receiver **200** in FIG. **1** includes a probe light source **111pr** that produces a probe beam of light **120pr** at a probe frequency. The probe light source **111pr** may be referred to as a third light source, and the probe beam of light **120pr** may be referred to as a third beam of light. The probe frequency may be referred to as a third frequency and may be represented by $\nu_{\text{sub.3}}$, $\nu_{\text{sub.pr}}$, $\omega_{\text{sub.3}}$, or $\omega_{\text{sub.pr}}$. The probe light source **111pr** in FIG. **1** may include one or more lasers. For example, the optical receiver **200** in FIG. **2** includes a probe laser **110pr** that produces a probe beam of light **120pr**.

[0050] The optical combiner **130b** in each of FIGS. **1-2** may be a dichroic or a non-dichroic beamsplitter that combines the Raman signal **160** and the probe beam **120pr** to produce a combined probe-Raman signal **210** that is directed to a detector **220**. An optical receiver **200** may include one or more optical detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with at least a portion of a probe beam **120pr** to produce an electronic signal. Each of the optical receivers **200** in FIGS. **1-2** includes one detector **220** that receives the combined probe-Raman signal **210**. The Raman signal **160** and the probe beam **120pr** are coherently mixed at the detector **220**, and this heterodyne mixing process produces an electronic signal, which in FIGS. **1-2** is indicated as analog photocurrent signal **i**. In FIGS. **1-2**, the detection electronics **230** receives the photocurrent signal **i** and produces a digital output signal **240** that corresponds to the photocurrent signal. Detection electronics **230** may include or may be referred to as an electronic circuit. The digital output signal **240** may be sent to a processor, and the processor may determine a characteristic of the analog photocurrent signal **i** based on the digital output signal **240**. For example, the characteristic of the analog photocurrent signal (or the characteristic of an analog voltage signal that corresponds to the photocurrent signal) determined by the processor may include one or more of: a peak amplitude (e.g., a peak amplitude of the photocurrent signal or a peak amplitude of the corresponding voltage signal), an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization (e.g., a polarization of a Raman signal). An analog photocurrent signal may be referred to as a photocurrent, a photocurrent signal, or a current signal, and an analog voltage signal may be

referred to as a voltage signal. A processor of a Raman spectroscopy system **100** may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. A processor may be similar to the computer system **1000** illustrated in FIG. **58** and described herein. In some embodiments, a processor or a portion of a processor may be located in the detection electronics **230**. For example, a processor located in the detection electronics **230** may receive a digital output signal **240** and perform some preprocessing of the digital signal or determine a characteristic of a photocurrent signal based on the digital signal.

[0051] In FIGS. **1-2**, the probe beam **120_{pr}** does not travel through the sample **150**, and the probe beam is combined with the Raman signal **160** after the Raman signal has exited the sample **150**. In other embodiments, a Raman spectroscopy system **100** may include a probe laser **110_{pr}** that produces a probe beam **120_{pr}** that is directed through the sample **150**. For example, the probe beam **120_{pr}** may be combined with the pump beam **120_{pu}** and the Stokes beam **120₅**, and all three beams may be directed to the sample **150**. The probe beam **120_{pr}** may travel through the sample **150** and may exit the sample along with the Raman signal **160**. The optical receiver **200** may not include a combiner **130_b**, since the probe beam **120_{pr}** and the Raman signal **160** may already be combined after they exit the sample **150**. The probe beam **120_{pr}** and the Raman signal **160** may be directed to a detector **220** without being transmitted or reflected by an optical combiner **130_b**.

[0052] The detection electronics **230** in FIG. **2** includes an electronic amplifier **232** and a digitizer **236**. The electronic amplifier **232** may include a transimpedance amplifier that amplifies the photocurrent signal *i* to produce an analog voltage signal **234** that corresponds to the photocurrent signal *i* (e.g., the photocurrent signal *i* and the analog voltage signal **234** may have similar temporal shapes or may include similar electronic frequency components). The electronic amplifier **232** may include an additional gain stage that further amplifies an intermediate voltage signal produced by the transimpedance amplifier to produce the voltage signal **234**. Additionally, the electronic amplifier **232** may include an electronic filter (e.g., a low-pass, high-pass, or band-pass filter) that filters the photocurrent signal or voltage signal. For example, the electronic amplifier **232** may include (i) a high-pass filter that removes a DC offset and low-frequency components (e.g., frequency components below 10 MHz) from the photocurrent signal or (ii) a band-pass filter that removes the DC and low-frequency components as well as high-frequency components (e.g., frequency components above 5 GHz). Herein, an electronic signal produced in response to coherent mixing may refer to a current signal (e.g., a photocurrent *i* produced by a detector **220**) or may refer to a corresponding voltage signal (e.g., a voltage signal **234** produced by an electronic amplifier that amplifies a photocurrent produced by a detector to produce the voltage signal). In FIG. **2**, the digitizer **236** receives the voltage signal **234** and produces a digital output signal **240** that includes a digital representation of the voltage signal **234**. The digital output signal **240** may be a time-domain digital representation of the voltage signal **234**. The digital output signal **240** may be referred to as corresponding to or representing the voltage signal **234** or the photocurrent signal *i*. The digitizer **236** may include an analog-to-digital converter (ADC) that produces a digital version of the voltage signal **234**. Additionally or alternatively, the digitizer **236** may include a peak detector that determines a peak value of the voltage signal **234**.

[0053] A Raman spectroscopy system **100** may include one or more optical detectors **220**. An optical detector **220** (which may be referred to as a detector, photodetector, or photodiode) may include a PN photodiode, PIN photodiode, avalanche photodiode (APD), single-photon avalanche diode (SPAD), silicon photomultiplier (SiPM), or photomultiplier tube (PMT). A PN photodiode refers to a photodiode structure formed by a p-doped semiconductor and an n-doped semiconductor, where the PN acronym refers to the structure having p-doped and n-doped regions. A PIN photodiode refers to a photodiode structure formed by an undoped intrinsic semiconductor region located between p-doped and n-doped regions, where the PIN acronym refers to the structure having p-doped, intrinsic, and n-doped regions.

[0054] A PN photodiode, PIN photodiode, APD, or SPAD may include any suitable semiconductor material, such as for example: silicon, germanium, gallium arsenide (GaAs), indium phosphide (InP), indium arsenide (InAs), aluminum arsenide (AlAs), indium antimonide (InSb), aluminum antimonide (AlSb), gallium antimonide (GaSb), aluminum gallium arsenide (AlGaAs), indium gallium arsenide (InGaAs), indium aluminum arsenide (InAlAs), indium arsenide antimonide (InAsSb), aluminum arsenide antimonide (AlAsSb), aluminum gallium antimonide (AlGaSb), gallium arsenide antimonide (GaAsSb), aluminum indium arsenide antimonide (AlInAsSb), indium gallium arsenide antimonide (InGaAsSb), indium gallium aluminum arsenide (InGaAlAs), aluminum gallium arsenide antimonide (AlGaAsSb), or silicon germanium (SiGe). For example, the Raman signal **160** and the probe beam **120_{pr}** in FIGS. **1-2** may each have a wavelength in the 400-1100 nanometer (nm) range, and the detector **220** may include a silicon PIN photodiode. As another example, the Raman and probe beams in FIGS. **1-2** may each have a wavelength in the 1000-1700 nm range, and the detector **220** may include an InGaAs PIN photodiode. As another example, a detector **220** may include an APD that includes a semiconductor material with antimonide (e.g., InSb, AlSb, GaSb, InAsSb, AlAsSb, AlGaSb, GaAsSb, AlInAsSb, InGaAsSb, or AlGaAsSb).

[0055] In order for a detector **220** to detect an optical signal, the wavelength of the optical signal must be within the detector's wavelength range of responsivity (e.g., approximately 400-1100 nm for a silicon detector, and approximately 1000-1700 nm for an InGaAs detector) and a frequency of an amplitude modulation of the optical signal must be within the electronic bandwidth of the detector. The electronic bandwidth (Δf) of a detector **220** refers to the range of electronic modulation frequencies over which a detector may detect an optical signal, where detection of the optical signal refers to (i) the detector producing a photocurrent signal i that corresponds to the optical signal and (ii) an electronic amplifier **232** producing a voltage signal **234** that corresponds to the photocurrent signal. If a silicon detector **220** has an electronic bandwidth that extends from 100 MHz to 5 GHz, then the detector may detect optical signals with (i) wavelengths between approximately 400 nm and 1100 nm and (ii) amplitude modulation between 100 MHz and 5 GHz. For example, a 900-nm optical signal with an amplitude modulation at a frequency between 100 MHz and 5 GHz may be detected by the silicon detector **220**. The silicon detector **220** may not detect a continuous-wave or substantially constant portion of a 900-nm optical signal (e.g., the substantially constant portion of the optical signal may produce a DC current in the detector that may be electronically filtered out), and the silicon detector **220** may not detect a portion of the optical signal with an amplitude modulation greater than approximately 5 GHz. Herein, reference to the electronic bandwidth (Δf) of a detector **220** may refer to (i) the electronic bandwidth of just the detector or (ii) the overall bandwidth of the detector in combination with an electronic amplifier **232**.

[0056] A detector **220** may have an electronic bandwidth Δf between approximately 100 MHz and approximately 50 GHz. For example, the detector **220** in each of FIGS. **1-2** may have an electronic bandwidth between 100 MHz and 10 GHz. As another example, the detector **220** in each of FIGS. **1-2** may have an electronic bandwidth that extends from a low-frequency cutoff to a high-frequency cutoff. The low-frequency cutoff may be approximately DC (i.e., zero hertz), 1 MHz, 10 MHz, 50 MHz, or 100 MHz, and the high-frequency cutoff may be approximately 500 MHz, 1 GHz, 2 GHz, 5 GHz, 10 GHz, 20 GHz, or 50 GHz. The electronic bandwidth of a detector **220** may refer to the bandwidth of only the detector **220**. For example, a detector **220** may have an electronic bandwidth that extends from DC to 10 GHz, and the detector may be referred to as having a 10-GHz bandwidth that extends from DC to 10 GHz. Alternatively, the electronic bandwidth of a detector **220** may refer to the overall bandwidth of the detector in combination with an electronic amplifier **232** that amplifies the photocurrent signal i produced by the detector. An electronic amplifier **232** may have a low-frequency cutoff (e.g., DC, 1 MHz, 10 MHz, 50 MHz, or 100 MHz) and a high-frequency cutoff (e.g., 500 MHz, 1 GHz, 2 GHz, 5 GHz, 10 GHz, 20 GHz, or 50 GHz),

and a detector-amplifier combination may be referred to as having an electronic bandwidth that extends from the low-frequency cutoff to the high-frequency cutoff. For example, if the detector bandwidth extends from DC to 10 GHz, and the electronic amplifier bandwidth extends from DC to 5 GHz, then the detector (or, the detector-amplifier combination) may be referred to as having an electronic bandwidth of 5 GHz that extends from DC to 5 GHz. As another example, if the detector bandwidth extends from DC to 10 GHz, and the electronic amplifier bandwidth extends from 100 MHz to 5 GHz, then the detector may be referred to as having an electronic bandwidth of 4.9 GHz that extends from 100 MHz to 5 GHz.

[0057] A Raman spectroscopy system **100** may include one or more optical waveplates that change or rotate the polarization of a beam of light. For example, a half-wave plate may be used to rotate a linearly polarized beam of light to a different polarization orientation (e.g., from vertically polarized to horizontally polarized), and a quarter-wave plate may be used to convert a linearly polarized beam of light to a circular or elliptical polarization. The pump laser **110_{pu}** in FIG. 2 may produce linearly polarized light, and the waveplate **132a** may be a half-wave plate that rotates the polarization of the pump beam **120_{pu}** prior to the pump beam being directed to the sample **150**. Alternatively, the waveplate **132a** may be a quarter-wave plate that converts the linearly polarized pump beam **120_{pu}** to a circular or elliptical polarization prior to the pump beam being directed to the sample **150**. Similarly, the Stokes laser **110_S** in FIG. 2 may produce linearly polarized light, and the waveplate **132b** may be (i) a half-wave plate that rotates the polarization of the Stokes beam **120_S** or (ii) a quarter-wave plate that converts the Stokes beam **120_S** to a circular or elliptical polarization.

[0058] The probe laser **110_{pr}** in FIG. 2 may produce linearly polarized light, and the waveplate **132c** may be (i) a half-wave plate that rotates the polarization of the probe beam **120_{pr}** or (ii) a quarter-wave plate that converts the probe beam **120_{pr}** to a circular or elliptical polarization prior to the probe beam being combined with the Raman signal **160**. Changing the polarization of the probe beam **120_{pr}** may allow the Raman signal **160** and the probe beam to be coherently mixed. The polarization of the probe beam **120_{pr}** can be changed so that it has both horizontal and vertical polarization components, which ensures that at least a portion of the probe beam **120_{pr}** and the Raman signal **160** have polarizations that are oriented in the same direction so that their electric fields may be added together.

[0059] Each of the optical waveplates **132** in FIG. 2 may be a free-space optical element, a fiber-optic component, a waveguide-based optical element, or a metamaterial-based optic. Additionally, each of the optical waveplates **132** in FIG. 2 may be a fixed waveplate or an adjustable waveplate. A fixed waveplate may have a fixed optical phase difference between the two axes of the waveplate (e.g., a quarter-wave plate may have a one-quarter wavelength phase difference, and a half-wave plate may have a one-half wavelength phase difference). An adjustable waveplate may allow for the phase difference between the two axes of the waveplate to be dynamically changed. For example, an electronically adjustable waveplate may include a Pockels cell, a liquid crystal device, or a photoelastic modulator that allows the phase difference to be adjusted electronically so that the waveplate can be dynamically configured to act as a waveplate having any suitable phase difference (e.g., a phase difference between zero wavelengths and one-half wavelength). An adjustable waveplate may switch between (i) applying no phase difference to incident light so that the transmitted light has the same polarization as the incident light and (ii) applying a one-quarter wavelength or one-half wavelength phase difference so that linearly polarized incident light is converted to circularly polarized light or is rotated to a different polarization. For example, the pump laser **110_{pu}** in FIG. 2 may produce vertically polarized light, and the waveplate **132a** may be an adjustable waveplate that switches between (i) applying no polarization rotation to the pump beam **120_{pu}** so that the pump beam remains vertically polarized and (ii) applying a 90-degree rotation to the pump-beam polarization so that the pump beam **120_{pu}** after the waveplate **132a** is horizontally polarized. Dynamically changing the polarization of the pump beam **120_{pu}** may allow

the Raman spectroscopy system **100** in FIG. 2 to perform measurements at two different pump-beam polarizations, which may produce additional data for characterization of the sample **150**.

[0060] In some embodiments, an optical waveplate **132** may be a metamaterial-based waveplate. A metamaterial refers to an engineered material having features or repeating patterns at scales smaller than the wavelength of light interacting with the metamaterial. A metamaterial may be configured to act as a mirror, lens, waveplate, diffractive optical element, optical combiner, or optical waveguide. A metamaterial-based waveplate may affect the polarization of a beam of light based on wavelength. For example, the Raman spectroscopy system **100** in FIG. 2 may include a metamaterial-based waveplate (not illustrated in FIG. 2) located after the combiner **130a** and before the sample **150**. The metamaterial waveplate may change the polarization of the pump beam **120pu** from linear to circular while preserving the polarization of the Stokes beam **120S** (e.g., the Stokes beam may remain linearly polarized and may not be significantly changed by the waveplate).

[0061] A Raman spectroscopy system **100** may include an optical filter that transmits light at one or more wavelengths and blocks light at one or more other wavelengths. The optical filter **134** in FIG. 2 is located between the sample **150** and the optical receiver **200** and may be configured to substantially transmit one or more optical wavelengths associated with the Raman signal **160** and substantially block one or more wavelengths associated with the pump beam **120pu** or the Stokes beam **120S**. For example, the optical filter **134** may transmit greater than 90% of the Raman signal **160** and may block greater than 90% of both the pump and Stokes beams. As another example, the optical filter **134** may transmit greater than 90% of the Raman signal **160** and the Stokes beam **120S**, and the optical filter **134** may block greater than 98% of the pump beam **120pu**.

[0062] A Raman spectroscopy system **100** may include an optical polarizer that transmits light having a particular polarization (e.g., horizontal) and blocks light having an orthogonal polarization (e.g., vertical). The optical polarizer **136** in FIG. 2 is located between the sample **150** and the optical receiver **200** and may be oriented to transmit light with a polarization associated with the Raman signal. Additionally, the polarizer **136** may block a polarization associated with the pump or Stokes beams. For example, the Stokes beam **120S** incident on the sample **150** may be vertically polarized, and the Raman signal **160** may be at least partially horizontally polarized. The polarizer **136** may be oriented to block vertically polarized light and transmit horizontally polarized light so that the Stokes beam **120S** is blocked and the Raman signal **160** is at least partially transmitted by the polarizer. A Raman spectroscopy system **100** may include both an optical filter **134** and an optical polarizer **136**. For example, the optical filter **136** in FIG. 2 may be configured to transmit the Raman signal **160** and the Stokes beam **120S** and block the pump beam **120pu**. Additionally, the Raman signal **160** and the Stokes beam **120S** may be orthogonally polarized, and the polarizer **136** may be oriented to transmit the Raman signal **160** and block the Stokes beam **120S**. Using a filter or polarizer located after the sample **150** to block light from the pump beam **120pu** or the Stokes beam **120S** may reduce noise in the system by reducing the amount of unwanted background light that reaches the detector **220**.

[0063] The sample **150** in FIGS. 1-2 may be a solid, liquid, or gas, or any combination thereof. The sample **150** may include a biological material, an organic material, an inorganic material, a crystalline material, an amorphous solid material, or any other suitable material or combination of suitable materials. For example, the sample **150** may include a drug, mineral, food, contaminant, or explosive material that produces a Raman signal **160** in response to excitation by the pump and Stokes beams. As another example, the sample **150** may be a biological material (e.g., blood, urine, saliva, sweat, or cerebrospinal fluid) and a component or molecule (e.g., glucose or cortisol) that is part of the biological material may produce a Raman signal **160**. As another example, the sample **150** may be water or wastewater that may include a contaminant, virus, bacteria, or an indicator of an infectious disease. The Raman signal **160** may be produced by coherent Raman scattering of the pump and Stokes beams within the sample **150**, and the frequency offset Ω between the pump and Stokes beams may be approximately equal to a vibrational frequency or an electronic-transition

frequency of a particular material that is part of the sample. The vibrational frequency of the particular material may correspond to a molecular vibration of a molecule, or in the case of a crystalline material, may correspond to a lattice vibration of a crystal. For example, the sample **150** may include glucose, and the frequency offset Ω may be approximately equal to a frequency of a molecular vibration of glucose.

[0064] The Raman spectroscopy systems **100** in FIGS. **1-2** may perform one or more measurements of a sample **150**, and each measurement may include determining a characteristic of an electronic signal that results from the coherent mixing of the Raman signal **160** and the probe beam **120pr**. The electronic signal may include a photocurrent i or a corresponding voltage signal **234**. The frequency offset Ω may be approximately equal to a vibrational frequency or electronic-transition frequency of a particular material, and based on the one or more measurements, a processor may determine whether the particular material is present in the sample **150**. Additionally or alternatively, the processor may determine an amount or a concentration of the particular material in the sample based on the measurements. For example, the frequency offset Ω may be approximately equal to a vibrational frequency of glucose, and, based on one or more optical heterodyne measurements of a Raman signal **160** produced by a sample **150**, the processor may determine (i) whether glucose is present in the sample or (ii) an amount or a concentration of glucose in the sample. The amount of glucose that is present in the sample **150** may be proportional to the amplitude of a photocurrent signal i produced by coherent mixing of the Raman signal **160** and the probe beam **120pr**. Based on the amplitude of the photocurrent signal i , the processor may determine the amount or concentration of glucose in the sample **150**.

[0065] A technical advantage of a coherent Raman spectroscopy system **100** as described herein is a higher spectral resolution or a better chemical sensitivity than a conventional Raman spectroscopy system. As such, a Raman spectroscopy system **100** as described herein may be referred to as a high-resolution Raman spectroscopy system or as a high-resolution coherent Raman spectroscopy system. In a conventional Raman spectroscopy system, a Raman signal produced by a sample may be measured in the optical domain using an optical spectrometer. A spectrometer typically uses a dispersive optical element (e.g., a diffraction grating) to separate the Raman signal into its various spectral components. However, this type of measurement performed in the optical domain typically has a spectral resolution on the order of 1 cm.sup.-1 (or, about 30 GHz). In contrast, the spectral resolution of a coherent Raman spectroscopy system with heterodyne detection, as described herein, is determined primarily by the spectral linewidth $\Delta\nu_{\text{sub.pr}}$ of the probe beam **120pr** that is coherently mixed with the Raman signal **160**. The probe laser **110pr** may include a wavelength-tunable laser diode with a linewidth of 200 MHz or less, which corresponds to a spectral resolution of the Raman spectroscopy system **100** of less than 200 MHz (or, less than 0.007 cm.sup.-1). This 200-MHz spectral resolution is more than 100 times better than the 30-GHz spectral resolution of a conventional Raman spectroscopy system. In some embodiments, the probe laser **110pr** may have a linewidth of 1 MHz or less, which corresponds to a spectral resolution of the Raman spectroscopy system **100** of less than 1 MHz (or, less than $33\times 10^{-6}\text{ cm.sup.-1}$). A related advantage of a coherent Raman spectroscopy system **100** is that the signal capture and analysis are performed in the electronic domain (e.g., at electronic frequencies between DC and 50 GHz) rather than in the optical domain (e.g., at optical frequencies between 60 THz and 1,000 THz). The coherent mixing of two optical signals (Raman signal **160** and probe beam **120pr**) produces an electronic signal which can be analyzed with relatively high resolution compared to an optical signal. This electronic signal analysis, along with the relatively narrow spectral linewidth of the probe laser **110pr**, provides a coherent Raman spectroscopy system **100** with a high spectral resolution. Additionally, the wavelength tunability of the probe laser **110pr** allows a Raman spectrum of a material to be determined at multiple frequencies with high spectral resolution.

[0066] The higher spectral resolution of a coherent Raman spectroscopy system **100** may provide a corresponding improvement in the ability of the coherent Raman spectroscopy system to sense

various chemical species. For example, a high-resolution coherent Raman spectroscopy system **100** may be able to distinguish between different chemical species that have Raman peaks located relatively close together, whereas a conventional Raman spectroscopy system may not be able to resolve spectral features below about 1 cm^{-1} . Additionally, the higher spectral resolution of a coherent Raman spectroscopy system **100** may allow for lower concentrations of materials to be detected, as compared to a conventional Raman spectroscopy system. For example, a coherent Raman spectroscopy system **100** may be able to detect small deviations in the chemical signature of a biological sample, which may indicate the presence of damage or a mutation, which in turn may be correlated with a disease or pathogen.

[0067] Another technical advantage of a coherent Raman spectroscopy system **100** as described herein is its relatively compact size. A coherent Raman spectroscopy system may be packaged in a relatively compact enclosure as compared to a conventional Raman spectroscopy system. Since the spectral resolution of an optical spectrometer scales inversely with the optical path length of the spectrometer (e.g., a longer path length provides better spectral resolution), an optical spectrometer with a spectral resolution around 1 cm^{-1} can be quite large or bulky. In contrast, since the spectral resolution of a coherent Raman spectroscopy system is determined primarily by the spectral linewidth of the probe laser **110pr**, a coherent Raman spectroscopy system does not require a long optical path length to provide high spectral resolution. Thus, an enclosure for a coherent Raman spectroscopy system may be significantly smaller than that for a conventional Raman spectroscopy system. In some embodiments, a coherent Raman spectroscopy system may be packaged as a compact device that may be referred to as a lab-on-a-chip or a spectrometer on a chip. For example, a coherent Raman spectroscopy system may be packaged as a wearable device that provides ongoing, continual monitoring for a person or an animal.

[0068] FIGS. 3-4 each illustrate an example Raman signal **160** produced by coherent Raman scattering. In a coherent Raman spectroscopy system **100**, the pump beam **120pu** and the Stokes beam **120S** are directed to a sample **150**, which produces a Raman signal **160** by coherent Raman scattering of the pump and Stokes beams within the sample. The frequency of the pump beam **120pu** is ν_i , and the frequency of the Stokes beam **120S** is ν_{s2} . The frequency offset Ω between the pump and Stokes beams equals $\nu_{s1} - \nu_{s2}$. The frequencies of the pump and Stokes beams may be set so that the frequency offset Ω is approximately equal to a vibrational frequency or an electronic-transition frequency of a particular material. A coherent Raman spectroscopy system **100** may measure a Raman signal **160** produced by a sample **150** to determine (i) whether the particular material is present in the sample or (ii) an amount or a concentration of the particular material within the sample.

[0069] The frequency offset Ω between the pump and Stokes beams may be any suitable fixed or adjustable value between approximately 5 terahertz (THz) and approximately 100 THz. Expressed in wavenumbers, this corresponds to the frequency offset (being between approximately 167 cm^{-1} and approximately 3336 cm^{-1}). For example, in a coherent Raman spectroscopy system **100**, the pump beam **120pu** may have a wavelength between approximately 1220 nanometers (nm) and approximately 1450 nm (which corresponds to a pump-beam frequency ν_i between approximately 246 THz and approximately 207 THz), and the Stokes beam **120S** may have a wavelength between approximately 1490 nm and approximately 1570 nm (which corresponds to a Stokes-beam frequency ν_{s2} between approximately 201 THz and approximately 191 THz). This system may produce pump and Stokes beams having a frequency offset Ω between approximately 5.6 THz and approximately 54.8 THz (or, in wavenumbers, between approximately 185 cm^{-1} and 1827 cm^{-1}). For example, if the pump beam **120pu** has a wavelength of 1330 nm (or equivalently, a frequency ν_i of 225.4 THz) and the Stokes beam **120S** has a wavelength of 1550 nm (or equivalently, a frequency ν_{s2} of 193.4 THz), then the frequency offset (between the pump and Stokes beams) is approximately 32 THz (or, in wavenumbers, 1067 cm^{-1}). The pump and Stokes beams in a Raman spectroscopy system **100**

may each have any suitable wavelength between approximately 300 nm and approximately 5,000 nm. For example, if the pump beam **120pu** has a wavelength of approximately 785 nm (or equivalently, a frequency $\nu_{\text{sub.1}}$ of 381.9 THz) and the Stokes beam **120S** has a wavelength of approximately 840 nm (or equivalently, a frequency $\nu_{\text{sub.2}}$ of 356.9 THz), then the frequency offset Ω between the pump and Stokes beams is approximately 25 THz (or, in wavenumbers, 834 cm^{-1}). As another example, the pump beam **120pu** may have a wavelength between approximately 700 nm and approximately 850 nm, between approximately 890 nm and approximately 920 nm, or between approximately 1000 nm and approximately 1100 nm.

[0070] The Raman signal **160** in each of FIGS. 3-4 is an optical signal with a spectral linewidth of $\Delta\nu_{\text{sub.R}}$. In FIG. 3, the Raman signal **160** has a center frequency approximately equal to $2\nu_{\text{sub.1}} - \nu_{\text{sub.2}}$ (which is equal to $\nu_{\text{sub.1}} + \Omega$, since $\Omega = \nu_{\text{sub.1}} - \nu_{\text{sub.2}}$). The center frequency of a Raman signal **160** may refer to the frequency of a central peak or the frequency of an approximate center of the Raman signal. Additionally or alternatively, the center frequency of a Raman signal **160** may correspond to a Raman peak of an associated Raman spectrum. The Raman signal **160** in FIG. 3 may be produced by coherent anti-Stokes Raman scattering (CARS) in which the pump and Stokes beams interact with a sample **150** to produce a Raman signal **160** at or around the frequency $2\nu_{\text{sub.1}} - \nu_{\text{sub.2}}$. For example, if the pump and Stokes beams have respective wavelengths of 1064 nm and 1550 nm (which corresponds to frequencies of approximately 281.8 THz and 193.4 THz), then the frequency offset Ω is 88.3 THz (or, 2947 cm^{-1}), and the anti-Stokes Raman signal **160** has a center wavelength of approximately 810 nm (which corresponds to a frequency of approximately 370 THz). The Raman signal **160** in FIG. 4, which overlaps the frequency of the Stokes beam **120S**, may be produced by stimulated Raman scattering. The Raman signal **160** in FIG. 4 may be centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S**. For example, the Raman signal **160** in FIG. 4 may have a center frequency that is within approximately 200 GHz (or, 6.7 cm^{-1}) of the Stokes-beam frequency $\nu_{\text{sub.2}}$.

[0071] FIG. 5 illustrates the example Raman signal **160** of FIG. 4 along with a probe beam **120pr**. The Raman signal **160** may be produced by stimulated Raman scattering of the pump beam **120pu** and Stokes beam **120S**, and the probe beam **120pr** may be used to measure the Raman signal. The Raman signal **160** is centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S**, and the frequency $\nu_{\text{sub.3}}$ of the probe beam of light **120pr** overlaps the Raman signal **160** and is relatively close to the frequency $\nu_{\text{sub.2}}$ of the Stokes beam of light **120S** (e.g., the probe-beam frequency $\nu_{\text{sub.3}}$ may be within 200 GHz of the Stokes-beam frequency $\nu_{\text{sub.2}}$). In a coherent Raman spectroscopy system **100**, the pump frequency $\nu_{\text{sub.1}}$, Stokes frequency $\nu_{\text{sub.2}}$, and probe frequency $\nu_{\text{sub.3}}$ may each be between approximately 60 THz and approximately 1,000 THz (which corresponds to a wavelength between approximately 5,000 nm and approximately 300 nm). For example, in FIG. 5 the pump frequency $\nu_{\text{sub.1}}$ may be 291 THz (which corresponds to a wavelength of approximately 1030 nm), the Stokes frequency $\nu_{\text{sub.2}}$ may be 240.03 THz (which corresponds to a wavelength of approximately 1249.00 nm), and the probe frequency $\nu_{\text{sub.3}}$ may be 240.00 THz (which corresponds to a wavelength of approximately 1249.14 nm).

[0072] Each of the light sources **111** of a coherent Raman spectroscopy system **100** may include a wavelength-tunable light source. A wavelength-tunable light source refers to a light source that can produce light at multiple different wavelengths within a range of wavelengths (or equivalently, at multiple different frequencies within a range of frequencies). For example, the probe light source **111pr** in FIG. 1 may include a wavelength-tunable light source where the wavelength of the probe beam **120pr** is adjustable over an 80-nm wavelength range from 1490 nm to 1570 nm (which corresponds to a 10.3 THz frequency range from approximately 201.2 THz to approximately 191.0 THz). At any given time, a wavelength-tunable light source **111** may operate at any one of the different wavelengths within its wavelength-tuning range. For example, during a first period of time, the probe light source **111pr** with a 1490-1570 nm wavelength-tuning range may produce a probe beam **120pr** at 1500 nm, and during a subsequent second period of time, the probe light

source may be tuned to operate at 1560 nm. A wavelength-tunable light source may be adjustable over a frequency range that corresponds to a wavelength range having a width or span between approximately 10 nm and approximately 100 nm. For example, the width of the wavelength-tuning range of a wavelength-tunable light source may be approximately 10 nm, 20 nm, 30 nm, 40 nm, 50 nm, 60 nm, 80 nm, or 100 nm. A wavelength-tuning range from 1490 nm to 1570 nm may be referred to as having an 80-nm width or an 80-nm span. Around 1550 nm, a 10-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 1.25 THz (or equivalently, 41.6 cm.sup.-1), and a 100-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 12.5 THz (or equivalently, 416 cm.sup.-1). Around 1050 nm, a 10-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 2.72 THz (or equivalently, 90.7 cm.sup.-1), and a 100-nm wavelength-tuning range corresponds to a frequency-tuning range of approximately 27.3 THz (or equivalently, 909 cm.sup.-1). A wavelength-tunable light source may be referred to as a frequency-tunable light source or a tunable light source.

[0073] Each of the light sources **111** of a coherent Raman spectroscopy system **100** may include one or more laser diodes, and each of the laser diodes may be a fixed-wavelength laser diode or a wavelength-tunable laser diode. A fixed-wavelength laser diode may operate at a single wavelength or within a relatively narrow wavelength range (e.g., within 0.1 nm of a particular wavelength). A fixed-wavelength laser diode may include a distributed feedback (DFB) laser diode, a distributed Bragg reflector (DBR) laser diode, a fiber-Bragg-grating (FBG) stabilized laser diode, a temperature-stabilized laser diode, or any other suitable fixed-wavelength laser diode. A wavelength-tunable laser diode may produce light at multiple different wavelengths within a range of wavelengths. For example, a wavelength-tunable laser diode may be configured to produce light at any wavelength within a wavelength range having a width between approximately 10 nm and approximately 100 nm. At any given time, a wavelength-tunable laser diode may operate at any one wavelength of multiple different wavelengths within a range of wavelengths. A wavelength-tunable laser diode may include an external-cavity laser diode, a thermally tuned laser diode, or a sampled-grating distributed Bragg reflector (SG-DBR) laser. For example, a wavelength-tunable SG-DBR laser may have a 40-nm wavelength-tuning range that extends from 1530 nm to 1570 nm, and the SG-DBR laser may be adjustable to operate at any single wavelength within the 40-nm wavelength range. A wavelength-tunable laser diode may be referred to as a frequency-tunable laser diode or a tunable laser diode. For example, a tunable laser with a 40-nm wavelength-tuning range that extends from 1530 nm to 1570 nm may also be referred to as a frequency-tunable laser with a 5.0-THz frequency-tuning range that extends from approximately 191 THz to approximately 196 THz.

[0074] Each of the light sources **111** of a coherent Raman spectroscopy system **100** may include one or more of the following: light-emitting diode (LED), super-luminescent light source, short-pulse laser, broadband light source, fiber laser, solid-state laser, quantum-cascade laser. For example, a light source that produces light over a relatively broad range of wavelengths (e.g., a super-luminescent light source, short-pulse laser, or broadband light source) may be used to investigate a sample over the broad range of wavelengths without having to use a wavelength-tunable light source.

[0075] FIG. **6** illustrates an expanded view of a portion of the Raman signal **160** of FIG. **5**. The portion of FIG. **5** enclosed by the dashed-line box is expanded in FIG. **6**. The peak of the Raman signal **160** in FIG. **6** is approximately coincident with the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S**, and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** overlaps the Raman signal.

[0076] The Raman signal **160** in FIG. **6** is an optical signal with a spectral linewidth of $\Delta\nu_{\text{sub.R}}$. The spectral linewidth of a Raman signal **160** may have a value between approximately 30 GHz and 300 GHz (or, in wavenumbers, between approximately 1 cm.sup.-1 and approximately 10 cm.sup.-1). The spectral linewidth $\Delta\nu_{\text{sub.R}}$ of a Raman signal **160** may represent the spectral width of a peak of the Raman signal (e.g., a full-width-at-half-maximum of the peak) or may represent an approximate extent or width of the full Raman signal. For example, the spectral

linewidth $\Delta\nu_{\text{sub.R}}$ of the Raman signal **160** in FIG. **13** corresponds to a full-width-at-half-maximum of a peak of the Raman signal. In FIG. **6**, the spectral linewidth $\Delta\nu_{\text{sub.R}}$ of the Raman signal **160** corresponds to an extent or width of the full Raman signal. For example, the spectral linewidth of a Raman signal **160** may equal a spectral width at which an envelope of the Raman signal has decreased to a particular level (e.g., to 50%, 20%, or 10% of a peak value). The envelope may be a curve that decreases monotonically away from a peak of the Raman signal and approximately follows an overall shape of the Raman signal. In FIG. **6**, the dashed-line curve that traces the peaks of the Raman signal **160** represents an envelope of the Raman signal, and the spectral linewidth $\Delta\nu_{\text{sub.R}}$ corresponds to a full-width-at-10% of the Raman-signal envelope (i.e., the points at which the envelope has decreased to 10% of its maximum value).

[0077] In a coherent Raman spectroscopy system **100**, the difference between the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be greater than a low-frequency limit $F_{\text{sub.1}}$ and less than a high-frequency limit $F_{\text{sub.2}}$ (i.e., $F_{\text{sub.1}} < |\nu_{\text{sub.2}} - \nu_{\text{sub.3}}| < F_{\text{sub.2}}$). For example, the low-frequency limit $F_{\text{sub.1}}$ may be approximately 1 MHz, 10 MHz, 50 MHz, 100 MHz, 200 MHz, 500 MHz, or 1 GHz, and the high-frequency limit $F_{\text{sub.2}}$ may be approximately 10 GHz, 20 GHz, 50 GHz, 100 GHz, 200 GHz, 500 GHz, or 1 THz. As another example, the low-frequency limit $F_{\text{sub.1}}$ may be related to the spectral linewidth $\Delta\nu_{\text{sub.pr}}$ of the probe beam **120pr** and the spectral linewidth $\Delta\nu_{\text{sub.s}}$ of the Stokes beam **120S** (e.g., $F_{\text{sub.1}}$ may be greater than $\Delta\nu_{\text{sub.pr}} + \Delta\nu_{\text{sub.s}}$). As another example, the high-frequency limit $F_{\text{sub.2}}$ may be related to the spectral linewidth $\Delta\nu_{\text{sub.R}}$ of a Raman signal **160** (e.g., $F_{\text{sub.2}}$ may be approximately equal $(0.5) \Delta\nu_{\text{sub.R}}$, $\Delta\nu_{\text{sub.R}}$, or $2\Delta\nu_{\text{sub.R}}$). As another example, $F_{\text{sub.1}}$ may be 100 MHz and $F_{\text{sub.2}}$ may be 200 GHz, which indicates that the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may differ by greater than 100 MHz and less than 200 GHz (i.e., $100 \text{ MHz} < |\nu_{\text{sub.2}} - \nu_{\text{sub.3}}| < 200 \text{ GHz}$). The frequencies $F_{\text{sub.1}}$ and $F_{\text{sub.2}}$ represent the frequency range with respect to the Stokes-beam frequency $\nu_{\text{sub.2}}$ over which the probe-beam frequency $\nu_{\text{sub.3}}$ may be scanned. The two hatched rectangles along the frequency axis in FIG. **6** represent the allowed frequency ranges for the probe beam **120pr**. The frequency of the probe beam **120pr** may be (i) between $\nu_{\text{sub.2}} - F_{\text{sub.2}}$ and $\nu_{\text{sub.2}} - F_{\text{sub.1}}$ or (ii) between $\nu_{\text{sub.2}} + F_{\text{sub.1}}$ and $\nu_{\text{sub.2}} + F_{\text{sub.2}}$. In an optical receiver **200**, the probe beam **120pr** and the Raman signal **160** may be coherently mixed together at an optical detector **220**. The frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be kept away from the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** by at least the low-frequency limit $F_{\text{sub.1}}$ (e.g., $|\nu_{\text{sub.2}} - \nu_{\text{sub.3}}| > F_{\text{sub.1}}$) to avoid mixing between the probe and Stokes beams, which could cause the detector to produce an unwanted electronic signal not related to the mixing between the probe beam and Raman signal. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be kept within the high-frequency limit $F_{\text{sub.2}}$ of the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** (e.g., $|\nu_{\text{sub.2}} - \nu_{\text{sub.3}}| < F_{\text{sub.2}}$), since measurements outside the high-frequency limit may not produce a significant electronic signal.

[0078] FIG. **7** illustrates an expanded view of a portion of the Raman signal **160** of FIG. **6**. The portion of FIG. **6** enclosed by the dashed-line box is expanded in FIG. **7**. The peak of the Raman signal **160** in FIG. **7** is located at frequency $\nu_{\text{sub.RP}}$, and the Stokes beam **120S** is located at the Stokes frequency $\nu_{\text{sub.2}}$. The Raman-signal peak frequency $\nu_{\text{sub.RP}}$ may correspond to a Raman peak of a material that is part of a sample **150** being measured by a Raman spectroscopy system **100**. For example, the difference between the pump-beam frequency $\nu_{\text{sub.1}}$ and the Raman-signal peak frequency $\nu_{\text{sub.RP}}$ may equal a vibrational frequency Q of the material (e.g., $\nu_{\text{sub.1}} - \nu_{\text{sub.RP}} = Q$). In some embodiments, a Stokes light source **1115** may be operated so that the Stokes-beam frequency $\nu_{\text{sub.2}}$ is approximately equal to the Raman-signal peak frequency $\nu_{\text{sub.RP}}$, and in other embodiments (e.g., as illustrated in FIG. **7**), the Stokes-beam frequency $\nu_{\text{sub.2}}$ may be slightly off-resonance or detuned with respect to the Raman-signal peak frequency $\nu_{\text{sub.RP}}$. For example, in FIG. **7**, the Stokes-beam frequency $\nu_{\text{sub.2}}$ may differ from the Raman-

signal peak frequency $\nu_{\text{sub.RP}}$ by less than or equal to 30 GHz, 10 GHz, 5 GHz, or 1 GHz.

[0079] The beams of light **120** produced by the pump, Stokes, and probe light sources **111** in a Raman spectroscopy system **100** may each have a spectral linewidth of less than 200 MHz. Additionally, one or more of the beams of light **120** may have a spectral linewidth of less than 1 MHz. For example, the spectral linewidth of a beam of light **120** may be less than 200 MHz, 100 MHz, 50 MHz, 10 MHz, 1 MHz, or 100 kHz. In FIG. 7, the spectral linewidth $\Delta\nu_{\text{sub.pr}}$ of the probe beam **120_{pr}** and the spectral linewidth $\Delta\nu_{\text{sub.s}}$ of the Stokes beam **120_S** may each be less than 200 MHz. In some embodiments, the probe light source **111_{pr}** may be configured to produce a probe beam **120_{pr}** having a relatively narrow spectral linewidth, which may allow a Raman spectroscopy system **100** to measure a Raman spectrum with a high degree of spectral resolution. For example, the spectral linewidth $\Delta\nu_{\text{sub.pr}}$ of the probe beam **120_{pr}** in FIG. 7 may be less than 1 MHz.

[0080] An optical receiver **200** of a coherent Raman spectroscopy system **100** may include one or more detectors **220**, where each detector is configured to coherently mix a portion of a Raman signal **160** with at least a portion of a probe beam of light **120_{pr}** to produce an electronic signal. For an optical receiver **200** with a single detector **220**, all or most of the probe beam of light **120_{pr}** may be mixed with the Raman signal **160**. For an optical receiver **200** with multiple detectors **220**, the probe beam of light **120_{pr}** may be split so that a portion of the probe beam of light is sent to each of the detectors. For example, in an optical receiver **200** with four detectors **220**, the probe beam **120_{pr}** may be split into four portions, and each detector may receive one of the four portions of the probe beam. Similarly, for an optical receiver **200** with a single detector **220**, all or most of the Raman signal **160** may be sent to the single detector, and for an optical receiver with multiple detectors **220**, the Raman signal **160** may be split so that a portion of the Raman signal is sent to each of the detectors.

[0081] Coherent mixing of a probe beam **120_{pr}** and a Raman signal **160**, which may be referred to as heterodyne detection, may occur when the two optical signals are optically combined and then detected by a detector **220**. Optically combining the probe beam **120_{pr}** and the Raman signal **160** may refer to combining the two optical signals so that their electric fields are summed together. For example, the probe beam and Raman signal may be combined (e.g., with an optical combiner **130**) so that the two signals are substantially coaxial and travel together in the same direction and along approximately the same optical path. Additionally, the probe beam and Raman signal may be combined so that at least a portion of their polarizations have the same orientation to allow at least a portion of their electric fields to be summed together. Once the probe beam and Raman signal are optically combined to produce a combined probe-Raman signal **210**, the probe beam and Raman signal may be coherently mixed at a detector **220**. The detector **220** may produce a photocurrent signal i corresponding to the coherent mixing of the probe beam **120_{pr}** and a portion of the Raman signal **160**.

[0082] The portion of a Raman signal **160** that is coherently mixed with a probe beam of light **120_{pr}** at a detector **220** to produce an electronic signal may refer to a spectral portion of the Raman signal. The spectral portion of a Raman signal **160** that is coherently mixed with a probe beam **120_{pr}** may include optical frequency components of the Raman signal that are within a particular frequency range of the frequency $\nu_{\text{sub.3}}$ of the probe beam of light, where the particular frequency range is based on or depends on the electronic bandwidth Δf of the optical detector **220**. In FIG. 7, the hatched region around the probe frequency $\nu_{\text{sub.3}}$ represents the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120_{pr}**. The particular frequency range illustrated by the hatched region extends from $\nu_{\text{sub.3}} - \Delta f$ to $\nu_{\text{sub.3}} + \Delta f$, where Δf is the electronic bandwidth of the detector. The particular frequency range includes the optical frequency components of the Raman signal **160** that are coherently mixed with the probe beam **120_{pr}** to produce an electronic signal.

[0083] The electronic signal produced by the detector **220** in response to the coherent mixing of the

portion of the Raman signal **160** and probe beam **120pr** may include one or more electronic frequency components, where each electronic frequency component has a frequency less than or equal to approximately Δf . For example, the electronic bandwidth Δf of the optical detector may be 10 GHz, and the electronic signal produced by the detector may include one or more electronic frequency components having frequencies less than or equal to approximately 10 GHz. Other optical frequency components of the Raman signal **160** that are outside the hatched region (e.g., optical frequencies less than $\nu_{\text{sub.3}} - \Delta f$ and greater than $\nu_{\text{sub.3}} + \Delta f$) may produce a coherent-mixing response in the detector **220**. However, since these Raman-signal optical frequency components would produce an electronic response at frequencies greater than Δf (which is outside of the electronic bandwidth of the detector), these optical frequency components will not result in any significant contribution to the electronic signal. The electronic bandwidth of the detector **220** effectively limits or filters the optical frequency components of the Raman signal **160** that are measured by the optical receiver **200** to optical frequency components that are within a particular frequency range of the probe frequency. Accordingly, the electronic bandwidth of the detector **220**, in combination with the relatively narrow spectral linewidth of the probe beam **120pr**, may allow a Raman spectroscopy system **100** to measure a Raman signal **160** with a high degree of spectral resolution. Herein, an optical frequency or an optical frequency component refers to a signal in the optical domain between approximately 60 THz and approximately 1,000 THz, and an electronic frequency or an electronic frequency component refers to a signal in the electronic domain between 0 Hz and approximately 50 GHz.

[0084] The electronic signal that results from coherent mixing of a Raman signal **160** and a probe beam **120pr** may include a coherent-mixing term that is proportional to a product of (i) $E_{\text{sub.R}}$, the amplitude of the electric field of the Raman signal and (ii) $E_{\text{sub.pr}}$, the amplitude of the electric field of the probe beam. The photocurrent signal i produced by a detector **220** in response to the coherent mixing of a Raman signal **160** and a probe beam **120pr** may be proportional to the square of the summed electric fields of the probe beam **120pr** and a spectral portion of the Raman signal **160**. This type of detector **220** that produces a photocurrent signal i that is proportional to the square of a received electric field may be referred to as a square-law detector. The photocurrent signal i may be expressed as $i(t) = k |\epsilon_{\text{sub.R}}(t) + \epsilon_{\text{sub.pr}}(t)|^2$, where k is a constant (e.g., k may account for the responsivity of the detector **220** as well as other constant parameters or conversion factors). For clarity, the constant k or other constants (e.g., conversion constants or factors of 2 or 4) may be excluded from expressions herein related to the photocurrent i or the voltage signal **234**. In the above expression for $i(t)$, $\epsilon_{\text{sub.R}}(t)$ is the electric field of the Raman signal **160**, and $\epsilon_{\text{sub.pr}}(t)$ is the electric field of the probe beam **120pr**. The electric field of the Raman signal **160** may be expressed as $E_{\text{sub.R}} \cos [2\pi\nu_{\text{sub.R}}t + \phi_{\text{sub.R}}]$, where $E_{\text{sub.R}}$ is the amplitude of the electric field of the Raman signal. The electric field of the probe beam **120pr** may be expressed as $E_{\text{sub.pr}} \cos [2\pi\nu_{\text{sub.3}}t + \phi_{\text{sub.pr}}]$, where $E_{\text{sub.pr}}$ is the amplitude of the electric field of probe beam. The frequency $\nu_{\text{sub.R}}$ is the optical frequency of the electric field of the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr**. The frequency $\nu_{\text{sub.R}}$ may include the optical frequency components of the Raman signal **160** from $\nu_{\text{sub.3}} - \Delta f$ to $\nu_{\text{sub.3}} + \Delta f$, where Δf is the electronic bandwidth of the detector. The frequency $\nu_{\text{sub.3}}$ is the optical frequency of the electric field of the probe beam **120pr**. The term $\phi_{\text{sub.R}}$ is the phase of the electric field of the Raman signal **160**, and the term $\phi_{\text{sub.pr}}$ is the phase of the electric field of the probe beam **120pr**.

[0085] The above expression for the photocurrent signal i may be expanded and written as $i(t) = E_{\text{sub.R}}^2 + E_{\text{sub.pr}}^2 + 2E_{\text{sub.R}}E_{\text{sub.pr}} \cos [2\pi(\nu_{\text{sub.R}} - \nu_{\text{sub.3}})t + \Delta\phi]$, where, for clarity, the constant k is not included. In this expanded expression for the photocurrent signal $i(t)$, the first term $E_{\text{sub.R}}^2$ corresponds to the optical power (PR) of the Raman signal **160**, and the second term $E_{\text{sub.pr}}^2$ corresponds to the optical power ($P_{\text{sub.pr}}$) of the probe beam **120pr**. The third term in the above expression is $2E_{\text{sub.R}}E_{\text{sub.pr}} \cos [2\pi(\nu_{\text{sub.R}} - \nu_{\text{sub.3}})t + \Delta\phi]$ and may

be referred to as a coherent-mixing term that represents coherent mixing between the electric fields of the Raman signal **160** and probe beam **120pr**. The phase difference $\Delta\phi$ is the phase difference between the electric fields of the Raman signal and the probe beam (e.g., $\Delta\phi = \phi_{\text{sub.R}} - \phi_{\text{sub.pr}}$). The coherent-mixing term is proportional to $E_{\text{sub.R}} \times E_{\text{sub.pr}}$, which is the product of the electric-field amplitudes of the Raman signal **160** and the probe beam **120pr**. Additionally, the coherent-mixing term includes a cosine function that varies in time based on the frequency difference ($\nu_{\text{sub.R}} - \nu_{\text{sub.3}}$) between the Raman signal **160** and the probe beam **120pr**. Since the spectral portion of the Raman signal **160** that is coherently mixed with the probe beam **120pr** includes the optical frequency components of the Raman signal from $\nu_{\text{sub.3}} - \Delta f$ to $\nu_{\text{sub.3}} + \Delta f$, the frequency-difference term ($\nu_{\text{sub.R}} - \nu_{\text{sub.3}}$) may include frequency components from zero to Δf . Accordingly, the coherent-mixing term may be referred to as including multiple electronic frequency components, where each electronic frequency component is proportional to $E_{\text{sub.R}} E_{\text{sub.pr}} \cos [2\pi f t + \Delta\phi]$. The frequency f , which may be referred to as an electronic frequency, is equal to the frequency difference ($\nu_{\text{sub.R}} - \nu_{\text{sub.3}}$) and has a value between zero and Δf . The coherent-mixing term may also be expressed as $2\sqrt{\{ \text{square root over } (P_{\text{sub.R}}) \}} \sqrt{\{ \text{square root over } (P_{\text{sub.pr}}) \}} \cos [2\pi f t + \Delta\phi]$, where $P_{\text{sub.R}}$ is the optical power of the Raman signal **160** and $P_{\text{sub.pr}}$ is the optical power of the probe beam **120pr**.

[0086] FIGS. **8-10** illustrate time-domain and frequency-domain plots of an example electronic signal resulting from coherent mixing of the Raman signal **160** and probe beam **120pr** of FIG. **7**. The electronic signal produced by a detector **220** may be a photocurrent signal i , and an electronic amplifier **232** may produce a voltage signal **234** that corresponds to the photocurrent signal i . The voltage signal **234** may be approximately proportional to the photocurrent signal i , and both the voltage and photocurrent signals may be proportional to $P_{\text{sub.R}} + P_{\text{sub.pr}} + 2\sqrt{\{ \text{square root over } (P_{\text{sub.R}}) \}} \sqrt{\{ \text{square root over } (P_{\text{sub.pr}}) \}} \cos [2\pi f t + \Delta\phi]$, where the electronic frequency f is equal to the frequency difference ($\nu_{\text{sub.R}} - \nu_{\text{sub.3}}$). The electronic frequency f may take on multiple values (or a continuous range of values) between zero and Δf , and so, the voltage signal **234** in the time domain may be viewed as a summation over these multiple frequency components. In the frequency domain, the voltage signal **234** may have frequency components that extend from DC to Δf (or, for an amplifier **232** with a low-frequency cutoff filter, from the low-frequency cutoff to Δf). The two terms $P_{\text{sub.R}}$ and $P_{\text{sub.pr}}$ represent the optical powers of the Raman signal and probe beam. Over the time duration of a measurement, the powers of the Raman signal and probe beam may be approximately constant, and so the contribution of $P_{\text{sub.R}} + P_{\text{sub.pr}}$ in the above expression may correspond to a DC offset with little or no time variation. An electronic amplifier **232** may include a high-pass or band-pass filter that removes or attenuates the DC or low-frequency components, resulting in an electronic signal with little or no DC offset (e.g., as illustrated by the time-domain voltage signal **234** in FIG. **8**).

[0087] The signal characteristic **162** in each of FIGS. **7-10** represents a characteristic of an electronic signal (e.g., a characteristic of a photocurrent signal i or a characteristic of a corresponding voltage signal **234**) associated with a Raman signal **160**. A signal characteristic **162** may be determined from a digital signal **240**, where the digital signal is a digital representation of a photocurrent signal i or a voltage signal **234**. A characteristic **162** of an electronic signal may be determined by a processor of a Raman spectroscopy system **100** and may include one or more data points **163**, where each data point represents: a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency (e.g., an electronic frequency or an optical frequency), a phase, or a polarization. A Raman spectroscopy system **100** may measure one or more signal characteristics **162** associated with one or more Raman signals **160**, and based on the measured characteristics, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0088] FIG. **8** illustrates an example time-domain plot of a voltage signal **234** resulting from

coherent mixing of the Raman signal **160** and probe beam **120_{pr}** of FIG. 7. A digitizer **236** (e.g., an ADC) may produce a digital signal **240** that represents the voltage signal **234** in the time domain, and a processor may determine a characteristic **162** of the electronic signal based on the digital signal. The signal characteristic **162** in FIG. 8 includes a single data point **163_t** which represents a peak amplitude $A_{\text{sub.t}}$ of the time-domain voltage signal **234**. The signal characteristic **162** may also include the optical frequency $\nu_{\text{sub.3}}$ that the probe beam **120_{pr}** was set to when the electronic signal was obtained. Other time-domain-type data points **163** associated with the signal characteristic **162** in FIG. 8 may include (i) an amplitude of the time-domain voltage signal **234** at a particular time (e.g., at a temporal center) or (ii) an area associated with the voltage signal. For an area associated with the voltage signal **234**, a processor may first convert the voltage signal into a non-negative signal (e.g., by taking the absolute value or by squaring the values of the voltage signal) and then integrate the non-negative signal to determine an area under the curve.

[0089] A processor of a Raman spectroscopy system **100** may determine a Fourier transform of a digital signal **240**. A voltage signal **234** may be a time-domain signal, and the digital signal **240** may be a time-domain digital representation of the voltage signal **234**. The Fourier transform of the digital signal **240** may produce a frequency-domain representation of the voltage signal **234**. From the Fourier transform, the processor may determine one or more electronic frequency components of the voltage signal **234**. Determining a frequency component of a voltage signal **234** may include determining an amplitude of the frequency-domain voltage signal at a particular frequency (e.g., at 4 GHz). FIGS. 9-10 each illustrate an example frequency-domain plot of a voltage signal **234** resulting from coherent mixing of the Raman signal **160** and probe beam **120_{pr}** of FIG. 7. The frequency-domain plots may be determined by taking a Fourier transform (e.g., a discrete Fourier transform or a fast Fourier transform) of a time-domain digital signal **240** that represents a time-domain voltage signal **234**. The x-axis of each of the frequency-domain plots in FIGS. 9-10 is labeled as “electronic frequency” to clarify and distinguish it from the x-axis of the frequency-domain plots in FIGS. 3-7. The x-axis of each of the frequency-domain plots in FIGS. 3-7 may be referred to as an “optical frequency,” where the pump beam **120_{pu}**, Stokes beam **120_S**, Raman signal **160**, and probe beam **120_{pr}** each have frequencies in the optical domain (e.g., in the range of 60-1,000 THz). In contrast, the x-axis of each of the frequency-domain plots in FIGS. 9-10 includes frequencies in the electronic domain (e.g., in the range of DC to 50 GHz). Coherent mixing of a Raman signal **160** and probe beam **120_{pr}** (which are signals in the optical domain) produces an electronic signal that can be detected and analyzed using electronic techniques. The y-axis of each of the plots in FIGS. 8-10 is labeled “electronic signal amplitude” and may have units of voltage, current, or electrical power (e.g., watts).

[0090] In FIG. 9, the frequency-domain voltage signal **234** has nonzero frequency components that extend from DC (i.e., zero hertz) to the detector-bandwidth frequency Δf . In FIG. 10, the frequency-domain voltage signal drops off before reaching zero hertz, indicating that the corresponding voltage signal **234** may have been AC-coupled or high-pass filtered to remove the DC or low-frequency components. The signal characteristic **162** in FIG. 9 includes a single data point **163_f** which represents a peak amplitude $A_{\text{sub.p}}$ of the frequency-domain voltage signal **234**. The data point **163_f** may include the peak amplitude $A_{\text{sub.p}}$ or the frequency $f_{\text{sub.p}}$ at which the peak amplitude is located. The signal characteristic **162** in FIG. 10 includes five data points: data point **163_a** at frequency $f_{\text{sub.a}}$, data point **163_b** at frequency $f_{\text{sub.b}}$, data point **163_c** at frequency $f_{\text{sub.c}}$, data point **163_d** at frequency $f_{\text{sub.d}}$, and data point **163_e** at frequency $f_{\text{sub.e}}$. Each data point **163** in FIG. 10 may include an amplitude value (e.g., amplitude A_a may be associated with data point **163_a**) or a frequency (e.g., frequency $f_{\text{sub.a}}$ may be associated with data point **163_a**). The frequencies associated with the data points **163** in FIG. 10 may have particular values. For example, the detector bandwidth Δf may be 5 GHz, and the frequencies $f_{\text{sub.a}}$, $f_{\text{sub.b}}$, $f_{\text{sub.c}}$, $f_{\text{sub.d}}$, and $f_{\text{sub.e}}$ may have respective values of approximately 1 GHz, 2 GHz, 3 GHz, 4 GHz, and 5 GHz. Other frequency-domain-type data points **163** associated with the signal characteristic

162 in FIG. **9** or **10** may include (i) an amplitude of the frequency-domain voltage signal **234** at a particular frequency or at a center frequency or (ii) an area or an average amplitude associated with the frequency-domain voltage signal **234**. Additionally, the signal characteristic **162** in FIG. **9** or **10** may include the optical frequency $\nu_{\text{sub.3}}$ that the probe beam **120_{pr}** was set to when the voltage signal **234** was obtained.

[0091] A processor of a coherent Raman spectroscopy system **100** may associate a determined signal characteristic **162** with a Raman frequency shift. For example, the signal characteristic **162** in FIG. **7** may be associated with a Raman frequency shift having a frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3}}$, which is the difference between the pump-beam frequency $\nu_{\text{sub.1}}$ and the probe-beam frequency $\nu_{\text{sub.3}}$. A sample **150** under investigation may include a material with a Raman spectrum having a peak at a frequency Ω , which is approximately equal to $\nu_{\text{sub.1}} - \nu_{\text{sub.RP}}$, the difference between the pump-beam frequency $\nu_{\text{sub.1}}$ and the Raman-signal peak frequency $\nu_{\text{sub.RP}}$. The peak frequency Ω may be approximately equal to a vibrational frequency of the material and may correspond to a Raman frequency shift of Ω (e.g., when excited with light at a frequency ν , the material may produce Raman-shifted light at the frequencies $\nu + \Omega$ and $\nu - \Omega$). In FIG. **7**, the probe beam **120_{pr}** is being used to measure the Raman signal **160** at the probe-frequency $\nu_{\text{sub.3}}$, and the resulting signal characteristic **162** that is determined may be associated with a Raman spectrum of the material at the frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3}}$, which may be referred to as a Raman frequency shift of $\nu_{\text{sub.1}} - \nu_{\text{sub.3}}$.

[0092] FIG. **11** illustrates an example Raman signal **160** that is measured at multiple probe frequencies $\nu_{\text{sub.3}}$. The probe beam **120_{pr}** is tuned to multiple different probe frequencies ($\nu_{\text{sub.3-1}}$, $\nu_{\text{sub.3-2}}$, $\nu_{\text{sub.3-3}}$, . . . $\nu_{\text{sub.3-n}}$) to measure multiple respective signal characteristics (**162-1**, **162-2**, **162-3**, . . . **162-n**) of the Raman signal **160**. The Raman signal **160** is produced by coherent Raman scattering of pump and Stokes beams within a sample **150**. Throughout the measurements of the multiple signal characteristics **162** in FIG. **11**, the pump-beam frequency $\nu_{\text{sub.1}}$ and the Stokes-beam frequency $\nu_{\text{sub.2}}$ may remain substantially constant so that the resulting Raman signal **160** also remains substantially constant (e.g., the Raman signal may exhibit substantially the same shape and amplitude throughout the measurements). The number n of signal characteristics **162** that are measured may be 1, 5, 10, 50, 100, 500, 1,000, or any other suitable number of signal characteristics. At each of the n probe frequencies, the probe beam **120_{pr}** may be coherently mixed with a spectral portion of the Raman signal **160** that is within a particular frequency range of the probe frequency $\nu_{\text{sub.3}}$ (e.g., within Δf of the probe frequency, as illustrated in FIG. **7**) to measure one signal characteristic **162**. Each signal characteristic **162** may provide information about the Raman signal **160** in the frequency region from $\nu_{\text{sub.3}} - \Delta f$ to $\nu_{\text{sub.3}} + \Delta f$. Based on one or more measured signal characteristics **162**, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0093] At each of the n probe frequencies, a single measurement may be performed, or multiple measurements may be performed. For example, with the probe beam **120_{pr}** in FIG. **11** set to the probe-beam frequency $\nu_{\text{sub.3-1}}$, an optical receiver **200** may measure a single photocurrent signal i and produce a corresponding single digital output signal **240**. From that one output signal **240**, a processor may determine a signal characteristic **162-1**. Alternatively, with the probe beam **120_{pr}** in FIG. **11** set to the probe-beam frequency $\nu_{\text{sub.3-1}}$, an optical receiver **200** may measure a series of multiple photocurrent signals i (e.g., 2, 4, 10, 20, 50, 100, or any other suitable number of signals) and produce multiple corresponding digital output signals **240**. A processor may determine one signal characteristic **162-1** from the multiple digital output signals. For example, the processor may average or otherwise combine the multiple digital output signals **240** to produce one signal characteristic **162-1**. Measurement of a series of multiple photocurrent signals i may improve the measurement accuracy by averaging out noise or removing outliers from the measurements.

[0094] By tuning the probe-beam frequency $\nu_{\text{sub.3}}$ to multiple frequencies across at least a portion

of a Raman signal **160**, a coherent Raman spectroscopy system **100** may measure the Raman signal at multiple points **162**. For example, the probe-beam frequency $\nu_{\text{sub.3}}$ in FIG. **6** may be tuned across at least a portion of the frequency range between $\nu_{\text{sub.2}}-F_{\text{sub.2}}$ and $\nu_{\text{sub.2}}-F_{\text{sub.1}}$ or between $\nu_{\text{sub.2}}+F_{\text{sub.1}}$ and $\nu_{\text{sub.2}}+F_{\text{sub.2}}$. Each of the different probe-beam frequencies to which the probe beam **120_{pr}** is changed may be offset from an adjacent probe-beam frequency by a frequency increment ΔF between approximately 10 MHz and approximately 10 GHz. For example, the probe-beam frequency in FIG. **11** may be changed in frequency increments ΔF of approximately 5 GHz as it is tuned across at least a portion of the Raman signal **160**. Each signal characteristic **162** in FIG. **11** is plotted along the x-axis at the probe frequency $\nu_{\text{sub.3}}$ at which the signal characteristic was obtained. The y-axis in FIG. **11** may represent an amplitude or area associated with the signal characteristics **162**. For example, each signal characteristic **162** in FIG. **11** may be plotted along the y-axis at a value corresponding to an amplitude or an area of an electronic signal from which the signal characteristic was obtained.

[0095] A probe light source **111_{pr}** of a coherent Raman spectroscopy system **100** may include a wavelength-tunable laser, where the frequency $\nu_{\text{sub.3}}$ of the probe beam **120_{pr}** is adjustable by changing the wavelength of light produced by the wavelength-tunable laser. For example, a wavelength-tunable laser may be adjustable over a wavelength range having a width between approximately 10 nm and approximately 100 nm. As another example, the wavelength-tuning range of a wavelength-tunable laser may be between approximately 1000 nm and approximately 1100 nm, between approximately 1490 nm and approximately 1570 nm, or between approximately 1600 nm and approximately 1690 nm. A wavelength-tunable laser may be continuously tunable over a wavelength-tuning range or may be tunable to multiple discrete wavelengths within a wavelength-tuning range. For example, a wavelength-tunable laser may be continuously tunable to any wavelength between 1530 nm and 1570 nm. Alternatively, a wavelength-tunable laser may be tunable to a set of approximately 10, 100, or 1,000 discrete wavelengths between 1530 nm and 1570 nm (e.g., the wavelengths may be separated from one another by approximately 4 nm, 0.4 nm, or 0.04 nm, respectively). A probe light source **111_{pr}** may include a wavelength-tunable laser that sequentially changes the probe-beam frequency $\nu_{\text{sub.3}}$ to multiple different frequencies. For example, the probe-beam frequency in FIG. **6** may be tuned to approximately 100 different frequencies between the frequencies $\nu_{\text{sub.2}}-F_{\text{sub.2}}$ and $\nu_{\text{sub.2}}-F_{\text{sub.1}}$. At each of the different probe-beam frequencies, the probe beam **120_{pr}** and a spectral portion of the Raman signal **160** may be coherently mixed at a detector **220** to produce a corresponding electronic signal, and a processor may determine a signal characteristic **162** based on the electronic signal.

[0096] In FIG. **11**, the probe beam **120_{pr}** may initially be set to the probe-beam frequency $\nu_{\text{sub.3-1}}$ and coherently mixed at a detector **220** with a spectral portion of the Raman signal **160** around the probe frequency $\nu_{\text{sub.3-1}}$. For example, the spectral portion of the Raman signal **160** may include optical frequency components of the Raman signal from $\nu_{\text{sub.3-1}}-\Delta f$ to $\nu_{\text{sub.3-1}}+\Delta f$, where Δf is the electronic bandwidth of the optical detector **220**. A processor may determine a signal characteristic **162-1** based on the electronic signal resulting from the coherent mixing of the probe beam **120_{pr}** at frequency $\nu_{\text{sub.3-1}}$ and the associated spectral portion of the Raman signal **160**. After measuring the Raman signal **160** around the probe frequency $\nu_{\text{sub.3-1}}$, the probe light source **111_{pr}** may change the probe-beam frequency by a frequency change ΔF to the frequency $\nu_{\text{sub.3-2}}$. The probe frequency $\nu_{\text{sub.3-2}}$ is equal to $\nu_{\text{sub.3-1}}+\Delta F$, and the frequency change ΔF between adjacent frequencies may be between approximately 10 MHz and approximately 10 GHz. The frequency change ΔF may be a fixed value or may be dynamically adjusted during a measurement of a Raman signal **160**. After the probe-beam frequency is changed to $\nu_{\text{sub.3-2}}$, the probe beam **120_{pr}** may be coherently mixed with the spectral portion of the Raman signal **160** around the probe frequency $\nu_{\text{sub.3-2}}$ (e.g., the spectral portion may include optical frequency components of the Raman signal from $\nu_{\text{sub.3-2}}-\Delta f$ to $\nu_{\text{sub.3-2}}+\Delta f$). A processor may determine a signal characteristic **162-2** based on the electronic signal resulting from the coherent mixing of the probe

beam **120pr** at frequency $\nu_{\text{sub.3-2}}$ and the associated spectral portion of the Raman signal **160**. After measuring the Raman signal **160** around the probe frequency $\nu_{\text{sub.3-2}}$, the probe light source **111pr** may change the probe-beam frequency by the frequency change ΔF to the frequency $\nu_{\text{sub.3-3}}$, and a measurement of the Raman signal **160** around the frequency $\nu_{\text{sub.3-3}}$ may be performed. The Raman spectroscopy system **100** may sequentially change the frequency $\nu_{\text{sub.3}}$ of the probe light source and measure a spectral portion of the Raman signal **160** at each frequency until reaching the final probe frequency $\nu_{\text{sub.3-n}}$. During the measurements, the probe-beam frequency $\nu_{\text{sub.3}}$ may be tuned so that it avoids overlapping with the frequency $\nu_{\text{sub.2}}$ of the Stokes beam to prevent mixing between the probe and Stokes beams.

[0097] FIG. **12** illustrates an example Raman spectrum corresponding to the Raman signal of FIG. **11**. A Raman spectroscopy system **100** may measure a sample **150** and determine the signal characteristics **162** of the Raman signal **160** in FIG. **11**, and the corresponding Raman spectrum in FIG. **12** may represent the Raman spectrum of one or more materials that are part of the sample **150**. For example, the sample **150** may include glucose and the peak frequency $\nu_{\text{sub.1}}-\nu_{\text{sub.RP}}$ of the Raman spectrum may be approximately equal to a frequency Ω of a molecular vibration of glucose. A processor of the Raman spectroscopy system **100** may determine the Raman spectrum in FIG. **12** based on the signal characteristics **162** (and the associated probe-beam frequencies $\nu_{\text{sub.3}}$) of the Raman signal **160** in FIG. **11**. Each signal characteristic **162** of a Raman signal **160** measured at a probe frequency $\nu_{\text{sub.3}}$ may be associated with a Raman frequency shift having a frequency $\nu_{\text{sub.1}}-\nu_{\text{sub.3}}$, where $\nu_{\text{sub.1}}$ is the pump-beam frequency. To determine the Raman spectrum, each signal characteristic **162** may be transformed from its Raman-signal frequency $\nu_{\text{sub.3}}$ to a corresponding Raman-shift frequency $\nu_{\text{sub.1}}-\nu_{\text{sub.3}}$. For example, the processor may associate the signal characteristic **162-1** at frequency $\nu_{\text{sub.3-1}}$ in FIG. **11** with a Raman shift having the frequency $\nu_{\text{sub.1}}-\nu_{\text{sub.3-1}}$ in FIG. **12**. Similarly, the signal characteristic **162-2** at frequency $\nu_{\text{sub.3-2}}$ in FIG. **11** may be associated with a Raman shift having the frequency $\nu_{\text{sub.1}}-\nu_{\text{sub.3-2}}$ in FIG. **12**, and the signal characteristic **162-3** at frequency $\nu_{\text{sub.3-3}}$ in FIG. **11** may be associated with a Raman shift having the frequency $\nu_{\text{sub.1}}-\nu_{\text{sub.3-3}}$ in FIG. **12**. Additionally, the peak frequency $\nu_{\text{sub.RP}}$ of the Raman signal **160** in FIG. **11** may correspond to the Raman-signal peak in FIG. **12** with a frequency of $\nu_{\text{sub.1}}-\nu_{\text{sub.RP}}$, which in turn may correspond to the vibrational frequency Ω of a particular material.

[0098] Based on the Raman spectrum in FIG. **12**, a processor may determine (i) whether a particular material is present in a sample **150** or (ii) an amount or a concentration of the particular material in the sample **150**. For example, the processor may compare one or more peaks, signal characteristics **162**, or other features of the Raman spectrum in FIG. **12** to a previously determined Raman spectrum for glucose. If one or more peaks or characteristics **162** of the Raman spectrum in FIG. **12** match or line up with peaks or characteristics of the Raman spectrum for glucose, then the processor may determine that glucose is present in the sample **150**. Alternatively, if the Raman spectrum in FIG. **12** is missing one or more peaks or characteristics of the Raman spectrum for glucose, then the processor may determine that little or no glucose is present in the sample **150**. As another example, the processor may determine the amount or concentration of glucose in the sample **150** based on the Raman spectrum in FIG. **12**. The concentration of glucose in the sample **150** may be related to the amplitude or height of one or more peaks of the Raman spectrum in FIG. **12** (e.g., the glucose concentration may be approximately proportional to the height of one or more Raman peaks). The concentration of glucose may then be determined based at least in part on the height of the Raman peak located at the Raman-shift frequency $\nu_{\text{sub.1}}-\nu_{\text{sub.RP}}$.

[0099] FIG. **13** illustrates another example Raman signal **160** that is measured at multiple probe frequencies. The probe beam **120pr** is tuned to multiple different probe frequencies ($\nu_{\text{sub.3-1}}$, $\nu_{\text{sub.3-2}}$, $\nu_{\text{sub.3-3}}$, . . . $\nu_{\text{sub.3-n}}$) to measure multiple respective signal characteristics (**162-1**, **162-2**, **162-3**, . . . **162-n**) of the Raman signal **160**. The Raman signal **160** in FIG. **13** has a single peak centered at frequency $\nu_{\text{sub.RP}}$ with a spectral linewidth of $\Delta\nu_{\text{sub.R}}$. The Raman signal **160**

in FIG. 11 has one main peak located at frequency $\nu_{\text{sub.RP}}$ along with multiple smaller peaks located on either side of the main peak. A Raman signal **160** may include a single peak (e.g., as illustrated in FIG. 13) or may include multiple peaks (e.g., as illustrated in FIG. 11).

[0100] FIGS. 14-15 each illustrate a second example Raman signal obtained by changing the frequency offset Ω between a pump beam **120_{pu}** and a Stokes beam **120_S**. A coherent Raman spectroscopy system **100** may include a pump light source **111_{pu}** or a Stokes light source **111_S** with a wavelength-tunable laser, and the frequency offset Ω may be adjustable by changing the wavelength of the wavelength-tunable laser. A wavelength-tunable pump or Stokes laser may be continuously tunable over a wavelength-tuning range or may be tunable to multiple discrete wavelengths within a wavelength-tuning range. For example, Stokes beam **120** in FIG. 14 may be produced by a continuously tunable laser diode that can produce light at the frequencies $\nu_{\text{sub.2}}$ and $\nu_{\text{sub.2}}$. As another example, a Stokes light source **111_S** may include two fixed-wavelength laser diodes that operate at the respective frequencies $\nu_{\text{sub.2}}$ and $\nu_{\text{sub.2}}$. Similarly, pump beam **120_{pu}** in FIG. 15 may be produced by a continuously tunable laser diode that can produce light at the frequencies $\nu_{\text{sub.1}}$ and ν , or a pump light source **111_{pu}** may include two fixed-wavelength laser diodes that operate at the respective frequencies $\nu_{\text{sub.1}}$ and $\nu'_{\text{sub.1}}$.

[0101] In FIG. 14, the pump frequency $\nu_{\text{sub.1}}$ is fixed, and the Stokes frequency is changed from $\nu_{\text{sub.2}}$ to $\nu'_{\text{sub.2}}$ to change the frequency offset from $\Omega_{\text{sub.1}}$ to $\Omega_{\text{sub.2}}$. Frequency offset $\Omega_{\text{sub.1}}$ equals $\nu_{\text{sub.1}} - \nu_{\text{sub.2}}$, and frequency offset $\Omega_{\text{sub.2}}$ equals $\nu_{\text{sub.1}} - \nu'_{\text{sub.2}}$. In FIG. 15, the Stokes frequency $\nu_{\text{sub.2}}$ is fixed, and the pump frequency is changed from $\nu_{\text{sub.1}}$ to $\nu'_{\text{sub.1}}$ to change the frequency offset from $\Omega_{\text{sub.1}}$ to $\Omega_{\text{sub.2}}$. Frequency offset $\Omega_{\text{sub.1}}$ equals $\nu_{\text{sub.1}} - \nu_{\text{sub.2}}$, and frequency offset $\Omega_{\text{sub.2}}$ equals $\nu'_{\text{sub.1}} - \nu_{\text{sub.2}}$. In other embodiments, both the pump and Stokes frequencies may be changed to change the frequency offset (from one value to another).

[0102] FIG. 14 includes two Raman signals **160_a** and **160_b**. Initially, a Stokes light source **111_S** may produce a Stokes beam **120_S** at the frequency $\nu_{\text{sub.2}}$ to produce the frequency offset $\Omega_{\text{sub.1}}$ between the Stokes beam and the pump beam **120_{pu}**, where $\Omega_{\text{sub.1}} = \nu_{\text{sub.1}} - \nu_{\text{sub.2}}$. For example, the Stokes beam **120_S** may have a frequency $\nu_{\text{sub.2}}$ of 193 THz, and the pump beam **120_{pu}** may have a frequency $\nu_{\text{sub.1}}$ of 207 THz, which corresponds to a frequency offset $\Omega_{\text{sub.1}}$ of 14 THz (or, 467 cm^{-1}). The Raman signal **160_a** is produced by coherent Raman scattering of the Stokes beam **120_S** and pump beam **120_{pu}** within a sample **150**. The Raman signal **160_a** is centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120_S**, and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120_{pr}** overlaps the Raman signal **160_a** and is relatively close to the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120_S** (e.g., the probe-beam frequency $\nu_{\text{sub.3}}$ may be within 200 GHz of the Stokes-beam frequency $\nu_{\text{sub.2}}$). The probe beam **120_{pr}** and a spectral portion of the Raman signal **160_a** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120_{pr}** may be tuned across at least a portion of the Raman signal **160_a** to measure multiple signal characteristics **162** associated with the Raman signal.

[0103] After measuring the first Raman signal **160_a**, the Stokes light source **111_S** may change the frequency of the Stokes beam to produce a Stokes beam **120_{S'}** at the frequency $\nu'_{\text{sub.2}}$, resulting in a frequency offset of $\Omega_{\text{sub.2}}$ between the Stokes beam **120_{S'}** and the pump beam **120_{pu}**, where $\Omega_{\text{sub.2}} = \nu_{\text{sub.1}} - \nu'_{\text{sub.2}}$. For example, the Stokes beam **120_S** may be changed to a frequency $\nu'_{\text{sub.2}}$ of 182 THz, and the pump-beam frequency $\nu_{\text{sub.1}}$ may remain at 207 THz, which corresponds to a frequency offset $\Omega_{\text{sub.1}}$ of 25 THz (or, 834 cm^{-1}). The second Raman signal **160_b** is produced by coherent Raman scattering of the Stokes beam **120_{S'}** and pump beam **120_{pu}** within the sample **150**. The Raman signal **160_b** is centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120_{S'}**. Additionally, a probe light source **111_{pr}** may change the frequency of the probe beam to produce a probe beam **120_{pr'}** at a frequency $\nu_{\text{sub.3}}$ that overlaps the Raman signal **160_b** and is relatively close to the frequency $\nu'_{\text{sub.2}}$ of the Stokes beam **120_{S'}** (e.g., the probe-

beam frequency $\nu_{\text{sub.3}}$ may be within 200 GHz of the Stokes-beam frequency $\nu'_{\text{sub.2}}$).

[0104] The probe beam **120pr** and a spectral portion of the Raman signal **160b** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu'_{\text{sub.3}}$ of the probe beam **120pr** may be tuned across at least a portion of the second Raman signal **160b** to measure multiple signal characteristics **162** associated with the Raman signal.

[0105] The two Raman signals **160a** and **160b** in FIG. **14** may correspond to two respective peaks of a Raman spectrum of a material that is in the sample **150**. One Raman peak, associated with Raman signal **160a**, is located at a frequency of approximately $\Omega_{\text{sub.1}}$, and the frequency $\Omega_{\text{sub.1}}$ may correspond to a vibrational frequency of the material. The other Raman peak, associated with Raman signal **160b**, is located at a frequency of approximately $\Omega_{\text{sub.2}}$, and the frequency $\Omega_{\text{sub.2}}$ may correspond to another vibrational frequency of the material. A Raman spectroscopy system **100** may perform measurements of signal characteristics **162** at 1, 2, 4, 10, 20, or 50 different values of the frequency offset Ω . The frequency offset Ω between the Stokes beam **120S** and the pump beam **120pu** may be set to each of the different frequency-offset values to produce an associated Raman signal **160**. At each value of Ω , the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal. Then, the frequency of the Stokes beam or pump beam may be adjusted to the next value of Ω , where another measurement of an associated Raman signal **160** is performed. Based on the signal characteristics **162** associated with each of the values of the frequency offset Ω , a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0106] FIG. **15** includes the Raman signal **160c**. A first Raman signal (similar to Raman signal **160a** in FIG. **14**) may be approximately overlapped with the second Raman signal **160c** in FIG. **15**, and for clarity, the first Raman signal is not included in FIG. **15**. Initially, a Stokes light source **111S** may produce a Stokes beam **120S** at the frequency $\nu_{\text{sub.2}}$, and a pump light source **111pu** may produce a pump beam **120pu** at the frequency $\nu_{\text{sub.1}}$, which results in the frequency offset $\Omega_{\text{sub.1}}$ between the Stokes and pump beams in FIG. **15**. For example, the Stokes beam **120S** may have a frequency $\nu_{\text{sub.2}}$ of 193 THz, and the pump beam **120pu** may have a frequency $\nu_{\text{sub.1}}$ of 207 THz, which corresponds to a frequency offset $\Omega_{\text{sub.1}}$ of 14 THz (or, 467 $\text{cm}^{\text{sup.}}\text{--}1$). A first Raman signal (similar to Raman signal **160a** in FIG. **14**) is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu** within a sample **150**. The first Raman signal may be centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S**, and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may overlap the first Raman signal and may be relatively close to the frequency $\nu_{\text{sub.2}}$ of the Stokes beam of light **120S**. The probe beam **120pr** and a spectral portion of the first Raman signal may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be tuned across at least a portion of the first Raman signal to measure multiple signal characteristics **162** associated with the first Raman signal.

[0107] In FIG. **15**, after measuring the first Raman signal, the pump light source **111pu** may change the frequency of the pump beam **120pu** from $\nu_{\text{sub.1}}$ to $\nu'_{\text{sub.1}}$. This results in a frequency offset of $\Omega_{\text{sub.2}}$ between the Stokes beam **120S** and the pump beam **120pu'**, where $\Omega_{\text{sub.2}} = \nu'_{\text{sub.1}} - \nu_{\text{sub.2}}$. For example, the pump beam may be changed to a frequency $\nu'_{\text{sub.1}}$ of 218 THz, and the Stokes-beam frequency $\nu_{\text{sub.2}}$ may remain at 193 THz, which corresponds to a frequency offset $\Omega_{\text{sub.1}}$ of 25 THz (or, 834 $\text{cm}^{\text{sup.}}\text{--}1$). The second Raman signal **160c** is produced by coherent Raman scattering of the Stokes beam **120S** and pump beam **120pu'** within the sample **150**. As with the first Raman signal, the Raman signal **160c** is centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S**, and the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** overlaps the Raman signal **160c** and is relatively close to the frequency $\nu_{\text{sub.2}}$ of the Stokes beam of light **120S** (e.g., the probe-beam frequency $\nu_{\text{sub.3}}$ may be within 200 GHz of the Stokes-beam

frequency $\nu_{\text{sub.2}}$). The probe beam **120pr** and a spectral portion of the Raman signal **160c** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr** may be tuned across at least a portion of the Raman signal **160c** to measure multiple signal characteristics **162** associated with the Raman signal.

[0108] In FIG. **15**, the first Raman signal (which may be similar to Raman signal **160a** in FIG. **14**) and the second Raman signal **160c** may correspond to two different peaks of a Raman spectrum of a material that is in the sample **150**. One Raman peak, associated with the first Raman signal, is located at a frequency of approximately $\Omega_{\text{sub.1}}$, and the frequency $\Omega_{\text{sub.1}}$ may correspond to a vibrational frequency of the material. The other Raman peak, associated with Raman signal **160c**, is located at a frequency of approximately $\Omega_{\text{sub.2}}$, and the frequency $\Omega_{\text{sub.2}}$ may correspond to another vibrational frequency of the material. Based at least in part on the signal characteristics **162** associated with the two frequency offsets $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample.

[0109] FIG. **16** illustrates an example Raman signal **160** along with two probe beams of light **120pr-1** and **120pr-2**. The Raman signal **160** may be produced by coherent Raman scattering of the pump beam **120pu** and Stokes beam **120S** within a sample **150**. The frequency of the pump beam **120pu** is $\nu_{\text{sub.1}}$, and the frequency of the Stokes beam **120S** is $\nu_{\text{sub.2}}$, which corresponds to a frequency offset Ω equal to $\nu_{\text{sub.1}} - \nu_{\text{sub.2}}$. The Raman signal **160** is centered at or near the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** (e.g., the center frequency of the Raman signal **160** may be within 200 GHz of $\nu_{\text{sub.2}}$). The first probe beam **120pr-1** may be used to measure the Raman signal **160**, and the second probe beam **120pr-2** may be used to measure the pump beam **120pu**.

[0110] The frequency $\nu_{\text{sub.3}}$ of first probe beam **120pr-1** overlaps the Raman signal **160** and is relatively close to the frequency $\nu_{\text{sub.2}}$ of the Stokes beam **120S** (e.g., $\nu_{\text{sub.3}}$ may be within 200 GHz of $\nu_{\text{sub.2}}$). The probe beam **120pr-1** and a spectral portion of the Raman signal **160** may be coherently mixed at a detector **220** to produce an electronic signal, from which a signal characteristic **162** may be determined. Additionally, the frequency $\nu_{\text{sub.3}}$ of the probe beam **120pr-1** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal.

[0111] The frequency $\nu_{\text{sub.4}}$ of the second probe beam **120pr-2** is relatively close to the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu** (e.g., $\nu_{\text{sub.4}}$ may be within 50 GHz of $\nu_{\text{sub.1}}$). For example, the frequency $\nu_{\text{sub.4}}$ of the second probe beam **120pr-2** may be offset from the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu** by approximately 10 GHz, 5 GHz, or 1 GHz. Alternatively, the frequency $\nu_{\text{sub.4}}$ of the second probe beam **120pr-2** may be approximately equal to the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu**. After the pump beam **120pu** has interacted with the sample, the probe beam **120pr-2** may be coherently mixed with the pump beam. For example, after the pump and Stokes beams have produced the Raman signal **160** and after the pump beam has exited the sample, the pump and probe beams may be coherently mixed together at a detector **220** to produce an electronic signal, from which a signal characteristic may be determined.

[0112] FIG. **17** illustrates an example optical receiver **200** for measuring the Raman signal **160** and pump beam of light **120pu** from FIG. **16**. The combined pump-Stokes beam **140** (which includes pump beam **120pu** and Stokes beam **120S**) is directed to a sample **150**, which produces a Raman signal **160** in response to the pump and Stokes beams. The Raman signal **160** may be collected by one or more optical elements and directed to the optical receiver **200**. Additionally, residual light from the pump beam **120pu** may be collected and directed to the optical receiver **200** as a residual pump beam **120pu-2**. The residual pump beam **120pu-2** may include light from the pump beam **120pu** after the pump beam has interacted with and exited the sample **150**. The optical receiver **200** in FIG. **17** may be referred to as a two-channel optical receiver that includes two parallel

measurement channels for separately detecting and measuring the Raman signal **160** and the residual pump beam **120pu-2**. The first measurement channel detects the Raman signal **160** and includes probe laser **110pr-1**, detector **220-1**, and detection electronics **230-1**. The second measurement channel detects the residual pump beam **120pu-2** and includes probe laser **110pr-2**, detector **220-2**, and detection electronics **230-2**.

[0113] The Raman signal **160** and the residual pump beam **120pu-2** are directed to the combiner **130c**, which may be a dichroic beamsplitter, and the combiner **130c** reflects the Raman signal **160** and transmits the residual pump beam **120pu-2**. The combiner **130c** also transmits at least a portion of the probe beam **120pr-1** produced by the probe laser **110pr-1** and combines the probe beam **120pr-1** with the Raman signal **160** to produce a combined probe-Raman signal **210-1**. The probe-Raman signal **210-1** is sent to the detector **220-1**, where the probe beam **120pr-1** and a spectral portion of the Raman signal **160** are coherently mixed to produce a photocurrent signal i.sub.1. The detection electronics **230-1** receives the photocurrent signal i.sub.1 and produces a digital output signal **240-1** that corresponds to the photocurrent signal i.sub.1.

[0114] The combiner **130d** (which may be a dichroic or a non-dichroic beamsplitter) reflects at least a portion of the residual pump beam **120pu-2** and transmits at least a portion of the probe beam **120pr-2** produced by the probe laser **110pr-2**. The combiner **130d** combines the probe beam **120pr-2** with the residual pump beam **120pu-2** to produce a combined probe-pump signal **210-2**, which is sent to the detector **220-2**. The probe beam **120pr-2** and the residual pump beam **120pu-2** are coherently mixed at the detector **220-2** to produce a photocurrent signal i.sub.2, and the detection electronics **230-2** produces a digital output signal **240-2** that corresponds to the photocurrent signal i.sub.2.

[0115] The two digital output signals **240-1** and **240-2** may be sent to a processor which determines a signal characteristic **162** of each of the photocurrent signals i.sub.1 and i.sub.2 based on the digital output signals. Additionally, the frequency of the first probe beam **120pr-1** may be tuned across at least a portion of the Raman signal **160** to measure multiple signal characteristics **162** associated with the Raman signal. If the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu** remains fixed, the frequency $\nu_{\text{sub.4}}$ of the second probe beam **120pr-2** may also remain fixed. Alternatively, if the frequency $\nu_{\text{sub.1}}$ of the pump beam **120pu** is changed (e.g., to switch to a different frequency offset Ω), the frequency of the probe beam **120pr-2** may also be switched to maintain a particular frequency offset between the pump and probe frequencies.

[0116] Measurement of the residual pump beam **120pu-2** may be performed two or more times to determine how the power of the pump beam changes when the Raman signal **160** is produced. For example, the residual pump beam **120pu-2** may be measured once when the Stokes beam **120S** is turned off (and no Raman signal **160** is produced) and another time when the Stokes beam is turned on (and the Raman signal **160** is produced). A processor may determine the change in the power of the residual pump beam **120pu-2** associated with the Stokes beam **120S** being turned off and on. Since at least part of the Raman signal **160** may be produced by Stokes-shifted photons from the pump beam **120pu**, a decrease in the power of the residual pump beam **120pu-2** may correspond to the power of the Raman signal.

[0117] In another embodiment of a two-channel optical receiver, the optical receiver may not include a second probe laser **110pr-2**. Instead, the residual pump beam **120pu-2** may be sent to a detector **220** for direct detection without mixing the residual pump beam with another signal.

[0118] FIG. **18** illustrates an example Raman spectroscopy system **100** for measuring a Raman signal **160** produced by spontaneous Raman scattering. Instead of producing a Raman signal by coherent Raman scattering of pump and Stokes beams within a sample (e.g., as illustrated in FIGS. **1-2**), the Raman signal **160** in FIG. **18** is produced by spontaneous Raman scattering of the pump beam of light **120pu** within the sample **150**. While the production of the Raman signal **160** in FIG. **18** is different from that of the Raman spectroscopy systems of FIGS. **1-2**, the optical receiver **200** and the Raman-signal detection technique in FIG. **18** is similar to that of FIGS. **1-2**. In FIG. **18**, the

Raman signal **160** is detected by coherently mixing the Raman signal with a probe beam of light **120_{pr}**.

[0119] The Raman spectroscopy system **100** in FIG. **18** includes a pump light source **111_{pu}** that produces a pump beam of light **120_{pu}** at a pump frequency $\nu_{\text{sub.1}}$. The pump beam **120_{pu}** is directed to a sample **150** (e.g., by one or more optical elements), and the sample **150** produces a Raman signal **160** by spontaneous Raman scattering of light from the pump beam. The spontaneous Raman signal **160** is collected (e.g., by one or more optical elements) and directed to the optical receiver **200**. The optical receiver **200** includes a probe light source **111_{pr}** that produces a probe beam of light **120_{pr}** at a probe frequency $\nu_{\text{sub.3}}$, where the probe frequency overlaps the Raman signal. The optical combiner **130_b**, which may be a dichroic or a non-dichroic beamsplitter, combines the Raman signal **160** and the probe beam **120_{pr}** to produce a combined probe-Raman signal **210** that is directed to a detector **220**. The detector **220** coherently mixes a spectral portion of the Raman signal **160** with the probe beam **120_{pr}** to produce a photocurrent signal *i*. The detection electronics **230** may produce (i) an analog voltage signal that corresponds to the photocurrent signal *i* and (ii) a digital output signal **240** that corresponds to the photocurrent signal or the voltage signal. The digital output signal **240** may be sent to a processor, and the processor may determine a signal characteristic **162** of the photocurrent signal or voltage signal based on the digital output signal **240**.

[0120] FIG. **19** illustrates an example Raman signal produced by the Raman spectroscopy system of FIG. **18**. The Raman signal **160** produced by spontaneous Raman scattering of the pump beam **120_{pu}** has a peak frequency of $\nu_{\text{sub.RP}}$. The frequency offset Ω between the pump beam and the peak frequency of the Raman signal equals $\nu_{\text{sub.1}} - \nu_{\text{sub.RP}}$, and the frequency offset Ω may correspond to a vibrational frequency of a material that is part of the sample **150**. In FIG. **19**, the probe beam **120_{pr}** may be coherently mixed with a spectral portion of the Raman signal **160** that is within a particular frequency range of the probe frequency $\nu_{\text{sub.3}}$ to measure one signal characteristic **162**. Additionally, the probe light source **111_{pr}** in FIG. **18** may include a wavelength-tunable laser that tunes the probe beam **120_{pr}** to multiple frequencies across at least a portion of the Raman signal **160**, and the optical receiver **200** may measure multiple respective signal characteristics **162** associated with the Raman signal.

[0121] FIG. **20** illustrates an example laser diode **110** that produces a free-space beam of light **120**. A free-space optical beam refers to a beam of light that propagates through a medium (e.g., air, liquid, or a dielectric material) without being optically confined within a physical waveguiding structure like an optical fiber or optical waveguide. The laser diode **110** in FIG. **20** may be part of a pump light source **111_{pu}**, a Stokes light source **111_S**, or a probe light source **111_{pr}**, and the free-space beam **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**. For example, the probe laser **110_{pr}** in FIG. **2** may be similar to the laser diode **110** in FIG. **20**, and the probe laser **110_{pr}** may produce a free-space probe beam **120_{pr}** that is combined with the Raman signal **160** by a free-space beam combiner **130_b**. In FIG. **20**, the electronic driver **112** supplies laser-diode drive current to the laser diode **110**, and the laser diode produces output light that is collimated by a lens **114** to produce a collimated free-space beam **120**. In other embodiments, a lens **114** may produce a focused free-space beam **120** (e.g., the lens may focus the free-space beam onto a sample **150**). The laser current/supplied to the laser diode **110** may be a substantially constant DC current resulting in an output beam of light **120** having a substantially constant optical power. Additionally or alternatively, the laser current/may include pulses of current resulting in an output beam **120** that includes corresponding pulses of light.

[0122] The output beam of light **120** produced by the laser diode **110** may have a spectral linewidth of less than approximately 200 MHz, 100 MHz, 50 MHz, 10 MHz, 1 MHz, or 100 kHz. The laser diode **110** in FIG. **20** may be a wavelength-tunable laser diode where the wavelength of the output beam **120** is adjustable over a wavelength range having a width between approximately 10 nm and approximately 100 nm. For example, the operating wavelength of the laser diode **110** may be

tunable over at least a portion of one of the following wavelength ranges: 1000 nm to 1100 nm; 1220 nm to 1450 nm; 1490 nm to 1570 nm; 1600 nm to 1690 nm. Alternatively, the laser diode **110** in FIG. **20** may be a fixed-wavelength laser diode. For example, the laser diode **110** in FIG. **20** may be a distributed feedback (DFB) laser diode with a spectral linewidth of less than 1 MHz, and the output beam **120** may have any suitable substantially fixed wavelength between approximately 600 nm and approximately 2000 nm.

[0123] FIG. **21** illustrates an example laser diode **110** that produces seed light **122** that is amplified by a semiconductor optical amplifier (SOA) **124**. Instead of directly emitting an output beam **120** (e.g., as illustrated in FIG. **20**), the light from a laser diode **110** may first be amplified by an optical amplifier. The laser diode **110** in FIG. **21** acts as a seed laser that produces seed light **122** that is coupled into the input end of the waveguide **125** of the SOA **124**. The SOA waveguide **125** in FIG. **21** is indicated by the cross-hatched region within the SOA **124**. The SOA **124** amplifies the seed light as it propagates within the waveguide from the input end to the output end, and the output beam of light **120** is emitted from the output end of the SOA. The optical gain provided by the SOA may come from electrical current that is supplied to the SOA by an electronic driver (not illustrated in FIG. **21**). For example, an electronic driver **112** may supply substantially constant DC electrical current to the laser diode **110** and to the SOA **124**, and the resulting output beam **120** may have substantially constant optical power. The output beam **120** may be a free-space beam, or the output beam **120** may be coupled into an optical fiber or into a waveguide of a photonic integrated circuit (PIC).

[0124] The laser diode **110** and the SOA **124** in FIG. **21** may be fabricated or integrated together on the same chip so that seed light **122** from the laser diode is directly coupled into the waveguide **125** of the SOA. The waveguide **125** of the SOA **124** may be a tapered optical waveguide (as illustrated in FIG. **21**) with a width that increases along a lateral direction from the input end that receives the seed light **122** to the output end that emits the output beam **120**. A light source that includes a seed laser diode **110** that supplies seed light **122** that is amplified by a SOA **124** (as illustrated in FIG. **21**) may be referred to as a master-oscillator power-amplifier laser (MOPA laser). The seed laser diode **110** may be referred to as a master oscillator, and the SOA **124** may be referred to as a power amplifier. The MOPA laser in FIG. **21** may be part of a pump light source **111_{pu}**, a Stokes light source **111₅**, or a probe light source **111_{pr}**, and the output beam of light **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**.

[0125] FIG. **22** illustrates an example laser diode **110** that produces seed light **122** that is amplified by a fiber-optic amplifier **126**. The laser diode **110** in FIG. **22** acts as a seed laser that produces seed light **122** that is coupled into an optical fiber **116**. The optical fiber **116** directs the seed light **122** to the fiber-optic amplifier **126**, and the fiber-optic amplifier amplifies the seed light as it propagates through an optical gain fiber of the fiber-optic amplifier. The optical gain fiber may be doped with rare-earth ions (e.g., neodymium, erbium, or ytterbium) or bismuth that provide the optical gain to the seed light. One or more pump lasers may optically pump the active material (e.g., rare-earth ions or bismuth ions) in the optical gain fiber, which in turn provide optical amplification to the seed light **122** propagating through the gain fiber. The amplified seed light produced by the fiber-optic amplifier **126** propagates in an optical fiber as a fiber-coupled output beam **120**. The laser diode **110** and fiber-optic amplifier **126** in FIG. **22** may be part of a pump light source **111_{pu}**, a Stokes light source **111₅**, or a probe light source **111_{pr}**, and the fiber-coupled output beam **120** may be directed to an optical combiner **130**, a sample **150**, or a detector **220**.

[0126] A pump light source **111_{pu}**, a Stokes light source **111₅**, or a probe light source **111_{pr}** may include a seed laser diode **110** followed by an optical amplifier. The seed laser diode **110** may produce seed light **122** that is amplified by the optical amplifier to produce an output beam of light **120**. An optical amplifier may include a SOA **124** (e.g., as illustrated in FIG. **21**) or a fiber-optic amplifier **126** (e.g., as illustrated in FIG. **22**). In some embodiments, an optical amplifier may include a SOA **124** followed by a fiber-optic amplifier **126**. For example, a seed laser diode **110**

may produce seed light **122** that is first amplified by a SOA **124** and then further amplified by a fiber-optic amplifier **126**.

[0127] In some embodiments, a pump light source **111_{pu}**, a Stokes light source **111_S**, or a probe light source **111_{pr}** may include a laser diode **110** and an optical fiber **116** and may not include a fiber-optic amplifier. For example, light produced by a laser diode **110** may be coupled into an optical fiber **116** to produce a fiber-coupled beam **120**, and the optical fiber may direct the laser-diode light to an optical combiner **130**, a sample **150**, or a detector **220**. A laser diode **110** that produces a fiber-coupled beam **120** may be referred to as a fiber-coupled laser diode.

[0128] FIG. **23** illustrates an example sampled-grating distributed Bragg reflector (SG-DBR) laser **110**. An SG-DBR laser **110** is a wavelength-tunable laser diode that produces an output beam **120** that can be tuned over a wavelength range having a width of between 20 nm and 50 nm. For example, an SG-DBR laser **110** may have a 40-nm wavelength-tuning range from approximately 1530 nm to approximately 1570 nm or from approximately 1630 nm to approximately 1670 nm. The SG-DBR laser **110** in FIG. **23** may be part of a pump light source **111_{pu}**, a Stokes light source **111_S**, or a probe light source **111_{pr}**, and the output beam **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**. For example, an SG-DBR laser **110** may be part of a probe light source **111_{pr}** that produces the probe beam **120_{pr}** in FIG. **6**, and the SG-DBR laser may tune the frequency $\nu_{\text{sub.3}}$ of the probe beam across at least a portion of the Raman signal **160**. An SG-DBR laser **110** may produce a free-space beam **120**, a fiber-coupled beam **120**, or a beam that is coupled into a waveguide of a PIC. Alternatively, an SG-DBR laser **110** may be integrated with a SOA **124** that amplifies the light produced by the SG-DBR laser (e.g., as illustrated in FIG. **21**).

[0129] The SG-DBR laser **110** in FIG. **23** includes a back mirror **180**, a phase section **182**, a gain section **184**, and a front mirror **186**, where the phase and gain sections are located between the front and back mirrors. The laser diode current I supplied to the SG-DBR laser **110** includes the following: current $I_{\text{sub.b}}$ supplied to the back mirror **180**, current $I_{\text{sub.p}}$ supplied to the phase section **182**, current $I_{\text{sub.g}}$ supplied to the gain section **184**, and current $I_{\text{sub.t}}$ supplied to the front mirror **186**. The gain current $I_{\text{sub.g}}$ provides optical gain to the active optical waveguide **188** of the SG-DBR laser **110**, and the other currents may be used to set the wavelength of the output beam **120** produced by the SG-DBR laser. The electronic driver **112** may supply particular combinations of electrical currents to the back mirror **180**, phase section **182**, gain section **184**, and front mirror **186**, where each particular combination of electrical currents causes the SG-DBR laser **110** to produce an output beam **120** at a particular wavelength. For example, an SG-DBR laser **110** may be part of a wavelength-tunable probe light source **111_{pr}** that produces the probe beam **120_{pr}** in FIG. **11**, and the electronic driver **112** may supply particular different combinations of the electrical currents $I_{\text{sub.b}}$, $I_{\text{sub.p}}$, $I_{\text{sub.g}}$, and $I_{\text{sub.t}}$ to produce the different probe frequencies $\nu_{\text{sub.3-1}}$, $\nu_{\text{sub.3-2}}$, $\nu_{\text{sub.3-3}}$, . . . and $\nu_{\text{sub.3-n}}$ to tune the probe beam across at least a portion of the Raman signal **160** in FIG. **11**.

[0130] FIG. **24** illustrates an example light source **111** with multiple laser diodes **110** and an optical multiplexer **118** that combines light produced by the laser diodes into a single output beam of light **120**. Each of the laser diodes **110-1**, **110-2**, . . . **110-N** produces a respective output beam **120-1**, **120-2**, . . . **120-N**, and the optical multiplexer **118** combines the output beams into the output beam of light **120**. The light source **111** may be configured to switch between operating the N laser diodes one at a time so that, at any given time, only one laser diode produces light, and the output beam **120** includes just the light produced by that one laser diode.

[0131] The optical multiplexer **118** may be a free-space device, a fiber-optic device, a waveguide-based device fabricated in a PIC, or a metamaterial-based device, and the multiplexer may combine N different wavelengths of light from the N laser diodes into a single output beam **120**. The optical multiplexer **118** may include one or more of the following: a free-space diffraction grating; an arrayed waveguide grating (AWG); a metamaterial that acts as a diffractive optical element; one or more optical filters; one or more optical combiners; one or more optical switches (e.g., thermo-

optic switches, liquid crystal switches, electro-optic switches, mechanical optical switches, or microelectromechanical systems (MEMS) switches); a series of two or more fiber Bragg gratings with optical circulators.

[0132] The light source **111** in FIG. **24** may be a pump light source **111_{pu}**, a Stokes light source **111_S**, or a probe light source **111_{pr}**, and the output beam **120** may be a pump beam **120_{pu}**, a Stokes beam **120_S**, or a probe beam **120_{pr}**. The output beam **120** may be a free-space beam, or the output beam **120** may be coupled into an optical fiber or into a waveguide of a photonic integrated circuit (PIC). For example, each of the laser diodes **110-1**, **110-2**, . . . **110-N** may produce a fiber-coupled beam **120-1**, **120-2**, . . . **120-N**, and the optical multiplexer **118** may be a fiber-optic device that produces a fiber-coupled output beam **120**. As another example, the optical multiplexer **118** may be a waveguide-based device fabricated in a PIC. The laser diodes **110** may be fabricated separately and then attached or connected to the PIC or a PIC substrate, or one or more of the laser diodes may be fabricated together with the PIC. The output beam **120** may be sent to an optical combiner **130**, a sample **150**, or a detector **220**. Alternatively, the light source **111** may include an optical amplifier (e.g., an SOA or a fiber-optic amplifier) located after the optical multiplexer **118**, and the output beam **120** may be coupled from the multiplexer to an optical amplifier that provides optical amplification to the output beam.

[0133] The light source **111** in FIG. **24** includes N laser diodes **110-1**, **110-2**, . . . **110-N**, where N is an integer greater than or equal to 2. Each of the N laser diodes **110** in FIG. **24** may be a wavelength-tunable laser diode or a fixed-wavelength laser diode. For example, the light source in FIG. **24** may include (i) N wavelength-tunable laser diodes, (ii) N fixed-wavelength laser diodes, or (iii) one or more wavelength-tunable laser diodes and one or more fixed-wavelength laser diodes. The light source **111** in FIG. **24** may be referred to as a wavelength-tunable light source, a frequency-tunable light source, or a tunable light source. A wavelength-tunable light source may include one or more continuously tunable laser diodes (e.g., SG-DBR laser diodes); multiple fixed-wavelength laser diodes (e.g., multiple DFB laser diodes), each laser diode operating at a different wavelength; or any combination thereof.

[0134] The light source **111** in FIG. **24** may be a pump light source **111_{pu}** or a Stokes light source **111_S** that includes N fixed-wavelength laser diodes, each laser diode having a different operating wavelength. The wavelength of the output beam **120** produced by the wavelength-tunable light source **111** in FIG. **24** may be adjustable to any wavelength of N different wavelengths by selecting one of the N fixed-wavelength laser diodes for operation. The frequency offset Ω between the pump beam **120_{pu}** and Stokes beam **120_S** may be adjustable by selecting one of the fixed-wavelength laser diodes for operation. For example, the Stokes beams **120_S** and **120_{S'}** in FIG. **14** may be produced by the light source **111** in FIG. **24**. Laser diode **110-1** may be a fixed-wavelength laser diode operating at the frequency $\nu_{sub.2}$, and laser diode **110-2** may be a fixed-wavelength laser diode operating at the frequency $\nu_{sub.2}$. Selecting laser diode **110-1** for operation produces the frequency offset $\Omega_{sub.1}$ in FIG. **14**, and selecting laser diode **110-2** for operation produces the frequency offset $\Omega_{sub.2}$.

[0135] The light source **111** in FIG. **24** may operate only one of the N laser diodes **110** at any given time. Each of the laser diodes **110** may operate at a particular wavelength or over a particular range of wavelengths, and one of the laser diodes may be selected for operation based on the wavelength that is needed to perform a particular measurement. For example, the pump beams **120_{pu}** and **120_{pu'}** in FIG. **15** may be produced by the light source **111** in FIG. **24**. Laser diode **110-1** may be a fixed-wavelength laser diode operating at the frequency $\nu_{sub.1}$, and laser diode **110-2** may be a fixed-wavelength laser diode operating at the frequency $\nu'_{sub.1}$. During a first measurement period, laser diode **110-1** may be operated to produce output beam **120-1** at the frequency $\nu_{sub.1}$, and the other laser diodes **110-2** to **110-N** may be turned off or otherwise configured to not produce light. The multiplexer **118** receives the output beam **120-1** from the laser diode **110-1** and directs it to the output of the multiplexer to produce the output beam **120** having a frequency $\nu_{sub.1}$. During

a second measurement period, the Raman signal **160c** in FIG. **15** may be measured, and laser diode **110-2** may be operated to produce output beam **120-2** at the frequency $\nu'_{\text{sub.1}}$. The other laser diodes (i.e., laser diodes **110-1** to **110-N**, excluding laser diode **110-2**) may be turned off or otherwise configured to not produce light. The multiplexer **118** receives the output beam **120-2** from the laser diode **110-2** and directs it to the output of the multiplexer to produce the output beam **120** having a frequency $\nu'_{\text{sub.1}}$.

[0136] The light source **111** in FIG. **24** may be a probe light source that includes N wavelength-tunable laser diodes. The probe light source may be configured to tune over one or more wavelength ranges having a total width between $p \cdot N \cdot \Delta\lambda_{\text{sub.av}}$ and $N \cdot \Delta\lambda_{\text{sub.av}}$, where $\Delta\lambda_{\text{sub.av}}$ is an average wavelength-tuning range of the N laser diodes **110**, and p is a wavelength-overlap parameter between 0.5 and 1. For example, if the overlap parameter p has a value of 0.7, then the combined wavelength-tuning range of the N wavelength-tunable laser diodes may be between $(0.7)N \cdot \Delta\lambda_{\text{sub.av}}$ and $N \cdot \Delta\lambda_{\text{sub.av}}$. The wavelength-overlap parameter p represents the amount of wavelength overlap between adjacent wavelength-tuning ranges (e.g., an overlap value p of 1 indicates that there is no wavelength overlap between the wavelength-tuning ranges). For example, the light source **111** in FIG. **24** may be a probe light source that includes three SG-DBR laser diodes having respective wavelength-tuning ranges of 1490-1530 nm, 1520-1560 nm, and 1550-1590 nm. Each of the SG-DBR laser diodes has a 40-nm tuning range with a 10-nm overlap between adjacent tuning ranges, which results in the light source **111** in FIG. **24** having a 100-nm wavelength-tuning range from 1490 nm to 1590 nm. In this case, the average wavelength-tuning range $\Delta\lambda_{\text{sub.av}}$ of the three laser diodes is 40 nm, and the total wavelength-tuning range has a width of 100 nm, which corresponds to the wavelength-overlap parameter p having a value of approximately 0.83. The probe light source **111** may produce an output beam **120** having any wavelength from 1490 nm to 1590 nm by selecting one of the three SG-DBR laser diodes for operation and tuning the laser diode to the desired wavelength. As another example, the light source **111** in FIG. **24** may be a probe light source **111** that includes three SG-DBR laser diodes having respective wavelength-tuning ranges of 1490-1530 nm, 1530-1560, and 1640-1680 nm. Each of the SG-DBR laser diodes has a 40-nm tuning range with no wavelength overlap between adjacent tuning ranges. The average wavelength-tuning range $\Delta\lambda_{\text{sub.av}}$ of the three laser diodes is 40 nm, and the total wavelength-tuning range has a width of 120 nm. This corresponds to the wavelength-overlap parameter p having a value of 1, indicating that there is no wavelength overlap between the tuning ranges of the three laser diodes.

[0137] FIG. **25** illustrates an example pump laser **110pu** and Stokes laser **110S** with a fiber-optic combiner **130** that produces a combined pump-Stokes beam **140** coupled into an optical fiber **116**. A Raman spectroscopy system **100** may include one or more optical elements that direct the pump and Stokes beams to a sample **150**. The optical elements may include a combiner **130** that combines the pump beam **120pu** and the Stokes beam **120S** to produce a combined pump-Stokes beam **140** that is directed to a sample **150**. In FIG. **1**, the combiner **130a** may be a free-space optical combiner, and the pump beam **120pu**, Stokes beam **120S**, and combined beam **140** may each be free-space beams. The combiner **130** in FIG. **25** is a fiber-optic combiner that receives the pump and Stokes beams via two input optical fibers **116** and combines the two beams into a combined pump-Stokes beam **140** that propagates in an output optical fiber **116**. The pump laser **110pu** may be a fiber-coupled laser diode that produces a pump beam **120pu** that is directed to the fiber-optic combiner **130** via an input optical fiber **116**. Similarly, the Stokes laser **110S** may be a fiber-coupled laser diode that produces a Stokes beam **120S** that is directed to the fiber-optic combiner **130** via another input optical fiber **116**. The pump laser **110pu** or the Stokes laser **110S** may be followed by an optical amplifier (not illustrated in FIG. **25**) that amplifies the pump beam **120pu** or Stokes beam **120S** prior to directing the light to the combiner **130**. After the fiber-optic combiner **130** combines the pump and Stokes beams, the output optical fiber **116** may direct the combined pump-Stokes beam **140** to a sample **150**.

[0138] The fiber-optic combiner **130** in FIG. **25** may include a fiber-optic wavelength division multiplexer (WDM) with two input optical fibers (for the pump and Stokes beam) and one output optical fiber for the output beam **120**. The WDM may include a dichroic beamsplitter or a fused fiber coupler. Each of the input or output optical fiber **116** may be a single-mode optical fiber or a multi-mode optical fiber.

[0139] In some embodiments, instead of using a single pump laser or a single Stokes laser (as illustrated in FIG. **25**), a Raman spectroscopy system may use a pump or Stokes light source **111** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. **24**). For example, the light source **111** in FIG. **24** may be a pump light source that produces a fiber-coupled output beam **120** that is coupled to an input fiber of the fiber-optic combiner **130** in FIG. **25**. One of the laser diodes of the pump light source may be selected for operation, and the light from the selected laser diode may be directed by the multiplexer **118** to an optical fiber that is coupled to the combiner **130**. Additionally or alternatively, the light source **111** in FIG. **24** may be a Stokes light source that produces a fiber-coupled output beam **120** that is coupled to an input fiber of the fiber-optic combiner **130** in FIG. **25**.

[0140] FIG. **26** illustrates an example laser diode **110** coupled to a waveguide **172** of a photonic integrated circuit (PIC) **170**. A PIC **170** (which may be referred to as a planar lightwave circuit (PLC), a waveguide-based device, an integrated-optic device, or an integrated optoelectronic device) may be fabricated from a material that includes silicon, silicon dioxide (which may be referred to as silica), glass, indium phosphide (InP), silicon nitride (SiN), a polymer, graphene, or an electro-optic material (e.g., lithium niobate (LiNbO₃) or lithium tantalate (LiTaO₃)). A PIC **170** may include one or more optical waveguides **172** that confine and guide a beam of light. An optical waveguide **172** that is part of a PIC **170** may be referred to as a PIC waveguide and may be a passive optical waveguide formed in the PIC, and the waveguide may convey light from one optical element to another with relatively low optical loss.

[0141] In FIG. **26**, light from the laser diode **110** is coupled into the PIC waveguide **172** to produce a waveguide-coupled beam **120**. The PIC waveguide **172** may convey the beam of light **120** from the laser diode **110** to another optical element (e.g., an optical combiner **130** or a detector **220**) or to a sample **150**. Light from the laser diode **110** in FIG. **26** may be coupled into the PIC waveguide **172** using one or more lenses, or the laser diode **110** may be butt-coupled to an input of the waveguide so that the light from the laser diode is directly coupled into the waveguide. The laser diode **110** may be mechanically attached or connected to the PIC **170** or to a substrate to which the PIC is also attached. For example, the laser diode **110** may be attached using epoxy, adhesive, or solder. Alternatively, the laser diode **110** in FIG. **26** may be located apart from the PIC **170**, and the laser diode may send a beam of light **120** to the PIC **170** via optical fiber. An output end of the optical fiber may be attached or connected to the PIC **170** so that the light is coupled into the PIC waveguide **172**. Light from the laser diode **110** may be amplified by an optical amplifier (not illustrated in FIG. **26**) prior to being coupled into the PIC waveguide **172**. For example, the laser diode **110** in FIG. **26** may be a MOPA laser similar to that illustrated in FIG. **21**, and the output beam **120** produced by the MOPA laser may be directly coupled into the PIC waveguide **172**. In some embodiments, a laser diode **110** and a PIC **170** may be fabricated separately, and the laser diode may then be attached or connected to the PIC or to a PIC substrate. In other embodiments, rather than fabricating the two devices separately, a laser diode **110** and a PIC **170** may be fabricated together so that the laser diode is directly integrated into the PIC during the fabrication process. For example, the PIC **170** and laser diode **110** in FIG. **26** may both be InP-based devices that are fabricated together on an indium phosphide wafer.

[0142] FIG. **27** illustrates an example pump laser **110_{pu}** and Stokes laser **110_S** with a photonic integrated circuit (PIC) **170** that produces a combined pump-Stokes beam **140** coupled into an optical waveguide **172** of the PIC. The PIC **170** includes a waveguide combiner **130** and three PIC waveguides **172** (two input waveguides for the pump and Stokes beams and one output waveguide

for the combined pump-Stokes beam **140**). The waveguide combiner **130** is a waveguide-based optical combiner that combines the pump beam **120_{pu}** and the Stokes beam **120_S** to produce a combined pump-Stokes beam **140**. The combined pump-Stokes beam **140** is coupled to an output PIC waveguide **172** of the PIC **170**, and the output waveguide may direct the beam to a sample **150**. The pump laser **110_{pu}** or the Stokes laser **110_S** may be a laser diode that is mechanically attached or connected to the PIC **170** or to a substrate to which the PIC is also attached.

Alternatively, the pump laser **110_{pu}** or the Stokes laser **110_S** may be a fiber-coupled laser diode that sends a beam of light to the PIC via optical fiber (e.g., an output end of the optical fiber may be attached to the PIC so that the light is coupled into an input PIC waveguide **172**). In other embodiments, the pump laser **110_{pu}** or the Stokes laser **110_S** may include a laser diode that is fabricated together with the PIC **170** so that the laser diode is directly integrated into the PIC during fabrication.

[0143] In some embodiments, instead of using a single pump laser or a single Stokes laser (as illustrated in FIG. 27), a Raman spectroscopy system may use a pump or Stokes light source **111** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. 24). For example, the light source **111** in FIG. 24 may be a pump or Stokes light source that produces a fiber-coupled output beam **120** that is coupled to an input PIC waveguide **172** of the PIC **170** in FIG. 27. Alternatively, the multiplexer **118** in FIG. 24 may be a waveguide-based device and the output beam **120** may propagate in a PIC waveguide that directs the light to an input optical waveguide **172** of the PIC **170** in FIG. 27.

[0144] FIG. 28 illustrates an example fiber-optic combiner **130** that combines a Raman signal **160** with a probe beam **120_{pr}**. An optical receiver **200** may include an optical combiner **130** that combines a Raman signal **160** and a probe beam of light **120_{pr}** to produce one or more combined probe-Raman signals **210** that are each directed to a detector **220**. In FIG. 1, the combiner **130_b** may be a free-space optical combiner, and the Raman signal **160**, probe beam **120_{pr}**, and the combined probe-Raman signal **210** may each be free-space beams. The combiner **130** in FIG. 28 is a fiber-optic combiner that receives the Raman signal **160** and the probe beam **120_{pr}** via two input optical fibers **116** and combines the two beams into a combined probe-Raman signal **210** that is directed to a detector **220** via an output optical fiber **116**. Each of the input or output optical fiber **116** in FIG. 28 may be a single-mode optical fiber or a multi-mode optical fiber.

[0145] The Raman signal **160** in FIG. 28 may be a free-space beam that is coupled into an input optical fiber **116** using one or more lenses. The probe laser **110_{pr}** may be a fiber-coupled laser diode that directs the probe beam **120_{pr}** to the fiber-optic combiner **130** via optical fiber **116**. The probe beam **120_{pr}** may be amplified by an optical amplifier (not illustrated in FIG. 28) prior to being directed to the combiner **130**. In some embodiments, instead of using a single probe laser **110_{pr}** (as illustrated in FIG. 28), an optical receiver **200** may use a light source **111** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. 24). For example, the light source **111** in FIG. 24 may be a probe light source that produces a fiber-coupled output beam **120** that is coupled to an input optical fiber **116** of the fiber-optic combiner **130** in FIG. 28.

[0146] FIG. 29 illustrates an example photonic integrated circuit (PIC) **170** with a waveguide combiner **130** that combines a Raman signal **160** with a probe beam **120_{pr}**. An optical combiner **130** may be a waveguide combiner **130** that is part of a PIC **170** and may combine a Raman signal **160** and a probe beam of light **120_{pr}** to produce one or more combined probe-Raman signals **210** that are each directed to a detector **220**. The waveguide combiner **130** in FIG. 29 receives the Raman signal **160** and the probe beam **120_{pr}** via two input PIC waveguides **172**. A waveguide combiner **130** may produce 1, 2, or 4 combined output beams **210**. The waveguide combiner **130** in FIG. 29 combines the Raman signal **160** and the probe beam **120_{pr}** to produce two combined probe-Raman signals **210_a** and **210_b** which are each directed to a respective detector **220_a** and **220_b** via two output PIC waveguides **172**. The PIC **170** in FIG. 29 may be part of an optical receiver **200**.

[0147] The Raman signal **160** in FIG. **29** may be a free-space beam that is coupled into an input PIC waveguide **172** using one or more lenses. The probe laser **110_{pr}** may be a laser diode that is mechanically attached or connected to the PIC **170** or to a substrate to which the PIC is also attached, or the probe laser may be fabricated together with the PIC so that the probe laser is directly integrated into the PIC during the fabrication process. Alternatively, the probe laser **110_{pr}** may be a fiber-coupled laser diode that sends the probe beam **120_{pr}** to the PIC **170** via optical fiber (e.g., an output end of the optical fiber may be attached to the PIC so that the light is coupled into an input PIC waveguide **172**). The probe beam **120_{pr}** may be amplified by an optical amplifier (not illustrated in FIG. **29**) prior to being coupled into an input PIC waveguide **172**. In some embodiments, instead of using a single probe laser **110_{pr}** (as illustrated in FIG. **29**), an optical receiver **200** may use a light source **111** with multiple laser diodes and an optical multiplexer **118** (e.g., as illustrated in FIG. **24**), and the probe beam **120_{pr}** may be delivered from the multiplexer to the PIC **170** via optical fiber or via a waveguide **172** of the PIC.

[0148] A Raman spectroscopy system **100** may include one or more optical elements that (i) direct a pump beam **120_{pu}** and a Stokes beam **120_S** to a sample **150** and (ii) direct a Raman signal **160** and a probe beam **120_{pr}** to one or more detectors **220**. The optical elements may include one or more PICs **170** that each include one or more optical waveguides **172**. One or more of the PIC waveguides **172** may direct the pump beam **120** and the Stokes beam **120_s** to the sample **150**. For example, a PIC **170** may include an optical combiner **130** that produces a combined pump-Stokes beam **140** that is directed to the sample by an optical waveguide **172** of the PIC **170**. One or more other PIC waveguides **172** may direct the Raman signal **160** and the probe beam **120_{pr}** to one or more detectors **220**. For example, a PIC **170** may include an optical combiner **130** that combines the Raman signal **160** and the probe beam **120_{pr}** to produce one or more combined probe-Raman signals **210** that are each directed to a detector **220** by an optical waveguide **172** of the PIC **170**.

[0149] FIGS. **30-35** each illustrate example frequency ranges of a pump beam **120_{pu}** and a Stokes beam **120_S**. The pump beam **120_{pu}** may be produced by a pump light source **111_{pu}**, and the Stokes beam **120_S** may be produced by a Stokes light source **111_S**. The pump and Stokes light sources may each include one or more fixed wavelength laser diodes or one or more wavelength-tunable laser diodes. In each of FIGS. **30-35**, the pump beam **120_{pu}** and the Stokes beam **120_S** each have one or more fixed frequencies or one or more frequencies that are adjustable over a particular frequency range. The corresponding frequency offset Ω between the pump and Stokes beams is indicated as a range of frequencies or a set of discrete frequencies that the frequency offset can be set to, based on the frequencies that are available to the pump and Stokes beams. The frequency offset Ω is determined from $v_{\text{sub.1}} - v_{\text{sub.2}}$, where $v_{\text{sub.1}}$ is the range or set of fixed frequencies for the pump beam **120_{pu}**, and $v_{\text{sub.2}}$ is the range or set of fixed frequencies for the Stokes beam **120_S**. The frequency offsets $\Omega_{\text{sub.1}}$, $\Omega_{\text{sub.2}}$, $\Omega_{\text{sub.3}}$, and $\Omega_{\text{sub.4}}$ in FIGS. **30-35** may have any suitable value between approximately 5 THz and approximately 100 THz. The frequency range $\Delta\Omega$ over which a frequency offset may be varied may have any suitable value between approximately 5 THz and approximately 80 THz.

[0150] In FIG. **30**, the pump beam **120_{pu}** has a single fixed frequency $v_{\text{sub.1}}$, and the Stokes beam **120_S** has a frequency $v_{\text{sub.2}}$ that is adjustable over a frequency range of width $\Delta v_{\text{sub.2}}$. The adjustable frequency range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120_S** extends from a low frequency $v_{\text{sub.2L}}$ to a high frequency $v_{\text{sub.2H}}$, where $\Delta v_{\text{sub.2}} = v_{\text{sub.2H}} - v_{\text{sub.2L}}$. The pump laser **110_{pu}** that produces the pump beam **120_{pu}** in FIG. **30** may be a fixed-wavelength laser diode, and the Stokes laser **110_S** that produces the Stokes beam **120_S** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. In FIG. **30**, the frequency range $\Delta\Omega$ is also equal to the frequency range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120_S**. When the Stokes beam **120_S** is set to the lower frequency $v_{\text{sub.2L}}$, the frequency offset between the pump and Stokes beams is $v_{\text{sub.1}} - v_{\text{sub.2L}}$, which is equal to $\Omega_{\text{sub.2}}$. Similarly,

when the Stokes beam **120S** is set to the upper frequency $v_{\text{sub.2H}}$, the frequency offset between the pump and Stokes beams is $v_{\text{sub.1}} - v_{\text{sub.2H}}$, which is equal to $\Omega_{\text{sub.1}}$.

[0151] For example, the pump beam **120pu** in FIG. **30** may have a frequency $v_{\text{sub.1}}$ of 250 THz (corresponding to a wavelength of approximately 1200 nm), and the Stokes beam **120S** may be adjustable from a low frequency $v_{\text{sub.2L}}$ of 195 THz to a high frequency $v_{\text{sub.2H}}$ of 200 THz. This corresponds to a frequency-tuning range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120S** of 5 THz (167 cm^{-1} in wavenumbers) and a 38-nm wavelength-tuning range from approximately 1499 nm to approximately 1537 nm. The resulting frequency offset Ω between the pump and Stokes beams may be set to a value between the lower value $\Omega_{\text{sub.1}}$ of 50 THz (1668 cm^{-1} in wavenumbers) and the upper value $\Omega_{\text{sub.2}}$ of 55 THz (1835 cm^{-1} in wavenumbers), corresponding to a frequency range $\Delta\Omega$ of 5 THz (167 cm^{-1} in wavenumbers).

[0152] In FIG. **31**, the pump beam **120pu** has a single fixed frequency $v_{\text{sub.1}}$, and the Stokes beam **120S** has a frequency $v_{\text{sub.2}}$ that is adjustable over a frequency range of width $\Delta v_{\text{sub.2}}$. The adjustable frequency range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120S** extends from a low frequency $v_{\text{sub.2L}}$ to a high frequency $v_{\text{sub.2H}}$, where $\Delta v_{\text{sub.2}} = v_{\text{sub.2H}} - v_{\text{sub.2L}}$. The pump laser **110pu** that produces the pump beam **120pu** in FIG. **31** may be a fixed-wavelength laser diode. The Stokes light source **111S** that produces the Stokes beam **120S** may include two wavelength-tunable laser diodes. For example, the Stokes light source **111S** may be similar to the light source in FIG. **24** where two wavelength-tunable laser diodes are combined by a multiplexer **118**. A first wavelength-tunable laser diode may operate from frequency $v_{\text{sub.2L}}$ to frequency $v_{\text{sub.2M}}$, and a second wavelength-tunable laser diode may operate from frequency $v_{\text{sub.2M}}$ to frequency $v_{\text{sub.2H}}$. The total tuning range $\Delta v_{\text{sub.2}}$ of the Stokes light source **111S** equals the sum of the tuning ranges $\Delta v_{\text{sub.2a}}$ and $\Delta v_{\text{sub.2b}}$ of the two wavelength-tunable laser diodes. In other embodiments, if the tuning ranges of the two wavelength-tunable laser diodes overlap, the total tuning range $\Delta v_{\text{sub.2}}$ of the Stokes light source **111S** will be reduced by the amount of frequency overlap between the two lasers.

[0153] In FIG. **31**, the frequency offset (between the pump and Stokes beams may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. When the first wavelength-tunable laser diode (with a frequency range from $v_{\text{sub.2L}}$ to $v_{\text{sub.2M}}$) is selected to operate, the frequency offset Ω may be set to any value between $\Omega_{\text{sub.M}}$ and $\Omega_{\text{sub.2}}$. For example, when the Stokes beam **120S** is set to the lower frequency $v_{\text{sub.2L}}$, the frequency offset between the pump and Stokes beams is $v_{\text{sub.1}} - v_{\text{sub.2L}}$, which is equal to $\Omega_{\text{sub.2}}$. When the second wavelength-tunable laser diode (with a frequency range from $v_{\text{sub.2M}}$ to $v_{\text{sub.2H}}$) is selected to operate, the frequency offset Ω may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.M}}$. For example, when the Stokes beam **120S** is set to the upper frequency $v_{\text{sub.2H}}$, the frequency offset between the pump and Stokes beams is $v_{\text{sub.1}} - v_{\text{sub.2H}}$, which is equal to $\Omega_{\text{sub.1}}$. The total frequency range $\Delta\Omega$ of the frequency offset Ω is equal to the sum of the two frequency ranges $\Delta\Omega_{\text{sub.2a}}$ and $\Delta\Omega_{\text{sub.2b}}$. The frequency range $\Delta\Omega$ is also equal to the overall frequency range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120S**.

[0154] In FIG. **32**, the pump beam **120pu** can be set to two fixed frequencies $v_{\text{sub.1a}}$ and $v_{\text{sub.1b}}$, and the Stokes beam **120S** has a frequency $v_{\text{sub.2}}$ that is adjustable over a frequency range of width $\Delta v_{\text{sub.2}}$. The adjustable frequency range $\Delta v_{\text{sub.2}}$ of the Stokes beam **120S** extends from a low frequency $v_{\text{sub.2L}}$ to a high frequency $v_{\text{sub.2H}}$, where $\Delta v_{\text{sub.2}} = v_{\text{sub.2H}} - v_{\text{sub.2L}}$. The pump light source **111pu** that produces the two pump beams **120pu-a** and **120pu-b** may include two fixed-wavelength laser diodes. For example, the pump light source **111pu** may be similar to the light source in FIG. **24** where two fixed-wavelength laser diodes are combined by a multiplexer **118**. The Stokes laser **110S** that produces the Stokes beam **120S** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. When the pump light source **111pu** produces the pump beam **120pu-a** at

frequency $\nu_{\text{sub.2H}}$, the Stokes beam **120S** may be tuned to a frequency between $\nu_{\text{sub.2L}}$ and $\nu_{\text{sub.2H}}$ to produce a frequency offset Ω between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.M}}$. For example, with the Stokes beam **120S** set to the upper frequency $\nu_{\text{sub.2H}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1a}} - \nu_{\text{sub.2H}}$, which is equal to $\Omega_{\text{sub.1}}$. When the pump light source **111pu** produces the pump beam **120pu-b** at frequency $\nu_{\text{sub.1b}}$, the Stokes beam **120S** may be tuned to a frequency between $\nu_{\text{sub.2L}}$ and $\nu_{\text{sub.2H}}$ to produce a frequency offset Ω between $\Omega_{\text{sub.M}}$ and $\Omega_{\text{sub.2}}$. For example, with the Stokes beam **120S** set to the lower frequency $\nu_{\text{sub.2L}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1b}} - \nu_{\text{sub.2L}}$, which is equal to $\Omega_{\text{sub.2}}$.

[0155] FIG. 33 is similar to FIG. 30, except in FIG. 33, the Stokes beam **120S** has a fixed frequency $\nu_{\text{sub.2}}$ and the frequency of the pump beam **120pu** is adjustable. The adjustable frequency range $\Delta\nu_{\text{sub.1}}$ of the pump beam **120pu** extends from a low frequency $\nu_{\text{sub.1L}}$ to a high frequency $\nu_{\text{sub.1H}}$, where $\Delta\nu_{\text{sub.1}} = \nu_{\text{sub.1H}} - \nu_{\text{sub.1L}}$. The Stokes laser **110S** that produces the Stokes beam **120pu** in FIG. 33 may be a fixed-wavelength laser diode, and the pump laser **110pu** that produces the pump beam **120pu** may be a wavelength-tunable laser diode. The frequency offset Ω between the pump and Stokes beams may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. In FIG. 33, the frequency range $\Delta\Omega$ is also equal to the frequency range $\Delta\nu_{\text{sub.1}}$ of the pump beam **120pu**. When the pump beam **120pu** is set to the lower frequency $\nu_{\text{sub.1L}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1L}} - \nu_{\text{sub.2}}$, which is equal to $\Omega_{\text{sub.1}}$. Similarly, when the pump beam **120pu** is set to the upper frequency $\nu_{\text{sub.1H}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1H}} - \nu_{\text{sub.2}}$, which is equal to $\Omega_{\text{sub.2}}$.

[0156] In FIG. 34, both the pump beam **120pu** and the Stokes beam **120S** have adjustable frequencies. The pump laser **110pu** that produces the pump beam **120pu** and the Stokes laser **110S** that produces the Stokes beam **120S** may each include a wavelength-tunable laser diode. The adjustable frequency range $\Delta\nu_{\text{sub.1}}$ of the pump beam **120pu** extends from a low frequency $\nu_{\text{sub.1L}}$ to a high frequency $\nu_{\text{sub.1H}}$, where $\Delta\nu_{\text{sub.1}} = \nu_{\text{sub.1H}} - \nu_{\text{sub.1L}}$. The adjustable frequency range $\Delta\nu_{\text{sub.2}}$ of the Stokes beam **120S** extends from a low frequency $\nu_{\text{sub.2L}}$ to a high frequency $\nu_{\text{sub.2H}}$, where $\Delta\nu_{\text{sub.2}} = \nu_{\text{sub.2H}} - \nu_{\text{sub.2L}}$. The frequency offset (between the pump and Stokes beams) may be set to any value between $\Omega_{\text{sub.1}}$ and $\Omega_{\text{sub.2}}$, and the frequency range $\Delta\Omega$ of the frequency offset is $\Omega_{\text{sub.2}} - \Omega_{\text{sub.1}}$. When the Stokes beam **120S** is set to the lower Stokes frequency $\nu_{\text{sub.2L}}$ and the pump beam **120pu** is set to the upper pump frequency $\nu_{\text{sub.1H}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1H}} - \nu_{\text{sub.2L}}$, which is equal to $\Omega_{\text{sub.2}}$. When the Stokes beam **120S** is set to the upper Stokes frequency $\nu_{\text{sub.2H}}$ and the pump beam **120pu** is set to the lower pump frequency $\nu_{\text{sub.1L}}$, the frequency offset between the pump and Stokes beams is $\nu_{\text{sub.1L}} - \nu_{\text{sub.2H}}$, which is equal to $\Omega_{\text{sub.1}}$.

[0157] In FIG. 35, both the pump beam **120pu** and the Stokes beam **120S** can be set to two different fixed frequencies. A pump light source **111pu** and a Stokes light source **111S** may each include two or more fixed wavelength laser diodes, and each light source may be similar to the light source in FIG. 24 where multiple fixed-wavelength laser diodes are combined by a multiplexer **118**. In FIG. 35, the pump light source **111pu** that produces the two pump beams **120pu-a** and **120pu-b** may include two fixed-wavelength laser diodes, and the Stokes light source **111S** that produces the two Stokes beams **120S-a** and **120S-b** may include two fixed-wavelength laser diodes. The frequencies of the pump and Stokes beams may be selected to produce four different frequency offsets $\Omega_{\text{sub.1}}$, $\Omega_{\text{sub.2}}$, $\Omega_{\text{sub.3}}$, and $\Omega_{\text{sub.4}}$. For example, selecting Stokes beam **120S-b** at frequency $\nu_{\text{sub.2b}}$ and pump beam **120pu-a** at frequency $\nu_{\text{sub.1a}}$ produces a frequency offset $\Omega_{\text{sub.1}}$, which is equal to $\nu_{\text{sub.1a}} - \nu_{\text{sub.2b}}$. As another example, selecting Stokes beam **120S-a** at frequency $\nu_{\text{sub.2a}}$ and pump beam **120pu-b** at frequency $\nu_{\text{sub.1b}}$ produces a frequency offset $\Omega_{\text{sub.4}}$, which is equal to $\nu_{\text{sub.1b}} - \nu_{\text{sub.2a}}$. The frequency offset Ω may be produced by selecting Stokes beam **120S-a** at frequency $\nu_{\text{sub.2a}}$ and pump beam **120pu-a** at frequency $\nu_{\text{sub.1a}}$, and the frequency offset $\Omega_{\text{sub.3}}$ may be produced by selecting Stokes beam

120S-b at frequency $v_{\text{sub.2b}}$ and pump beam **120pu-b** at frequency $v_{\text{sub.1b}}$.

[0158] FIG. **36** illustrates an example optical receiver **200** with two detectors **220a** and **220b**. The optical receivers **200** in FIGS. **29** and **36** are similar, except one difference is that the optical receiver in FIG. **29** is a waveguide-based optical receiver, while the optical receiver in FIG. **36** is a free-space optical receiver **200**. The optical combiner **130** in FIG. **36** may be a 50/50 free-space beamsplitter that reflects approximately 50% of an incident beam of light and transmits approximately 50% of the beam. The optical combiner **130** splits the Raman signal **160** and the probe beam **120pr** into two beams to produce two combined probe-Raman signals **210a** and **210b**. The combined probe-Raman signal **210a** is directed to detector **220a** and includes a transmitted portion of the probe beam **120pr** and a reflected portion of the Raman signal **160** (e.g., approximately 50% of the probe beam and approximately 50% of the Raman signal). Similarly, the combined probe-Raman signal **210b** is directed to detector **220b** and includes a reflected portion of the probe beam **120pr** and a transmitted portion of the Raman signal **160**. The portions of the probe beam **120pr** and the Raman signal **160** that make up the combined probe-Raman signal **210a** may be coherently mixed at detector **220a** to produce the photocurrent signal $i_{\text{sub.a}}$. Similarly, the portions of the probe beam **120pr** and the Raman signal **160** that make up the combined probe-Raman signal **210b** may be coherently mixed at detector **220b** to produce the photocurrent signal $i_{\text{sub.b}}$.

[0159] The two detectors **220a** and **220b** are arranged so that their respective photocurrents $i_{\text{sub.a}}$ and $i_{\text{sub.b}}$ are subtracted. The anode of detector **220a** is electrically connected to the cathode of detector **220b**, and the subtracted photocurrent signal $i_{\text{sub.a}} - i_{\text{sub.b}}$ from the anode-cathode connection is sent to the detection electronics **230**, which produces a digital output signal **240** that corresponds to the subtracted photocurrent signal. The subtracted photocurrent signal may be expressed as $i_{\text{sub.a}} - i_{\text{sub.b}} = 2E_{\text{sub.R}} E_{\text{sub.pr}} \cos [2\pi(v_{\text{sub.R}} - v_{\text{sub.3}})t + \Delta\phi]$, which corresponds to the coherent-mixing term discussed herein. The subtracted photocurrent signal does not include the terms $E_{\text{sub.R}}^2$ and $E_{\text{sub.pr}}^2$ corresponding to the respective optical powers of the Raman signal **160** and the probe beam **120pr**. By subtracting the two photocurrents $i_{\text{sub.a}}$ and $i_{\text{sub.b}}$, the common-mode terms $E_{\text{sub.R}}^2$ and $E_{\text{sub.pr}}^2$ (as well as common-mode noise) that appear in each of the photocurrent signals $i_{\text{sub.a}}$ and $i_{\text{sub.b}}$ are substantially removed, leaving the coherent-mixing term, which is the quantity of interest. Since subtraction may remove common-mode noise, the subtracted photocurrent signal $i_{\text{sub.a}} - i_{\text{sub.b}}$ may have a reduced noise compared to each of the photocurrent signals $i_{\text{sub.a}}$ and $i_{\text{sub.b}}$ alone. The dual-detector arrangement in FIG. **36** in which the photocurrents are subtracted may be referred to as a balanced optical detector. A balanced detector may be implemented as a free-space device, a fiber-optic-based device, or a waveguide-based device.

[0160] FIG. **37** illustrates an example optical receiver **200** configured for polarization-sensitive detection of a Raman signal **160**. A polarization-sensitive optical receiver **200** may be used to determine the polarization of a Raman signal **160**. The polarization of a Raman signal **160** may be determined by a processor based on one or more digital output signals **240** produced by the polarization-sensitive optical receiver **200**. Determining the polarization of a Raman signal **160** may include determining a relative size or ratio of two orthogonal polarization components of the Raman signal (e.g., horizontal and vertical polarization components of the Raman signal). For example, if the ratio of the horizontal and vertical polarization components of a Raman signal **160** is 100:1, then the Raman signal may be determined to be substantially horizontally polarized. As another example, if the ratio of the horizontal and vertical polarization components of a Raman signal **160** is 1:1, then the horizontal and vertical polarization components of the Raman signal may be determined to be approximately equal (e.g., the Raman signal may be circularly polarized or linearly polarized at a 45-degree angle to the horizontal and vertical directions).

[0161] A polarization-sensitive optical receiver **200** may include a polarization beamsplitter (PBS) **135** that splits an input beam into two output beams, where one output beam is horizontally

polarized, and the other output beam is vertically polarized. The horizontally polarized output beam includes the horizontal polarization component of the input beam, and the vertically polarized output beam includes the vertical polarization component of the input beam. The Raman signal **160** in FIG. **37** is directed to a Raman-signal PBS **135R** that splits the Raman signal into a horizontal-polarization Raman signal **160-h** and a vertical-polarization Raman signal **160-v**. Similarly, the probe beam **120pr** is directed to a probe-beam PBS **135pr** that splits the probe beam into a horizontal-polarization probe beam **120pr-h** and a vertical-polarization probe beam **120pr-v**. [0162] A polarization-sensitive optical receiver **200** may include a waveplate **132** that changes the polarization of the probe beam **120pr** so that the probe beam is split into two polarization components. The two polarization components may each have approximately one-half the power of the probe beam **120pr**. The waveplate **132c** in FIG. **37** may be (i) a half-wave plate that rotates the polarization of the probe beam **120pr** or (ii) a quarter-wave plate that converts the probe beam **120pr** to a circular or elliptical polarization. For example, the probe laser **110pr** may produce a probe beam **120pr** that is vertically polarized, and the waveplate **132c** may be a half-wave plate that rotates the probe-beam polarization by 45 degrees so that the horizontal-polarization and vertical-polarization probe beams **120pr-h** and **120pr-v** each have approximately equal optical powers.

[0163] The optical combiner **130h** in FIG. **37** combines the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** to produce a horizontal probe-Raman signal **210h** that is directed to a horizontal-polarization optical receiver **200h**. Similarly, the optical combiner **130v** combines the vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** to produce a vertical probe-Raman signal **210v** that is directed to a vertical-polarization optical receiver **200v**. The horizontal-polarization optical receiver **200h** may include one or more optical detectors **200**, where each detector is configured to coherently mix at least a portion of the horizontal-polarization Raman signal **160-h** and at least a portion of the horizontal-polarization probe beam **120pr-h** to produce a horizontal-polarization electronic signal. Similarly, the vertical-polarization optical receiver **200v** may include one or more optical detectors **200**, where each detector is configured to coherently mix at least a portion of the vertical-polarization Raman signal **160-v** and at least a portion of the vertical-polarization probe beam **120pr-v** to produce a vertical-polarization electronic signal. The horizontal-polarization optical receiver **200h** and the vertical-polarization optical receiver **200v** may each include: (i) a single detector **220** (e.g., similar to the optical receiver **200** in FIG. **2**), (ii) two detectors **220** (e.g., similar to the balanced optical detector arrangement in FIG. **36**), or (iii) four detectors **220** (e.g., similar to the arrangement in FIG. **39**). The electronic signals may include a photocurrent signal **i**, and each optical receiver may include an electronic amplifier **232** that produces a corresponding voltage signal **234**. The horizontal-polarization optical receiver **200h** may produce a digital output signal **240-h** corresponding to the horizontal-polarization electronic signal, and the vertical-polarization optical receiver **200v** may produce a digital output signal **240-v** corresponding to the vertical-polarization electronic signal.

[0164] A processor may determine one or more characteristics of the horizontal-polarization and vertical-polarization electronic signals based on the digital output signals **240-h** and **240-v**. Additionally, a processor may determine a polarization of the Raman signal **160** based on the characteristics of the horizontal-polarization and vertical-polarization electronic signals. For example, the characteristics of the electronic signals may include an amplitude or an area associated with the electronic signals, and the polarization of the Raman signal **160** may be expressed as a relative size or ratio of the amplitudes or areas associated with the horizontal and vertical polarization components of the Raman signal. If the horizontal digital output signal **240-h** includes an amplitude characteristic with value 100 and the vertical digital output signal **240-v** includes a corresponding amplitude characteristic with value 1, then the Raman signal **160** may be determined to be substantially horizontally polarized. If the horizontal and vertical digital output signals each include amplitude characteristics having approximately equal values, then the Raman

signal **160** may be determined to have approximately equal horizontal and vertical polarization components.

[0165] A polarization-sensitive optical receiver **200** as illustrated in FIG. **37** may be implemented with free-space optical elements, fiber-optic components, waveguide-based optical elements, a metamaterial-based device, or any suitable combination thereof. For example, the two PBSs **135** in FIG. **37** may be free-space polarization beamsplitter cubes, and the Raman signal **160** and the probe beam **120_{pr}** may be free-space optical beams. Alternatively, the two PBSs **135** may be fiber-optic components, and the Raman signal **160** and the probe beam **120_{pr}** may be conveyed to the PBSs **135** via optical fiber (e.g., single-mode optical fiber or polarization-maintaining optical fiber). Additionally, the horizontally and vertically polarized probe-Raman signals **210_h** and **210_v** may be conveyed to the respective h-polarization and v-polarization optical receivers via polarization-maintaining optical fiber. The h-polarization and v-polarization optical receivers may each preserve the polarization of the respective horizontally and vertically polarized probe-Raman signals. For example, the h-polarization and v-polarization optical receivers may each include polarization-maintaining optical fiber that maintains the polarization of the beams. Alternatively, the h-polarization and v-polarization receivers may each include a PIC with optical waveguides configured to maintain the polarization of the beams.

[0166] FIG. **38** illustrates an example optical receiver **200** configured to detect in-phase and quadrature components of a Raman signal **160**. The optical receiver **200** includes a 90-degree optical hybrid **250** and four detectors **220I⁺**, **220I⁻**, **220Q⁺**, and **220Q⁻**. A 90-degree optical hybrid **250** is an optical-combiner component with two input ports and four output ports. Input light received at each of the two input ports is split, combined, and directed to each of the four output ports, and a 90-degree phase shift is imparted to one of the split beams before the Raman signal **160** and probe beam **120_{pr}** are combined. The 90-degree optical hybrid **250** in FIG. **38** combines a Raman signal **160** and a probe beam **120_{pr}** to produce four combined output beams: two in-phase combined beams **210I⁺** and **210I⁻**, and two quadrature combined beams **210Q⁺** and **210Q⁻**. Each of the four combined beams **210** may include a portion of the Raman signal **160** and a portion of the probe beam **120_{pr}**, and each of the combined beams is directed to one of the four detectors of the optical receiver **200**. In FIG. **38**, each of the four detectors produces a photocurrent signal that corresponds to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120_{pr}**.

[0167] A 90-degree optical hybrid **250** may be configured so that the combined beams directed to each of the output ports have approximately the same optical power or energy. For example, the 90-degree optical hybrid **250** in FIG. **38** may split the Raman signal **160** into four approximately equal portions and direct each of the Raman-signal portions to one of the detectors. Similarly, the probe beam **120_{pr}** may be split into four approximately equal portions directed to each of the four detectors. In the example of FIG. **38**, the combined beam **210I⁺**, which is directed to detector **220I⁺**, may include approximately one-quarter of the power of the Raman signal **160** and approximately one-quarter of the power of the probe beam **120_{pr}**. Similarly, each of the three other combined beams (**210I⁻**, **210Q⁺**, **210Q⁻**) in FIG. **38** may also include approximately one-quarter of the Raman signal **160** and approximately one-quarter of the probe beam **120_{pr}**.

[0168] A 90-degree optical hybrid **250** may be implemented as a waveguide-based device in a PIC. The 90-degree optical hybrid **250** in FIG. **38** is a waveguide-based optical device that includes two waveguide-based optical splitters (**252a**, **252b**) and two waveguide-based optical combiners (**130I**, **130Q**). Splitter **252a** may split the Raman signal **160** into two portions having substantially equal optical power, a first portion directed to combiner **130I** and a second portion directed to combiner **130Q**. Similarly, splitter **252b** may split the probe beam **120_{pr}** into two portions having substantially equal power, a first portion directed to combiner **130I** and a second portion directed to combiner **130Q**. Each optical combiner **130** combines a portion of the Raman signal **160** with a portion of the probe beam **120_{pr}**, and the combined portions are split into a first combined beam

(e.g., combined beam **210I+**) and a second combined beam (e.g., combined beam **210I-**). The combined beam **210I+** is directed to detector **220I+** and includes portions of the Raman signal **160** and the probe beam **120pr** (e.g., approximately 25% of the Raman signal **160** and approximately 25% of the probe beam **120pr**). The combined beam **210I-** is directed to detector **220I-** and may include approximately 25% of the Raman signal **160** and approximately 25% of the probe beam **120pr**.

[0169] In other embodiments, all or part of a 90-degree optical hybrid **250** may be implemented as a free-space optical device. For example, a free-space 90-degree optical hybrid **250** may include one or more free-space beamsplitters or combiners that receive the Raman signal **160** and probe beam **120pr** as free-space beams and produce four free-space combined beams (**210I+**, **210I-**, **210Q+**, **210Q-**). Alternatively, all or part of a 90-degree optical hybrid **250** may be implemented as a fiber-optic device. For example, a 90-degree optical hybrid **250** may be contained in a package with two input optical fibers that direct the Raman signal **160** and probe beam **120pr** into the package and four output optical fibers that direct the four combined beams to four respective detectors.

[0170] A 90-degree optical hybrid **250** may include an optical phase shifter **254** that imparts a 90-degree phase change ($\Delta\phi$) to a portion of the probe beam **120pr** or to a portion of the Raman signal **160**. The phase shifter **254** may apply the 90-degree phase change after a beam of light is split by an optical splitter **252** and prior to combining the Raman signal with the probe beam at an optical combiner **130**. For example, a splitter **252a** may split the Raman signal **160** into two portions, and a phase shifter **254** may impart a 90-degree phase change to one portion of the Raman signal with respect to the other portion, after which the two portions are sent to two different optical combiners. As another example, a splitter **252b** may split the probe beam **120pr** into two portions, and a phase shifter **254** may impart a 90-degree phase change to one portion of the probe beam with respect to the other portion. In FIG. 38, the phase shifter is located after the splitter **252b** and before the combiner **130Q**. The splitter **252b** splits the probe beam **120pr** into two portions, and the phase shifter **254** imparts a 90-degree phase change to the probe-beam portion directed to combiner **130Q**. The other portion of the probe beam **120pr** directed to combiner **130I** does not pass through the phase shifter **254** and does not receive a phase shift from the phase shifter **254**.

[0171] An optical phase shifter **254** may be implemented as a part of a waveguide-based 90-degree optical hybrid **250**. For example, a phase shifter **254** may be implemented as part of an optical waveguide that only one portion of the probe beam **120pr** propagates through. That part of the optical waveguide may be temperature controlled to adjust the refractive index of the waveguide and produce a relative phase delay of approximately 90 degrees between two portions of the probe beam **120pr**. Additionally or alternatively, the 90-degree optical hybrid **250** as a whole may be temperature controlled to set and maintain a 90-degree phase delay. As another example, a phase shifter **254** may be implemented by applying an external electric field to part of an optical waveguide to change the refractive index of the waveguide and produce a 90-degree phase delay. In other embodiments, a phase shifter **254** may be implemented as a part of a free-space or fiber-coupled 90-degree optical hybrid **250**. For example, the input and output beams in a free-space 90-degree optical hybrid **250** may be reflected by or transmitted through the optical surfaces of a free-space optical hybrid **250** so that a relative phase shift of 90 degrees is imparted to one portion of the probe beam **120pr** with respect to another portion of the probe beam.

[0172] In FIG. 38, each of the four detectors produces a photocurrent signal that corresponds to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120pr**. The photocurrents are subtracted in a manner similar to that illustrated in FIG. 36. The photocurrents $i_{\text{sub.I}+}$ and $i_{\text{sub.I}-}$ from detectors **220I+** and **220I-** are subtracted to produce the subtracted in-phase photocurrent signal $i_{\text{sub.I}}$ which is equal to $i_{\text{sub.I}+} - i_{\text{sub.I}-}$. Similarly, the photocurrents $i_{\text{sub.Q}+}$ and $i_{\text{sub.Q}-}$ from detectors **220Q+** and **220Q-** are subtracted to produce the subtracted quadrature photocurrent signal $i_{\text{sub.Q}}$ which is equal to $i_{\text{sub.Q}+} - i_{\text{sub.Q}-}$. Each of the subtracted

photocurrent signals represents a coherent-mixing term corresponding to the coherent mixing of a portion of the Raman signal **160** and a portion of the probe beam **120pr**. The two subtracted photocurrent signals *i.sub.I* and *i.sub.Q* are similar, except the in-phase photocurrent signal *i.sub.I* includes a cosine function, while the quadrature photocurrent signal *i.sub.Q* includes a sine function. This difference between the two subtracted photocurrent signals arises from the 90-degree phase shift provided by the phase shifter **254**. Because a 90-degree phase shift is imparted to the probe beam **120pr** directed to the combiner **130Q**, the subtracted quadrature photocurrent signal *i.sub.Q* includes a sine function (which has a 90-degree phase offset with respect to a cosine function).

[0173] Each of the subtracted photocurrent signals *i.sub.I* and *i.sub.Q* may be sent to detection electronics **230** that produce voltage signals and digital output signals corresponding to the subtracted photocurrent signals. Based on the digital output signals (which result from the four photocurrent signals *i.sub.I+*, *i.sub.I-*, *i.sub.Q+*, and *i.sub.Q-*), a processor may determine an in-phase portion *IP* associated with the Raman signal **160** and a quadrature portion *Q* associated with the Raman signal. Additionally or alternatively, the processor may determine a phase associated with the Raman signal **160**. For example, the processor may determine a phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr**. A phase difference may be referred to as a phase offset or a relative phase between the Raman signal **160** and the probe beam **120pr**.

[0174] The in-phase portion *IP* associated with the Raman signal **160** may be determined from a characteristic (e.g., an amplitude or an area) of an electronic signal associated with the in-phase photocurrent signal *i.sub.I*, and the quadrature portion *Q* may be determined from a characteristic associated with the quadrature photocurrent signal *i.sub.Q*. The in-phase portion *IP* may correspond to an amount of the Raman signal **160** that is in-phase with the probe beam **120pr**, and the quadrature portion *Q* may represent an amount of the Raman signal that is out of phase (i.e., 90-degrees phase-shifted) with the probe beam. For example, the in-phase portion *IP* and the quadrature portion *Q*, may each have values from -1 to 1 . If the Raman signal **160** is in-phase with the probe beam **120pr**, then the in-phase portion *IP* may have a value of approximately 1 , and the quadrature portion *Q* may have a value of approximately 0 . Similarly, if the Raman signal **160** is out of phase by ± 90 degrees with respect to the probe beam **120pr**, then the in-phase portion *IP* may have a value of 0 , and the quadrature portion *Q* may have a value of ± 1 . The phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr** may be determined from the expression $\Delta\phi = \arctan(Q/IP)$. For example, if *Q* is 0 and *IP* is 1 , then the Raman signal and the probe beam are substantially in phase, with a phase difference $\Delta\phi$ of 0 degrees. As another example, if *Q* is 1 and *IP* is 0 , then the Raman signal and the probe beam are substantially out of phase, with a phase difference $\Delta\phi$ of 90 degrees.

[0175] FIG. **39** illustrates an example optical receiver **200** configured to detect polarization as well as in-phase and quadrature components of a Raman signal **160**. The optical receiver **200** in FIG. **39** is similar the optical receiver **200** in FIG. **37**, where the horizontal-polarization optical receiver **200h** and the vertical-polarization optical receiver **200v** each includes a 90-degree optical hybrid **250**. Each of the 90-degree optical hybrids in FIG. **39** may be similar to the 90-degree optical hybrid **250** in FIG. **38**. In FIG. **39**, the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** are directed to a horizontal-polarization optical receiver **200h** that includes a 90-degree optical hybrid **250h** and four detectors **220h-I+**, **220h-I-**, **220h-Q+**, and **220h-Q-**. The vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-h** are directed to a vertical-polarization optical receiver **200v** that includes a 90-degree optical hybrid **250v** and four detectors **220v-I+**, **220v-I-**, **220v-Q+**, and **220v-Q-**. The h-polarization optical receiver **200h** may be used to determine the relative size of the horizontal-polarization Raman signal **160-h** as well as the in-phase and quadrature components of the horizontal-polarization Raman signal **160-h**. Similarly, the v-polarization optical receiver **200h** may be used to determine the relative size of the vertical-polarization Raman signal **160-v** as well as the

in-phase and quadrature components of the vertical-polarization Raman signal **160-v**.

[0176] The 90-degree optical hybrid **250h** in FIG. **39** combines the horizontal-polarization Raman signal **160-h** and the horizontal-polarization probe beam **120pr-h** to produce four horizontally polarized combined output beams: two in-phase combined beams **210h-I+** and **210h-I-**, and two quadrature combined beams **210h-Q+** and **210h-Q-**. Each of the four horizontally polarized combined beams includes a portion of the horizontal-polarization Raman signal **160-h** and a portion of the horizontal-polarization probe beam **120pr-h**. The four combined beams are directed to four respective detectors (**220h-I+**, **220h-I-**, **220h-Q+**, **220h-Q-**), and each detector produces a respective photocurrent signal (i.sub.h-I+, i.sub.h-I-, i.sub.h-Q+, i.sub.h-Q-) that corresponds to the coherent mixing of a portion of the horizontal Raman signal **160-h** and a portion of the horizontal probe beam **120pr-h**. Each of the four photocurrents i.sub.h-I+, i.sub.h-I-, i.sub.h-Q+, and i.sub.h-Q- may be referred to as a horizontal-polarization electronic signal. The photocurrents from the detectors are subtracted to produce a subtracted horizontal in-phase photocurrent signal i.sub.h-I in which is equal to i.sub.h-I+-i.sub.h-I- and a subtracted horizontal quadrature photocurrent signal i.sub.h-Q which is equal to i.sub.h-Q+-i.sub.h-Q-.

[0177] The 90-degree optical hybrid **250v** in FIG. **39** combines the vertical-polarization Raman signal **160-v** and the vertical-polarization probe beam **120pr-v** to produce four vertically polarized combined output beams: two in-phase combined beams **210v-I+** and **210v-I-**, and two quadrature combined beams **210v-Q+** and **210v-Q-**. Each of the four vertically polarized combined beams includes a portion of the vertical-polarization Raman signal **160-v** and a portion of the vertical-polarization probe beam **120pr-v**. The four combined beams are directed to four respective detectors (**220v-I+**, **220v-I-**, **220v-Q+**, **220v-Q-**), and each detector produces a respective photocurrent signal (i.sub.v-I+, i.sub.v-I-, i.sub.v-Q+, i.sub.v-Q-) that corresponds to the coherent mixing of a portion of the vertical Raman signal **160-v** and a portion of the vertical probe beam **120pr-v**. Each of the four photocurrents i.sub.v-I+, i.sub.v-I-, i.sub.v-Q+, and i.sub.v-Q- may be referred to as a vertical-polarization electronic signal. The photocurrents from the detectors are subtracted to produce a subtracted vertical in-phase photocurrent signal i.sub.v-I which is equal to i.sub.v-I+-i.sub.v-I- and a subtracted vertical quadrature photocurrent signal i.sub.v-Q which is equal to i.sub.v-Q+-i.sub.v-Q-.

[0178] Each of the subtracted photocurrent signals i.sub.h-I, i.sub.h-Q, i.sub.v-I, and i.sub.v-Q may be sent to detection electronics **230** that produces voltage signals and digital output signals corresponding to the subtracted photocurrent signals. Based on the digital output signals (which are determined from the four horizontal-polarization electronic signals and the four vertical-polarization electronic signals), a processor may determine (i) the polarization of the Raman signal **160** and (ii) a phase associated with the Raman signal (e.g., a phase difference $\Delta\phi$ between the Raman signal **160** and the probe beam **120pr**). Determining the polarization of a Raman signal **160** may include determining a relative size or ratio of the horizontal and vertical polarization components of the Raman signal. For example, the relative size of the horizontal polarization component of the Raman signal **160** may be determined by adding characteristics (e.g., areas or amplitudes) associated with the two horizontal photocurrent signals i.sub.h-I and i.sub.h-Q. Similarly, the relative size of the vertical polarization component of the Raman signal **160** may be determined by adding characteristics associated with the two vertical photocurrent signals i.sub.v-I and i.sub.v-Q. As an example, if the relative size of the horizontal polarization component of the Raman signal **160** is 1 and the relative size of the vertical polarization component of the Raman signal **160** is 100, then the Raman signal **160** may be determined to be substantially vertically polarized.

[0179] Based on the digital output signals, a processor may determine a phase associated with the Raman signal. For example, the processor may determine (i) a phase difference $\Delta\phi_{\text{sub.h}}$ between the horizontal Raman signal **160-h** and the horizontal probe beam **120pr-h** and (ii) a phase difference $\Delta\phi_{\text{sub.v}}$ between the vertical Raman signal **160-v** and the vertical probe beam **120pr-v**.

Based on the digital output signals, a processor may determine in-phase and quadrature portions associated with each of the horizontal Raman signal **160-h** and vertical Raman signal **160-v**. The phase difference $\Delta\phi_{\text{sub.h}}$ between the horizontal Raman signal **160-h** and the horizontal probe beam **120pr-h** may be determined from the expression $\Delta\phi_{\text{sub.h}} = \arctan(Q_{\text{sub.h}}/IP_{\text{sub.h}})$, where $Q_{\text{sub.h}}$ and $IP_{\text{sub.h}}$ are the quadrature and in-phase portions associated with the horizontal Raman signal. The phase difference $\Delta\phi_{\text{sub.v}}$ between the vertical Raman signal **160-v** and the vertical probe beam **120pr-v** may be determined from the expression $\Delta\phi_{\text{sub.v}} = \arctan(Q_{\text{sub.v}}/IP_{\text{sub.v}})$, where $Q_{\text{sub.v}}$ and $IP_{\text{sub.v}}$ are the quadrature and in-phase portions associated with the vertical Raman signal.

[0180] An optical receiver **200** may include one or more detectors **220**. An optical receiver **200** may include one detector **220** (e.g., as illustrated in FIGS. **1**, **2**, and **18**), or an optical receiver **200** may include multiple detectors **220** (e.g., as illustrated in FIGS. **17**, **36**, **38**, and **39**). An optical receiver **200** with multiple detectors **220** may include 2, 3, 4, 8, 16, or any other suitable number of detectors. For example, an optical receiver **200** may include two detectors **220** arranged so that their respective photocurrents are subtracted (e.g., as illustrated in FIG. **36**). As another example, an optical receiver **200** may include four detectors **220** (e.g., as illustrated in FIG. **38**) or eight detectors **220** (e.g., as illustrated in FIG. **39**). In an optical receiver **200** with multiple detectors **220**, portions of a probe beam **120pr** and a Raman signal **160** may be coherently mixed together at one or more of the multiple detectors **220**, and each of these one or more detectors may produce a photocurrent signal i corresponding to the coherent mixing of the probe beam and the Raman signal. Any of the optical receivers **200** described herein as having a single detector **220** may also be configured to have two or more detectors. For example, the optical receiver in FIG. **1** (which includes one detector **220**) may include a second detector (not illustrated in FIG. **1**), and the detection electronics **230** may be configured to receive and process photocurrent signals from each of the two detectors.

[0181] FIGS. **40-43** each illustrate an example Raman spectroscopy system **100** that includes a photonic integrated circuit (PIC) **170**. In each of FIGS. **40-43**, the PIC **170** includes multiple optical waveguides **172**, and a pump beam **120pu** and a Stokes beam **120S** are directed to a sample **150** by one or more of the waveguides. A Raman signal **160** is produced by coherent Raman scattering of the pump and Stokes beams of light at the sample **150**, and the Raman signal is received and directed to an optical receiver **200** by one or more of the waveguides **172**. The optical receiver **200** in each of FIGS. **40-43** includes a probe light source **111pr** or probe laser **110pr**, a detector **220**, and detection electronics **230**. The probe light source **110pr** or probe laser **110pr** produces a probe beam of light **120pr** that is combined with the Raman signal **160**, and the optical detector **220** produces a photocurrent signal i corresponding to coherent mixing between the Raman signal and the probe beam of light. The detection electronics **230** receives the photocurrent signal i and produces a digital output signal **240** corresponding to the photocurrent signal. Each of the Raman spectroscopy systems **100** in FIGS. **40-43** may be referred to as a Raman spectroscopy system with a PIC, a PIC-based Raman spectroscopy system, a portable Raman spectroscopy system, or a wearable Raman spectroscopy system.

[0182] A Raman spectroscopy system **100** may include a pump, Stokes, or probe light source **111** or may include a pump, Stokes, or probe laser **110**. The Raman spectroscopy system **100** in FIG. **40** includes a pump light source **111pu**, a Stokes light source **111S**, and a probe light source **111pr**. The Raman spectroscopy system **100** in each of FIGS. **41-43** includes a pump laser **110pu**, a Stokes laser **110S**, and a probe laser **110pr**. A Raman spectroscopy system **100** that includes a light source **111** (which may include multiple fixed-wavelength or tunable laser diodes) may instead include a single laser **110** (which may be a fixed-wavelength or tunable laser diode), and vice versa. For example, the Raman spectroscopy system **100** in FIG. **40** may include a pump laser **110pu** instead of a pump light source **111pu**, a Stokes laser **110S** instead of a Stokes light source **111S**, or a probe laser **110pr** instead of a probe light source **111pr**. As another example, the Raman spectroscopy

system **100** in each of FIGS. **41-43** may include a pump light source **111_{pu}** instead of a pump laser **110_{pu}**, a Stokes light source **111_S** instead of a Stokes laser **110_S**, or a probe light source **111_{pr}** instead of a probe laser **110_{pr}**. A pump, Stokes, or probe light source may be similar to the light source **111** illustrated in FIG. **24**.

[0183] The PICs in each of FIGS. **40-43** include multiple optical waveguides **172** that are used to guide light from one location or device to another location or device. An optical waveguide **172** of a PIC **170** may confine and guide light based on the higher refractive index of the waveguide with respect to surrounding cladding material. A PIC **170** may include silicon-on-insulator (SOI), silica-on-silicon, indium phosphide (InP), silicon nitride (SiN), an electro-optic material (e.g., lithium niobate or lithium tantalate), graphene, or a polymer, and the optical waveguides of a PIC may be fabricated into the PIC material. For example, a SOI PIC may be fabricated from a wafer that includes a silicon substrate, a layer of silicon dioxide (referred to as an insulator or silica layer), and a layer of silicon. The waveguides are silicon waveguides formed (e.g., using photolithography) in the silicon layer so that light propagates in a silicon waveguide that is at least partially surrounded by silicon dioxide, which forms a cladding. As another example, in a silica-on-silicon PIC, light propagates in a silica waveguide that is doped to increase its refractive index with respect to the surrounding silica material, which forms a cladding. As another example, an InP PIC may have waveguides made from a quaternary compound (e.g., InGaAsP or InGaAlAs) that are at least partially surrounded by lower-refractive-index InP, which forms a cladding. Additionally, an InP PIC may include an InP-based laser diode that is integrated into and fabricated along with the PIC. As another example, a polymer PIC may be made from a polymer material (e.g., polyimide, an acrylic polymer (e.g., an acrylate or methacrylate), an epoxy-based material (e.g., SU-8), a polycarbonate, or a fluorinated polymer), and waveguides may be fabricated using photolithography, laser direct writing, imprint lithography, or reactive ion etching. The waveguides in a polymer PIC may include a higher refractive index polymer material that is at least partially surrounded by one or more lower refractive index polymer materials, which form a cladding.

[0184] A PIC **170** may include or may be made from a flexible material. For example, a PIC **170** may be made from a material having a flexural modulus between approximately 0.002 gigapascals (GPa) and approximately 4 GPa. As another example, a PIC **170** may be made from a polymer material that is flexible.

[0185] In some embodiments, a PIC **170** may be attached to a substrate **320** (e.g., using an epoxy or adhesive). The substrate **320** may provide protection or mechanical support for the PIC **170**, or the substrate **320** may be used for attachment or support of other components that are part of a PIC-based Raman spectroscopy system **100**. For example, the PIC **170** in each of FIGS. **40-43** may be attached to the substrate **320** using epoxy or adhesive, and one or more of the other components (e.g., light source **111**, laser **110**, detector **220**, detection electronics **230**, computer **1000**, power-management system **300**, battery **310**) may be attached to the substrate using epoxy, adhesive, or solder.

[0186] In some embodiments, a substrate **320** may include or may be made from a rigid material (e.g., a metal (e.g., aluminum or copper), a ceramic (e.g., aluminum nitride), or a rigid plastic material). A rigid substrate **320** may be attached to a PIC **170** that includes a material having low flexibility (e.g., silicon, InP, SiN, or an electro-optic material), and the substrate may provide mechanical support to prevent the PIC from damage caused by flexing. In other embodiments, a substrate **320** may include or may be made from a flexible material. For example, a PIC **170** may be made from a flexible polymer material, and the substrate **320** may be made from a flexible material such as polyimide, polyethylene naphthalate (PEN), silicone, a cyclic olefin polymer, or a metal foil. A flexible substrate **320** attached to a flexible PIC **170** may provide a protective cover for the PIC or the components coupled to the PIC while also allowing the PIC to flex. A Raman spectroscopy system **100** that includes a flexible PIC **170** or a flexible substrate **320** may be attached to a surface that moves or flexes. For example, a PIC-based Raman spectroscopy system

100 may be affixed to the skin of a patient's body, and the flexibility of the PIC **170** or substrate **320** may allow the system to operate and remain attached to the patient while the patient moves. A substrate **320** may include a thermally conductive material (e.g., a metal, a ceramic, or a silicone-based material combined with a thermally conductive material) to dissipate heat produced by a laser **110** or other electronic component.

[0187] The Raman signal **160** in each of FIGS. **40-43** is produced by coherent Raman scattering of the pump and Stokes beams of light at the sample **150**. The Raman signal **160** may be produced by coherent Raman scattering that occurs within the sample **150**, at the surface of the sample, or near the sample. For example, a Raman signal **160** may be produced while the pump and Stokes beams propagate through the sample as free-space optical beams. The beams may propagate along one or more optical waveguides **172** as guided optical beams and may then be emitted as free-space optical beams that propagate through the sample. An optical waveguide **172** may have an output face with a curved shape or a lens may be attached to or positioned near the output face so that a beam is emitted as a collimated or focused free-space optical beam. The sample **150** may be located within a sample holder (e.g., a cuvette), and the pump and Stokes beams may be emitted from the output face of an optical waveguide **172** and directed to propagate into the sample holder and then through the sample. A sample holder may be integrated into, attached to, or located near a PIC **170** or a substrate **320**. As another example, a Raman signal **160** may be produced while the pump and Stokes beams propagate through a channel or a hollow optical waveguide that contains the sample. The pump and Stokes beams may propagate along an optical waveguide **172** and may then be coupled into a channel or hollow waveguide which contains a sample. The channel or hollow waveguide may be integrated into, attached to, or located near a PIC **170** or a substrate **320**. As another example, a Raman signal **160** may be produced while the pump and Stokes beams propagate near a sample **150**. The pump and Stokes beams may propagate along an evanescent waveguide located near the sample, and evanescent portions of the pump and Stokes beams that extend into the sample may interact with the sample to produce a Raman signal **160**.

[0188] In a PIC-based Raman spectroscopy system **100**, a pump beam **120_{pu}** and a Stokes beam **120_S** may be directed to a sample **150** together as a combined optical beam or separately as two optical beams. In FIG. **40**, the pump and Stokes beams are combined at a waveguide combiner **130_a** to produce a combined pump-Stokes beam **140**. The combined pump-Stokes beam **140** includes the pump and Stokes beams, which propagate to the sample **150** along the same optical waveguide **172**. The pump and Stokes beams, which propagate together in the same direction and along the same optical waveguide **172**, may be referred to as co-propagating optical beams. In FIG. **41**, the pump and Stokes beams propagate to the sample **150** as two separate optical beams that are directed to the sample by two different optical waveguides **172**. The pump and Stokes beams in FIG. **41** enter the sample **150** from opposite sides and may be referred to as counter-propagating optical beams that propagate to the sample in opposite directions.

[0189] The Raman spectroscopy system **100** in each of FIGS. **40-43** includes a battery **310** that supplies electrical power to one or more active components of the system. For example, the battery **310** may supply power to a light source **111**, laser **110**, detector **220**, detection electronics **230**, or computer **1000**. The battery **310** may be a rechargeable battery that is charged by wired charging or wireless charging. A Raman spectroscopy system **100** may also include a power-management system **300** that filters, regulates, conditions, or modifies a voltage or current supplied by the battery **310**. For example, the battery **310** may have a nominal output voltage of 12 volts, and the power-management system **300** may include a current source (e.g., a current regulator or a laser-diode driver) powered by the battery that supplies an electrical current of approximately 100 mA to a laser diode. As another example, the detector **220** may operate with a reverse-bias voltage of 5 volts, and the power-management system **300** may include a voltage regulator that converts the battery voltage to a 5-volt bias voltage. As another example, the power-management system **300** may include a filter or protection circuit that filters out electrical noise or suppresses current or

voltage transients.

[0190] The Raman spectroscopy system **100** in each of FIGS. **40-43** includes a computer **1000** (which may be referred to as a computer system, a controller, a computing device, a computing system, a processor, or a data-processing apparatus). The computer **1000** may receive a digital output signal **240** from detection electronics **230** and determine a characteristic of the corresponding photocurrent signal *i* based on the digital output signal. For example, the characteristic of a photocurrent signal *i* (or the characteristic of a voltage signal **234** that corresponds to the photocurrent signal) determined by the computer **1000** may include one or more of: a peak amplitude (e.g., a peak amplitude of the photocurrent signal or a peak amplitude of the corresponding voltage signal), an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, and a polarization (e.g., a polarization of a Raman signal). A computer **1000** of a Raman spectroscopy system **100** may communicate with an external computer using a wired connection or a wireless connection (e.g., via BLUETOOTH or a WI-FI network). For example, a computer **1000** may send a digital output signal **240** to an external computer via a wired or wireless connection, and the external computer may process the signal to determine a characteristic of the corresponding photocurrent signal *i*. Alternatively, a computer **1000** may perform some initial data analysis of a digital output signal **240** (e.g., to determine a characteristic of a photocurrent signal) and send the result to an external computer, and the external computer may store the result or perform additional analysis.

[0191] A Raman spectroscopy system **100** may include a display (which may be referred to as a screen, monitor, or visual output device). The display may be used to display a value, image, or graph associated with a sample **150** measured by the Raman spectroscopy system **100**. For example, a display may show an amount or concentration of a particular material in the sample **150** or may show a graph of the amount or concentration of the material versus time. Additionally or alternatively, the display may present an alert from the Raman spectroscopy system **100**. For example, an alert may indicate that the battery **310** needs to be charged. As another example, an alert may indicate that the amount or concentration of a particular material in the sample **150** is above or below a particular target (e.g., a patient's blood glucose concentration is too high or too low). The display may be a touchscreen that allows a user to interact with the Raman spectroscopy system **100** (e.g., to change an operating parameter or to prompt the system to perform a particular measurement).

[0192] In each of FIGS. **40-43**, a light source **111** or laser **110** produces an optical beam **120** that is coupled into an optical waveguide **172** of a PIC **170**. For example, the Stokes light source **111S** in FIG. **40** may include multiple laser diodes and an optical multiplexer (e.g., similar to the light source **111** illustrated in FIG. **24**), and the Stokes beam **120S** may be coupled from the optical multiplexer into the optical waveguide **172** of the PIC **170** as a free-space optical beam (e.g., using one or more lenses to couple the Stokes light into the waveguide), a fiber-coupled optical beam, or a waveguide-coupled beam (e.g., the optical multiplexer may include an optical waveguide that is directly coupled to the optical waveguide of the PIC). As another example, the pump laser **110_{pu}** in FIG. **41** may be a laser diode (e.g., similar to that illustrated in FIG. **26**), and the pump beam **120_{pu}** may be coupled into the optical waveguide **172** of the PIC **170** as a free-space optical beam (e.g., using one or more lenses to couple the pump light from the laser to the waveguide), by butt-coupling the laser diode to the input of the waveguide, or via optical fiber. A laser diode may be butt-coupled to an optical waveguide **172** by positioning the output face of the laser diode so that it is attached to, in contact with, or located near (e.g., less than 1 mm from) the input face of the optical waveguide. For example, the output face of the laser diode may be attached to the input face of the optical waveguide using an optically clear adhesive. Similarly, light from an optical fiber may be coupled into an optical waveguide **172** by positioning the end face of the optical fiber so that it is attached to, in contact with, or located near the input face of the optical waveguide.

Alternatively, a laser diode may be fabricated together with a PIC so that the laser diode is directly integrated into the PIC during the fabrication process. For example, the PIC **170** in FIG. **41** may be an InP-based PIC, and the Stokes laser **110S** may be an InP-based laser diode that is fabricated together with the PIC.

[0193] A PIC-based Raman spectroscopy system **100** may include one or more optical detectors **220**. For example, a PIC-based Raman spectroscopy system **100** may include a detector **220** for detecting a probe-Raman signal **210**. Additionally, a system may include one or more additional detectors, for example, to act as a reference detector in a balanced-detection configuration (e.g., to detect a portion of the Stokes beam **120S** or probe beam **120pr**). Each of the Raman spectroscopy systems **100** in FIGS. **40-43** includes a detector **220** configured to receive a combined probe-Raman signal **210** and produce a photocurrent signal *i* corresponding to coherent mixing of the Raman signal **160** and the probe beam **120pr**. In each of FIGS. **40-43**, a waveguide **172** directs the probe-Raman signal **210** to the detector **220**, and the detector may be mechanically attached or connected to the PIC **170** or to the substrate **320** (e.g., using epoxy, adhesive, or solder). The detector **220** or a package that houses the detector may be located at or near (e.g., less than 1 mm from) the output face of the waveguide **172**. For example, the detector **220** or a package that houses the detector may be attached to the output face of the waveguide **172** (e.g., using an optically clear adhesive) to directly receive the probe-Raman signal **210**. Alternatively, one or more lenses may receive the probe-Raman signal **210** from the waveguide as a free-space optical beam and focus or collimate the probe-Raman signal to couple the signal to the detector **220**.

[0194] The PIC **170** in FIG. **40** includes a waveguide combiner **130a** that combines the pump beam **120pu** and the Stokes beam **120S** to produce a combined pump-Stokes beam **140** that is directed to the sample **150** by an optical waveguide **172**. The pump-Stokes waveguide combiner **130a** in FIG. **40** is similar to the waveguide combiner **130** illustrated in FIG. **27**. In FIG. **40**, the pump-Stokes beam **140** is directed to a sample **150** and interacts with the sample to produce a Raman signal **160** by coherent Raman scattering of the pump and Stokes beams. The Raman signal **160** is received by an optical waveguide **172**, which directs the Raman signal to another waveguide combiner **130b**. The optical waveguide **172** that directs the pump-Stokes beam **140** to the sample **150** may have a curved output face or a lens may be attached to or positioned near the output face to collimate or focus the pump-Stokes beam into the sample **150**. Similarly, the optical waveguide **172** that receives the Raman signal **160** may have a curved input face or a lens may be attached to or positioned near the input face to collect and focus the Raman signal into the waveguide. The waveguide combiner **130b** combines the Raman signal **160** and the probe beam of light **120pr** to produce a combined probe-Raman signal **210** that is directed to a detector **220** by an optical waveguide **172**.

[0195] The PIC **170** in FIG. **41** includes a waveguide combiner **130** that is similar to the waveguide combiner **130** illustrated in FIG. **29**. The waveguide combiner **130** in FIG. **41(i)** directs at least a portion of the pump beam **120pu** to the sample **150** and (ii) combines the Raman signal **160** and the probe beam **120pr** to produce a combined probe-Raman signal **210** that is directed to a detector **220** by an optical waveguide **172**. The pump beam **120pu** is directed from the pump laser **110pu** to the waveguide combiner **130** via an optical waveguide **172**, and at least a portion of the pump beam exits the waveguide combiner via another optical waveguide **172**, which directs the pump beam to the sample **150**. The pump and Stokes beams are directed to the sample **150** by different optical waveguides **172**, and the Raman signal **160** produced by the pump and Stokes beams at the sample is coupled into the waveguide in which the pump beam **120pu** propagates. The pump beam **120pu** and the Raman signal **160** propagate in opposite directions along the same optical waveguide **172**, and the Raman signal is directed along that optical waveguide from the sample **150** to the waveguide combiner **130**. In other embodiments, the locations of the pump and Stokes lasers may be reversed so that the Stokes beam **120S** propagates through the waveguide combiner **130** to the sample **150**, and the Stokes beam and the Raman signal **160** propagate in opposite directions along

the same optical waveguide **172**. In FIG. **41**, the Raman signal **160** and the probe beam **120_{pr}** are directed to the waveguide combiner **130** via different optical waveguides **172**, and the waveguide combiner combines the Raman signal and the probe beam to produce a combined probe-Raman signal **210**. The combined probe-Raman signal **210** exits the waveguide combiner **130** via another optical waveguide **172**, which directs the probe-Raman signal to a detector **220**.

[0196] A PIC-based Raman spectroscopy system **100** may include an optical circulator **131** that directs the pump beam **120_{pu}**, Stokes beam **120_S**, or Raman signal **160** from one waveguide to another. An optical circulator **131** is a three-port optical component that receives input light at one port and directs that light to exit the circulator from another port. For example, light that enters the optical circulator **131** at port **1** is directed to exit the circulator from port **2**, and light that enters the circulator at port **2** is directed to exit the circulator from port **3**. An optical circulator **131** may be fabricated into a PIC **170**, or an optical circulator may be fabricated separately and then integrated into a PIC. Each of the Raman spectroscopy systems **100** in FIGS. **42** and **43** includes an optical circulator **131** with an optical waveguide **172** coupled to each of its three ports.

[0197] In FIG. **42**, a waveguide combiner **130_a** combines the pump beam **120_{pu}** and the Stokes beam **120_S** to produce a combined pump-Stokes beam **140** that is directed to an optical circulator **131** by an optical waveguide **172**. The circulator **131** receives the pump and Stokes beams at port **1** and directs the pump and Stokes beams to exit the circulator via port **2**. Port **2** is coupled to an optical waveguide **172** that directs the pump and Stokes beams to the sample **150**. The Raman signal **160** produced by the pump and Stokes beams at the sample **150** is directed to port **2** along the same optical waveguide **172** as the pump and Stokes beams. The circulator **131** receives the Raman signal **160** at port **2** and directs the Raman signal to exit the circulator via port **3**. Port **3** is coupled to an optical waveguide **172** that directs the Raman signal **160** to the optical receiver **200**, where the Raman signal is combined with a probe beam **120_{pr}** and directed to a detector **220**.

[0198] In FIG. **43**, the pump and Stokes beams are sent to the sample **150** separately. The pump beam **120_{pu}** is directed to port **1** of the circulator **131**, and the circulator directs the pump beam to exit the circulator via port **2**. Port **2** is coupled to an optical waveguide **172** that directs the pump beam **120_{pu}** to the sample **150**, and the Stokes beam **120_S** is directed to the sample via a separate optical waveguide. The Raman signal **160** produced by the pump and Stokes beams at the sample **150** is directed to port **2** along the same optical waveguide **172** as the pump beam **120_{pu}**. The circulator **131** receives the Raman signal **160** at port **2** and directs the Raman signal to exit the circulator via port **3**. Port **3** is coupled to an optical waveguide **172** that directs the Raman signal **160** to the optical receiver **200**, where the Raman signal is combined with a probe beam **120_{pr}** and directed to a detector **220**. In other embodiments, the pump and Stokes lasers may be reversed so that the Stokes beam **120_S** propagates through an optical circulator **131** and then to the sample **150**, and the pump beam **120_{pu}** is directed to the sample via a separate optical waveguide.

[0199] FIGS. **44-45** illustrate an example Raman spectroscopy system **100** that includes a photonic integrated circuit (PIC) **170** with an evanescent waveguide **400**. FIG. **44** illustrates a top view of the Raman spectroscopy system **100**, and FIG. **45** illustrates a side view of the system. The Raman spectroscopy system **100** in FIGS. **44-45** is similar to the Raman spectroscopy system **100** in FIG. **40**, except in FIGS. **44-45** the pump and Stokes beams interact with the sample **150** via an evanescent waveguide **400**, while in FIG. **40** the pump and Stokes beams propagate directly into or through the sample **150**. Any of the Raman spectroscopy systems **100** in FIGS. **40-43** may include an evanescent waveguide **400** (e.g., similar to that illustrated in FIGS. **44-45**), and the evanescent waveguide may allow the pump and Stokes beams to interact with a sample **150** to produce a Raman signal **160** without the pump and Stokes beams traveling directly into or through the sample.

[0200] An evanescent waveguide **400** is a type of optical waveguide in which a portion of the electromagnetic field of light propagating in the waveguide extends beyond the boundary of the waveguide core. For example, an optical waveguide may include a core waveguide region having a

higher refractive index than a surrounding cladding region. Light propagating in the waveguide may be primarily confined to the waveguide core, with an evanescent-field portion of the light located in the cladding. The portion of light located outside the waveguide core may be referred to as the evanescent-field portion. The dashed-line inset in FIG. 45 is an example plot of the intensity versus position along the z-axis for a combined pump-Stokes beam 140 while propagating in the evanescent waveguide 400. The region labeled 400 represents the core of the evanescent waveguide, and the region labeled 151 represents the evanescent-field portion of the pump and Stokes beams, which extend beyond the waveguide boundary and into the sample 150. The interaction region 152 in FIG. 45 represents the region where the evanescent-field portion 151 overlaps the sample 150 and corresponds to the part of the sample from which the Raman signal 160 is produced. The design parameters (e.g., size, shape, refractive index) of an evanescent waveguide 400 may be selected to increase the intensity or extent of the evanescent-field portion 151 as compared to a conventional optical waveguide. Additionally, an evanescent waveguide 400 may have an extended length to provide a sufficient distance for the pump and Stokes beams to interact with a sample 150 and produce a significant Raman signal 160. An evanescent waveguide 400 may have a spiral shape (e.g., as illustrated in FIG. 44), a serpentine shape, or any other suitable shape.

[0201] In FIG. 44, the pump and Stokes beams are combined at the waveguide combiner 130a to produce a combined pump-Stokes beams 140 that propagates through the evanescent waveguide 400. As illustrated in FIG. 45, the evanescent waveguide 400 may be positioned at or near the bottom edge of the PIC 170, and the sample 150 may be located near or adjacent to the evanescent waveguide so that a significant part of the evanescent-field portion 151 extends into the sample. For example, the Raman spectroscopy system 100 may be affixed to the skin of a patient's body, and the sample 150 may include sweat, the patient's skin, or interstitial fluid or blood located just under the surface of the skin. In FIG. 45, the Raman signal 160 is produced by the evanescent-field portions 151 of the pump and Stokes beams that extend into the sample 150 while the pump and Stokes beams propagate along the evanescent waveguide 400. At least a portion of the Raman signal 160 is coupled into the evanescent waveguide 400, and the Raman signal 160 propagates along the evanescent waveguide 400 and is then coupled to an optical waveguide 172. In FIG. 44, the Raman signal 160 and the probe beam 120 are combined at the waveguide combiner 130b to produce a combined probe-Raman signal 210, which is directed to a detector 220.

[0202] FIG. 46 illustrates an example Raman spectroscopy system 100 that includes a photonic integrated circuit (PIC) 170 and a hollow waveguide 410. A hollow waveguide 410, which may be referred to as a hollow optical waveguide or a channel waveguide, is a type of optical waveguide that provides a confined region for a combined pump-Stokes beam 140 to interact with a sample 150. A hollow waveguide 410 may include a hollow interior space into which a sample 150 may flow and through which a pump-Stokes beam 410 may propagate. The interior surface of a hollow waveguide 410 may include a reflective optical coating (e.g., a reflective dielectric or metallic coating) that reflects and confines the pump-Stokes beam 140 while propagating through the waveguide. A hollow waveguide 410 may be fabricated into a PIC 170, or a hollow waveguide may be fabricated separately and then integrated into or coupled to a PIC. Any of the Raman spectroscopy systems 100 in FIGS. 40-43 may include a hollow waveguide 410 similar to that illustrated in FIG. 46.

[0203] In FIG. 46, the pump-Stokes beam 140 propagates along an optical waveguide 172 of a PIC 170 and is then directed into a hollow waveguide 410 that contains a sample 150. The pump-Stokes beam 140 propagates along the hollow waveguide 410 and through the sample 150 contained in the waveguide, and the pump-Stokes beam interacts with the sample to produce a Raman signal 160 by coherent Raman scattering of the pump and Stokes beams. The Raman signal 160 propagates along the hollow waveguide 410 and is then coupled into an optical waveguide 172. The optical waveguide 172 that directs the pump-Stokes beam 140 into the hollow waveguide 410 may have a

curved output face or a lens may be attached to or positioned near the output face to collimate or focus the pump-Stokes beam into the hollow waveguide. Similarly, the optical waveguide **172** that receives the Raman signal **160** from the hollow waveguide **410** may have a curved input face or a lens may be attached to or positioned near the input face to collect and focus the Raman signal into the optical waveguide **172**.

[0204] A hollow waveguide **410** may include 1, 2, 3, or any other suitable number of sample ports **420**. A sample port **420** may be a hole located along a side or at an end of a hollow waveguide **410**, and a sample **150** may enter or exit the waveguide through a sample port. For example, the sample **150** may be a fluid that flows into or out of the hollow waveguide **410** via one or more sample ports **420**. In FIG. **46**, the hollow waveguide **410** includes a sample port **420** that allows the sample **150** to flow into or out of the waveguide. In other embodiments, a hollow waveguide **410** may include two sample ports **420**, and a sample **150** may flow into the hollow waveguide through one of the ports and out of the hollow waveguide through the other port. In some embodiments, a hollow waveguide **410** may be coupled to a device (e.g., a piezoelectric transducer, a vibration motor, an acoustic actuator, or a microfluidic pump) that imparts a vibration or a pumping action to the waveguide or the sample **150** to cause the sample to flow through the waveguide. A vibration or pumping device may allow multiple portions of a fluid sample **150** to be measured by preventing a small portion of the sample from stagnating and remaining substantially still within the hollow waveguide **410**.

[0205] A PIC-based Raman spectroscopy system **100** that includes a hollow waveguide **100** may be used to measure a fluid or gas sample **150** (e.g., sweat, saliva, blood, or breath). For example, a PIC-based Raman spectroscopy system **100** that includes a hollow waveguide **410** may be affixed to the skin of a patient's body, and the sample **150** may include sweat that flows into the hollow waveguide. As another example, a PIC-based Raman spectroscopy system **100** that includes a hollow waveguide **410** may be integrated into a dental appliance that fits into a patient's mouth, and the sample **150** may include saliva that flows into the hollow waveguide. As another example, a patient may exhale into a hollow waveguide **410** of a PIC-based Raman spectroscopy system **100**, and the system may perform a Raman-signal measurement of the patient's breath that flows through the hollow waveguide.

[0206] A pump-Stokes beam **140** and a corresponding Raman signal **160** may be substantially overlapped or co-propagating while propagating through a sample **150**. In FIG. **46**, the pump-Stokes beam **140** and the Raman signal **160** are illustrated as being offset laterally for clarity to allow each beam to be visualized. Other figures described herein may include similar lateral offsets between optical beams for clarity to allow the beams to be visualized.

[0207] FIGS. **47-49** each illustrate an example Raman spectroscopy system **100** that includes a photonic integrated circuit (PIC) **170** with a grating **430**. A PIC **170** may be a planar device having optical waveguides **172** arranged along the plane of the PIC, and an optical waveguide may include a grating **430** that directs an optical beam out of the PIC along an angle with respect to the plane of the PIC. A grating **430**, which may be referred to as a diffraction grating, waveguide grating, or grating coupler, may include a periodic variation in refractive index that is fabricated or written into an optical waveguide **172** of a PIC **170**. For example, a grating **430** may include a physical corrugation, a variation in material composition, or stressed-induced changes in refractive index applied at or near an optical waveguide **172**. A grating **430** may diffract light that propagates in an optical waveguide **172** of a PIC **170** to direct the light along a direction that is at an angle Θ with respect to the plane of the PIC. The diffraction angle Θ may have any suitable value, for example, between approximately 5° and approximately 80° . The parameters of a grating **430** (e.g., grating period, duty cycle, number of periods) may be selected to produce a particular diffraction angle Θ or a particular wavelength range over which the grating operates.

[0208] In each of FIGS. **47-49**, an optical waveguide **172** includes a grating **430** that directs a pump-Stokes beam of light **140** to a sample **150** and along a direction that is at an angle Θ with

respect to the plane of the PIC **170**. The plane of the PIC **170** corresponds to the x-y plane, and the pump-Stokes beam **140** is directed out of the x-y plane in the $-z$ direction along an angle Θ with respect to the x-y plane. The diffracted pump-Stokes beam **140** is directed to a sample **150** located external and adjacent to the Raman spectroscopy system **100**. A PIC-based Raman spectroscopy system **100** with a grating **430** may be used to perform a Raman-signal measurement of a sample **150** located external to the system. For example, the Raman spectroscopy system **100** may be a wearable device that is affixed to the skin of a patient's body, and the sample **150** may include sweat, the patient's skin, or interstitial fluid or blood located under the surface of the skin. As another example, the Raman spectroscopy system **100** may be configured as a portable handheld device that is brought close to or in contact with a sample **150** to perform a Raman-signal measurement of the sample. Any of the Raman spectroscopy systems **100** in FIGS. **40-43** may include an optical waveguide **172** with a grating **430** to perform Raman-signal measurements of a sample **150** located external to the system.

[0209] In FIG. **47**, a combined pump-Stokes beam **140** propagates along an optical waveguide **172** and is directed by a grating **430** along a direction that is at an angle θ with respect to the x-y plane of the PIC **170**. The pump-Stokes beam **140** is directed to a sample **150** located external to the Raman spectroscopy system **100**, and a Raman signal **160** is produced by coherent Raman scattering of the pump and Stokes beams while propagating through the sample. The Raman signal **160** is directed to the grating **430**, which diffracts the Raman signal to couple the Raman signal into the same optical waveguide **172** in which the pump-Stokes beam **140** propagates. The Raman signal **160** propagates in the opposite direction as the pump-Stokes beam **140** (e.g., similar to the arrangement in FIG. **42**), and the Raman signal may be directed to an optical receiver for detection.

[0210] The Raman spectroscopy system **100** in FIG. **48** includes an output optical waveguide **172a** that directs a pump-Stokes beam **140** to a sample **150** and an input optical waveguide **172b** that receives a Raman signal **160** (e.g., similar to the arrangement in FIG. **40**). The output optical waveguide **172a** includes an output grating **430a**, and the input optical waveguide **172b** includes an input grating **430b**. The combined pump-Stokes beam **140** propagates along the output optical waveguide **172a** and is directed by the output grating **430a** along an angle θ and to a sample **150** located external to the system **100**. A Raman signal **160** is produced by coherent Raman scattering of the pump and Stokes beams while propagating through the sample **150**. The Raman signal **160** is directed to the input grating **430b**, which receives the Raman signal and diffracts the Raman signal to couple the Raman signal into the input optical waveguide **172b**. The input grating **430b** may direct the Raman signal **160** to propagate along the input optical waveguide **172b**, and the Raman signal may propagate along the waveguide to an optical receiver for detection.

[0211] In FIG. **49**, a combined pump-Stokes beam **140** propagates along an optical waveguide **172** and is directed by a grating **430** along a direction that is at an angle θ with respect to the x-y plane of the PIC **170**. The pump-Stokes beam **140** is directed to a sample **150** located external to the Raman spectroscopy system **100**, and a Raman signal **160** is produced by coherent Raman scattering of the pump and Stokes beams while propagating through the sample. The Raman signal **160** is a distributed Raman signal that includes multiple Raman-signal portions produced while the pump and Stokes beams propagate through the sample. Each portion of the Raman signal **160** is produced from a different part of the sample **150**, and the Raman spectroscopy system **100** includes multiple detectors **220**, where each detector receives one of the Raman-signal portions. Detector **220a** receives Raman-signal portion **160a**, produced by a part of the sample **150** located relatively close to the Raman spectroscopy system **100**. Detector **220b** receives Raman-signal portion **160b**, produced by a part of the sample **150** located farther from the Raman spectroscopy system **100**. Detector **220c** receives Raman-signal portion **160c**, produced by a part of the sample **150** located farther still from the Raman spectroscopy system **100**. The Raman spectroscopy system **100** in FIG. **49** may be used to perform Raman-signal measurements of a sample **150** as a function of depth or distance from the system.

[0212] FIG. 50 illustrates an example Raman spectroscopy system **100** that includes a photonic integrated circuit (PIC) **170** and a mirror **460**. The mirror is located at the end of the waveguide **172** and reflects the pump-Stokes beam **140** to direct the pump-Stokes beam to a sample **150** and along a direction that is at an angle α with respect to the plane of the PIC **170**. The reflection angle α may have any suitable value, for example, between approximately 5° and approximately 175° . The mirror **460** may be fabricated into the PIC **170**, or the mirror may be fabricated separately and then integrated into or coupled to the PIC. For example, the mirror **460** may be fabricated into the PIC **170** by applying a reflective optical coating to an angled end face of the optical waveguide **170**. Alternatively, the mirror **460** may be fabricated separately and then attached to the PIC **170** or to the end face of the optical waveguide **170**. Any of the Raman spectroscopy systems **100** in FIGS. 40-43 may include a mirror **460** that directs a pump-Stokes beam **140** to a sample **150** and along a direction that is at an angle with respect to the plane of the PIC **170**.

[0213] In FIG. 50, a combined pump-Stokes beam **140** propagates along an optical waveguide **172** and is reflected by a mirror **460** along a direction that is at an angle α with respect to the x-y plane of the PIC **170**. The pump-Stokes beam **140** is directed to a sample **150** located external to the Raman spectroscopy system **100**, and a Raman signal **160** is produced by coherent Raman scattering of the pump and Stokes beams while propagating through the sample. The Raman signal **160** is directed to the mirror **460**, which reflects the Raman signal to couple the Raman signal into the same optical waveguide **172** in which the pump-Stokes beam **140** propagates. The Raman signal **160** propagates in the opposite direction as the pump-Stokes beam **140** (e.g., similar to the arrangement in FIG. 42), and the Raman signal may be directed to an optical receiver for detection. In other embodiments, a Raman spectroscopy system **100** may include an output optical waveguide with a mirror and an input optical waveguide with a mirror (e.g., similar to the arrangement in FIG. 48, with mirrors used in place of the gratings).

[0214] FIG. 51 illustrates an example Raman spectroscopy system **100** that is affixed to a patient **650**. A Raman spectroscopy system **100** may be affixed to a patient **650** using adhesive, tape, a hook-and-loop fastener, or any other suitable attachment technique. The patient **650** may be a human or an animal, and the Raman spectroscopy system **100** may be affixed to any suitable part of the patient's body (e.g., arm, leg, abdomen, or head). In FIG. 51, the Raman spectroscopy system **100** is affixed so that it is in contact with the skin **600** of the patient **650**.

[0215] The Raman spectroscopy system **100** in FIG. 51 may be referred to as a wearable Raman spectroscopy system and may be configured to remain affixed to the patient's body and perform Raman-signal measurements over an extended period of time (e.g., over a period of approximately one hour, one day, one week, or one month). The Raman-signal measurements may be performed continuously, periodically, or on demand when initiated by the patient **650**. For example, the Raman spectroscopy system **100** may perform a Raman-signal measurement (or a series of measurements) and then go into a low-power mode for some amount of time (e.g., one minute, 10 minutes, or one hour) before performing a subsequent measurement. The Raman spectroscopy system **100** may be powered by a rechargeable battery **310**, and performing measurements periodically with low-power operation between the measurements may allow the system to operate for an extended period of time before needing to charge the battery.

[0216] The Raman spectroscopy system **100** in FIG. 51 may be a PIC-based Raman spectroscopy system, and the sample **150** measured by the system may include sweat, skin **600**, interstitial fluid, blood, saliva, or any other suitable fluid or part of a patient's body. For example, the Raman spectroscopy system **100** may include a PIC **170** with an evanescent waveguide **400** (e.g., as illustrated in FIGS. 44-45), and the system may produce pump and Stokes beams that interact with a sample **150** (e.g., sweat, skin **600**, interstitial fluid, or blood) via the evanescent waveguide **400**. As another example, the Raman spectroscopy system **100** may include a hollow waveguide **410** (e.g., as illustrated in FIG. 46), and the system may produce pump and Stokes beams that interact with a sample **150** (e.g., sweat or saliva) that is contained in the hollow waveguide. As another

example, the Raman spectroscopy system **100** may include a grating **430** (e.g., as illustrated in FIGS. **47-49**) or mirror **460** (e.g., as illustrated in FIG. **50**) that directs the pump and Stokes beams into the patient's body, and the sample **150** may include skin **600**, interstitial fluid, or blood. As another example, the Raman spectroscopy system **100** may include one or more optical fibers **116** that are inserted into a patient's body (e.g., as illustrated in FIGS. **52-54**), and the sample **150** may include interstitial fluid, blood, an organ, or other tissue. One or more of the optical fibers **116** may be used to (i) direct the pump and Stokes beams into the patient's body and (ii) collect and direct a Raman signal to the system.

[0217] FIGS. **52-54** each illustrate an example Raman spectroscopy system **100** that includes a photonic integrated circuit (PIC) **170** and one or more optical fibers **116**. A PIC-based Raman spectroscopy system **100** may include an optical fiber **116** coupled to an optical waveguide **172**, where the optical fiber receives a pump-Stokes beam **140** from the optical waveguide and directs the pump-Stokes beam to a sample **150**. Additionally, the optical fiber **116** may receive a Raman signal **160** produced by the pump-Stokes beam **140**, or the system may include a second optical fiber that receives the Raman signal. Any of the Raman spectroscopy systems **100** in FIGS. **40-43** may include one or more optical fibers **116** configured to direct a pump-Stokes beam **140** to a sample **150** or receive a Raman signal **160**.

[0218] In each of FIGS. **52-54**, a Raman spectroscopy system **100** is affixed to a patient's body, and one or more optical fibers **116** are inserted through the skin **600** and extend into the patient's body. The optical fibers **116** may be inserted using a needle which is withdrawn after the optical fiber is installed. Each of the optical fibers **116** includes a coating **450**, which may include a protective jacket or a biocompatible coating. For example, a biocompatible coating **450** may be applied to the outer surface of an optical fiber **116** to prevent an immune rejection or inflammatory response (e.g., to allow the optical fiber to remain installed in a patient's body for an extended period of time) or to provide mechanical protection for the optical fiber. A biocompatible coating **450** may include silicone, polyurethane, polytetrafluoroethylene (PTFE), parylene, or an anticoagulant substance. The Raman spectroscopy system **100** in each of FIGS. **52-54** may be a wearable Raman spectroscopy system that remains affixed to the patient's body and performs Raman-signal measurements over an extended period of time. The sample measured by the Raman spectroscopy system **100** may include interstitial fluid, blood, an organ, or other tissue.

[0219] Each of the Raman spectroscopy systems **100** in FIGS. **52-54** includes an optical fiber **116** coupled to an optical waveguide **172**, where the optical fiber receives a pump-Stokes beam **140** from the optical waveguide and directs the pump-Stokes beam to a sample located in the body of a patient **650**. In FIG. **52**, the optical waveguide **172** includes a grating **430** that diffracts the pump-Stokes beam **140** to couple the pump-Stokes beam into the optical fiber **116**, which directs the pump-Stokes beam into the patient's body. The resulting Raman signal **160** is received by the optical fiber **116**, which directs the Raman signal to the same optical waveguide **172** in which the pump-Stokes beam **140** propagates. The Raman signal **160** propagates in the opposite direction as the pump-Stokes beam **140** (e.g., similar to the arrangement in FIG. **42**), and the Raman signal may be directed to an optical receiver for detection. The Raman spectroscopy system **100** in FIG. **53** includes two optical fibers: an output optical fiber **116a** coupled to an output optical waveguide **172a** and an input optical fiber **116b** coupled to an input optical waveguide **172b**. The output optical waveguide **172a** directs the pump-Stokes beam **140** to the output optical fiber **116a** via a mirror **460**, and the output optical fiber **116a** directs the pump-Stokes beam into the patient's body. The input optical fiber **116b** receives the resulting Raman signal **160** and directs the Raman signal to the input optical waveguide **172b** via a mirror **460**, and the Raman signal may propagate along the input optical waveguide to an optical receiver for detection. The Raman spectroscopy system **100** in FIG. **53** uses two different optical waveguides **172** for the pump-Stokes beam **140** and the Raman signal **160** (e.g., similar to the arrangement in FIG. **40**). In FIG. **54**, the pump-Stokes beam **140** is directly coupled from the optical waveguide **172** into the optical fiber **116**. For example, the

end face of the optical fiber **116** may be attached to, in contact with, or located near the output face of the optical waveguide **172**. The optical fiber **116** is installed into a blood vessel **610** of the patient **650** and directs the pump-Stokes beam **140** to blood **620** located in the blood vessel. The resulting Raman signal **160** is received by the optical fiber **116**, which directs the Raman signal to the same optical waveguide **172** in which the pump-Stokes beam **140** propagates. The Raman spectroscopy system **100** in FIG. **54** may be used to perform Raman-signal measurements of a patient's blood **620** over an extended period of time.

[0220] FIG. **55** illustrates an example Raman spectroscopy system **100** that includes a multi-chip PIC **170** with two secondary PICs **170a** and **170b**. A PIC-based Raman spectroscopy system **100** may include a PIC **170** that is a multi-chip PIC that includes two or more secondary PICs. Each of the secondary PICs may be a separate PIC that includes one or more optical waveguides **172** that (i) direct a pump beam **120_{pu}** to a sample **150**, (ii) direct a Stokes beam **120_S** to a sample, (iii) receive a Raman signal **160**, or (iv) direct a Raman signal to an optical receiver **200**. A multi-chip PIC and a secondary PIC may each be referred to as a PIC. In FIG. **55**, the secondary PIC **170a** includes waveguides **172** that receive the pump and Stokes beams and direct the combined pump-Stokes beam **140** to the sample **150**. The secondary PIC **170b** includes waveguides **172** that receive the Raman signal **160** and direct the Raman signal to the optical receiver **200**. Additionally, the secondary PIC **170b** includes waveguides **172** that receive the probe beam **120_{pr}** and direct the combined probe-Raman signal **210** to the detector **220**. Each of the secondary PICs of a multi-chip PIC **170** may include or may be fabricated from any suitable material (e.g., silicon, silicon dioxide, glass, InP, SiN, a polymer, graphene, or an electro-optic material), and the secondary PICs of a multi-chip PIC may include or may be fabricated from the same material or different materials. For example, both of the secondary PICs **170a** and **170b** may be silicon-based PICs. As another example, the secondary PIC **170a** may be an InP-based PIC, and the secondary PIC **170b** may be fabricated from an electro-optic material. One or more of the PICs **170** described herein may be configured as a multi-chip PIC that includes two or more secondary PICs. For example, one of more of the PIC-based Raman spectroscopy systems in FIGS. **40-54** may be configured to have a multi-chip PIC that includes two or more secondary PICs.

[0221] FIG. **56** illustrates an example Raman spectroscopy system **100** that includes multiple optical fibers **116**. The Raman spectroscopy system **100** includes fiber-optic components, and the pump-Stokes beam **140** and the Raman signal **160** are directed to and from the sample **150** via optical fiber **116**. The Raman spectroscopy system **100** in FIG. **56** may be referred to as a fiber-optic Raman spectroscopy system, a fiber-based Raman spectroscopy system, a portable Raman spectroscopy system, or a wearable Raman spectroscopy system.

[0222] In one example embodiment, a fiber-based Raman spectroscopy system **100** includes: a pump light source **111_{pu}** or a pump laser **110_{pu}** configured to produce a pump beam of light **120_{pu}** at a pump frequency; and a Stokes light source **111_S** or a Stokes laser **110_S** configured to produce a Stokes beam of light **120_S** at a Stokes frequency, where the pump and Stokes frequencies are offset by a frequency offset Ω . The system further includes multiple optical fibers **116** configured to: direct the pump and Stokes beams of light to a sample **150**; receive a Raman signal **160** produced by coherent Raman scattering of the pump and Stokes beams of light at the sample; and direct the Raman signal to an optical receiver **200** configured to detect the Raman signal. The optical receiver **200** includes: a probe light source **111_{pr}** or a probe laser **110_{pr}** configured to produce a probe beam of light **120_{pr}** at a probe frequency; an optical detector **220** configured to coherently mix a portion of the Raman signal **160** with at least a portion of the probe beam of light **120_{pr}** to produce a corresponding photocurrent signal i ; and an electronic circuit **230** configured to produce a digital output signal **240** corresponding to the photocurrent signal i .

[0223] In FIG. **56**, the pump light source **110_{pu}** and the Stokes light source **111_S** are each fiber-coupled devices that direct the respective pump beam **120_{pu}** and Stokes beam **120_S** to a fiber-optic combiner **130a** via an optical fiber **116**. The fiber-optic combiner **130a** (which is similar to the

combiner **130** in FIG. 25) combines the pump and Stokes beams to produce a combined pump-Stokes beam **140** that is directed to a sample **150** via an optical fiber **116**. The pump and Stokes beams interact with the sample **150** to produce a Raman signal **160** that is coupled into another optical fiber **116**, which directs the Raman signal to an optical receiver **200** for detection. The optical fiber **116** that directs the pump-Stokes beam **140** to the sample **150** may have a lensed face or a lens may be attached to or positioned near the end face to collimate or focus the pump-Stokes beam into the sample **150**. Similarly, the optical fiber **116** that receives the Raman signal **160** may have a lensed face or a lens may be attached to or positioned near the end face to collect and focus the Raman signal into the optical fiber. The fiber-optic combiner **130b** combines the Raman signal **160** and the probe beam of light **120pr** to produce a combined probe-Raman signal **210** that is directed to a detector **220** by an optical fiber **116**.

[0224] In a fiber-based Raman spectroscopy system **100**, the pump and Stokes beams may interact with a sample **150** and a Raman signal **160** may be collected in a manner similar to a PIC-based Raman spectroscopy system. For example, the pump beam **120pu** and Stokes beam **120S** may be directed to a sample **150** by the same optical fiber **116** or by two different optical fibers. Additionally, the Raman signal **160** may be collected by an optical fiber **116** through which the pump or Stokes beam propagates or may be collected by a separate optical fiber. As another example, a sample **150** may be located within a sample holder (e.g., a cuvette), and the pump and Stokes beams may be emitted from the end face of an optical fiber **116** and directed to propagate into the sample holder and then through the sample. As another example, a fiber-based Raman spectroscopy system **100** may include an evanescent optical fiber (e.g., similar to the evanescent waveguide **400** in FIGS. 44-45), and the pump and Stokes beams may interact with a sample **150** via the evanescent optical fiber. As another example, a fiber-based Raman spectroscopy system **100** may include a hollow waveguide **410** (e.g., similar to that illustrated in FIG. 46), and the pump and Stokes beams may interact with a sample **150** that is contained in the hollow waveguide. As another example, a fiber-based Raman spectroscopy system **100** may include a grating or a mirror that directs the pump and Stokes beam to a sample **150** located external to the system. As another example, one or more optical fibers **116** of a fiber-based Raman spectroscopy system **100** may direct the pump and Stokes beam to a sample **150** located external to the system (e.g., the optical fibers may be inserted into a patient's body).

[0225] Some technical advantages of a PIC-based or a fiber-based Raman spectroscopy system **100** include: the compact size of the system, the ability to use the system as a wearable or portable device, and the ability to operate autonomously over extended periods of time. The integration of PICs **170** or optical fiber **116** into a Raman spectroscopy system **100** allows for deployment in space-constrained environments where conventional spectroscopy equipment would be impractical. This miniaturization facilitates wearable or portable applications, permitting real-time, continuous monitoring directly on a patient's body for medical diagnostics. Additionally, a PIC-based or fiber-based Raman spectroscopy system **100** may operate autonomously for extended periods without requiring an external power source or an external cable connection.

[0226] A PIC-based or a fiber-based Raman spectroscopy system **100** may include any suitable technique for pump and Stokes beams to interact with a sample **150**. For example, a Raman spectroscopy system **100** may include a sample holder that contains a sample **150**, an evanescent waveguide **400**, an evanescent optical fiber, a hollow waveguide **410**, a grating **430** or mirror **460** that directs the pump and Stokes beam to a sample located external to the system, or one or more optical fibers **116** that may be inserted into a patient's body. In some embodiments, a Raman spectroscopy system may include multiple techniques for interacting with one or more samples **150**. For example, a Raman spectroscopy system may include an evanescent waveguide **400** for measurement of sweat and a grating **430** for measurement of interstitial fluid. As another example, a Raman spectroscopy system may include a grating **430** for measurement of skin, interstitial fluid, or blood and an optical fiber **116** that is installed in a blood vessel **610** for measurement of blood

620 located in the blood vessel.

[0227] FIG. **57** illustrates an example method **700** for measuring a Raman signal **160**. The method may be performed by a Raman spectroscopy system as described herein. For example, the method may be performed by one or more of the Raman spectroscopy systems **100** illustrated in FIGS. **1**, **2**, and **40-56**. The method may begin at step **710** by producing a first beam of light at a first frequency ($v_{\text{sub.1}}$). At step **720**, a second beam of light at a second frequency ($v_{\text{sub.2}}$) is produced. The first beam of light may be referred to as a pump beam of light **120_{pu}**, and the second beam of light may be referred to as a Stokes beam of light **120_S**. The first and second frequencies may be offset by a frequency offset Ω , where $\Omega = v_{\text{sub.1}} - v_{\text{sub.2}}$. For example, the first and second frequencies may each correspond to a wavelength between approximately 300 nm and approximately 5,000 nm, and the frequency offset Ω may be between approximately 5 THz and approximately 100 THz. The first and second beams of light may each be produced by a light source that includes one or more laser diodes, where each laser diode is a fixed-wavelength laser diode or a wavelength-tunable laser diode.

[0228] At step **730**, the first and second beams of light are directed to a sample **150**. At step **740**, a Raman signal **160** produced by the sample in response to the first and second beams of light is received, and the Raman signal is directed to an optical receiver **200**. For example, the Raman signal may be produced by coherent Raman scattering of the first and second beams of light at the sample. The first and second beams of light may be directed to the sample by one or more optical elements that include a free-space optical element, an optical fiber, or an optical waveguide. Similarly, one or more optical elements (which may include a free-space optic, an optical fiber, or an optical waveguide) may receive the Raman signal and direct the Raman signal to an optical receiver. For example, a Raman spectroscopy system **100** may include a photonic integrated circuit (PIC) **170** with multiple optical waveguides **172** that (i) direct the first and second beams of light to the sample, (ii) receive the Raman signal, and (iii) direct the Raman signal to an optical receiver. As another example, a Raman spectroscopy system **100** may include multiple optical fibers **116** that (i) direct the first and second beams of light to the sample, (ii) receive the Raman signal, and (iii) direct the Raman signal to an optical receiver.

[0229] At step **750**, the Raman signal is detected. Detection of the Raman signal **160**, which may be performed by an optical receiver **200**, includes steps **752**, **754**, and **756**. At step **752**, a third beam of light at a third frequency $v_{\text{sub.3}}$ is produced. The third beam of light (which may be referred to as a probe beam of light **120_{pr}**) may be produced by a wavelength-tunable laser, where the third frequency is adjustable by changing a wavelength of light produced by the wavelength-tunable laser. At step **754**, a portion of the Raman signal is coherently mixed with at least a portion of the third beam of light to produce a photocurrent signal i . The Raman signal and the third beam of light may be coherently mixed at a detector of an optical receiver. The portion of the Raman signal that is coherently mixed with the third beam of light may include optical frequency components of the Raman signal that are within a particular frequency range of the third frequency, and the particular frequency range may depend on the electronic bandwidth of the detector. For example, the particular frequency range may extend from approximately $v_{\text{sub.3}} - \Delta f$ to approximately $v_{\text{sub.3}} + \Delta f$, where $v_{\text{sub.3}}$ is the third frequency, and Δf is the electronic bandwidth of the detector. At step **756**, a digital output signal **240** corresponding to the photocurrent signal is produced. For example, an electronic amplifier **232** may produce an analog voltage signal **234** corresponding to the photocurrent signal, and a digitizer **236** may receive the analog voltage signal and produce a digital output signal **240** that corresponds to the analog voltage signal. The method **700** may end after step **750**, or the method may continue with the additional step **760**.

[0230] At step **760**, a characteristic of the photocurrent signal i is determined, at which point the method may end. For example, a Raman spectroscopy system **100** may include a computer **1000** that receives a digital output signal **240** from detection electronics **230** and determines a characteristic of the photocurrent signal based on the received digital signal. Alternatively, a Raman

spectroscopy system **100** may communicate with an external computer **1000** (e.g., via a wired or wireless connection) to send a digital output signal **240** to the external computer, and the external computer may determine a characteristic of the photocurrent signal based on the received digital signal. The characteristic of the photocurrent signal may be associated with the Raman signal **160** and may include a peak amplitude, an average amplitude, an amplitude at a particular frequency, an amplitude at a particular time, an amplitude at a frequency center, an amplitude at a temporal center, a DC offset, an area, a frequency, a phase, or a polarization. Additionally, the characteristic of the photocurrent signal may be associated with a Raman shift at a frequency $\nu_{\text{sub.1}} - \nu_{\text{sub.3}}$, where $\nu_{\text{sub.1}}$ is the first frequency, and $\nu_{\text{sub.3}}$ is the third frequency. Based on one or more determined signal characteristics **162**, a processor may determine (i) whether a particular material is present in a sample or (ii) an amount or a concentration of the particular material in the sample. [0231] FIG. **58** illustrates an example computer system **1000**. One or more computer systems **1000** may perform one or more steps of one or more methods described or illustrated herein. One or more computer systems **1000** may provide functionality described or illustrated herein. Software running on one or more computer systems **1000** may perform one or more steps of one or more methods described or illustrated herein or may provide functionality described or illustrated herein. A computer system **1000** may include or may be referred to as a processor, a controller, a computing device, a computing system, a computer, or a data-processing apparatus. Herein, reference to a computer system may encompass one or more computer systems, where appropriate. A processor of a Raman spectroscopy system **100** may include or may be referred to as a computer system, a controller, a computing device, a computing system, a computer, or a data-processing apparatus.

[0232] Computer system **1000** may take any suitable physical form. As an example, computer system **1000** may be an embedded computer system, a system-on-chip (SOC), an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), a single-board computer system (SBC), a desktop computer system, a laptop or notebook computer system, a graphics processing unit (GPU), a mainframe, a mesh of computer systems, a server, a tablet computer system, or any suitable combination of two or more of these. As another example, all or part of computer system **1000** may be combined with, coupled to, or integrated into a variety of devices, including, but not limited to, a medical device, wearable device, camera, camcorder, personal digital assistant (PDA), mobile telephone, smartphone, electronic reading device (e.g., an e-reader), game console, smart watch, clock, calculator, television monitor, flat-panel display, computer monitor, eyewear, or head-mounted display. Where appropriate, computer system **1000** may include one or more computer systems **1000**; be unitary or distributed; span multiple locations; span multiple machines; span multiple data centers; or reside in a cloud, which may include one or more cloud components in one or more networks. Where appropriate, one or more computer systems **1000** may perform without substantial spatial or temporal limitation one or more steps of one or more methods described or illustrated herein. As an example, one or more computer systems **1000** may perform in real time or in batch mode one or more steps of one or more methods described or illustrated herein. One or more computer systems **1000** may perform at different times or at different locations one or more steps of one or more methods described or illustrated herein, where appropriate.

[0233] As illustrated in the example of FIG. **58**, computer system **1000** may include a processor **1010**, memory **1020**, storage **1030**, an input/output (I/O) interface **1040**, a communication interface **1050**, or a bus **1060**. Computer system **1000** may include any suitable number of any suitable components in any suitable arrangement.

[0234] Processor **1010** may include hardware for executing instructions, such as those making up a computer program. As an example, to execute instructions, processor **1010** may retrieve (or fetch) the instructions from an internal register, an internal cache, memory **1020**, or storage **1030**; decode and execute them; and then write one or more results to an internal register, an internal cache,

memory **1020**, or storage **1030**. A processor **1010** may include one or more internal caches for data, instructions, or addresses. A processor **1010** may include one or more internal registers for data, instructions, or addresses. Processor **1010** may include any suitable number of any suitable internal registers, where appropriate. Where appropriate, processor **1010** may include one or more arithmetic logic units (ALUs); may be a multi-core processor; or may include one or more processors **1010**.

[0235] Memory **1020** may include main memory for storing instructions for processor **1010** to execute or data for processor **1010** to operate on. As an example, computer system **1000** may load instructions from storage **1030** or another source (such as, for example, another computer system **1000**) to memory **1020**. Processor **1010** may then load the instructions from memory **1020** to an internal register or internal cache. To execute the instructions, processor **1010** may retrieve the instructions from the internal register or internal cache and decode them. During or after execution of the instructions, processor **1010** may write one or more results (which may be intermediate or final results) to the internal register or internal cache. Processor **1010** may then write one or more of those results to memory **1020**. One or more memory buses (which may each include an address bus and a data bus) may couple processor **1010** to memory **1020**. Bus **1060** may include one or more memory buses. Memory **1020** may include random access memory (RAM). This RAM may be volatile memory, where appropriate. Where appropriate, this RAM may be dynamic RAM (DRAM) or static RAM (SRAM). Memory **1020** may include one or more memories **1020**, where appropriate.

[0236] Storage **1030** may include mass storage for data or instructions. As an example, storage **1030** may include a hard disk drive (HDD), a floppy disk drive, flash memory, an optical disc, a magneto-optical disc, magnetic tape, or a Universal Serial Bus (USB) drive or a combination of two or more of these. Storage **1030** may include removable or non-removable (or fixed) media, where appropriate. Storage **1030** may be internal or external to computer system **1000**, where appropriate. Storage **1030** may be non-volatile, solid-state memory. Storage **1030** may include read-only memory (ROM). Where appropriate, this ROM may be mask ROM (MROM), programmable ROM (PROM), erasable PROM (EPROM), electrically erasable PROM (EEPROM), flash memory, or a combination of two or more of these. Storage **1030** may include one or more storage control units facilitating communication between processor **1010** and storage **1030**, where appropriate. Where appropriate, storage **1030** may include one or more storages **1030**.

[0237] I/O interface **1040** may include hardware, software, or both, providing one or more interfaces for communication between computer system **1000** and one or more I/O devices. Computer system **1000** may include one or more of these I/O devices, where appropriate. One or more of these I/O devices may enable communication between a person and computer system **1000**. As an example, an I/O device may include a keyboard, keypad, microphone, monitor, mouse, printer, scanner, speaker, camera, stylus, tablet, touch screen, trackball, another suitable I/O device, or any suitable combination of two or more of these. An I/O device may include one or more sensors. Where appropriate, I/O interface **1040** may include one or more device or software drivers enabling processor **1010** to drive one or more of these I/O devices. I/O interface **1040** may include one or more I/O interfaces **1040**, where appropriate.

[0238] Communication interface **1050** may include hardware, software, or both providing one or more interfaces for communication (such as, for example, packet-based communication) between computer system **1000** and one or more other computer systems **1000** or one or more networks. As an example, communication interface **1050** may include a network interface controller (NIC) or network adapter for communicating with an Ethernet or other wire-based network or a wireless NIC (WNIC); a wireless adapter for communicating with a wireless network, such as a WI-FI network; or an optical transmitter (e.g., a laser or a light-emitting diode) or an optical receiver (e.g., a photodetector) for communicating using fiber-optic communication or free-space optical communication. Computer system **1000** may communicate with an ad hoc network, a personal area

network (PAN), a local area network (LAN), a wide area network (WAN), a metropolitan area network (MAN), or one or more portions of the Internet or a combination of two or more of these. One or more portions of one or more of these networks may be wired or wireless. As an example, computer system **1000** may communicate with a wireless PAN (WPAN) (such as, for example, a BLUETOOTH WPAN), a WI-FI network, a Worldwide Interoperability for Microwave Access (WiMAX) network, a cellular telephone network (such as, for example, a Global System for Mobile Communications (GSM) network), or other suitable wireless network or a combination of two or more of these. As another example, computer system **1000** may communicate using fiber-optic communication based on 100 Gigabit Ethernet (100 GbE), 10 Gigabit Ethernet (10 GbE), or Synchronous Optical Networking (SONET). Computer system **1000** may include any suitable communication interface **1050** for any of these networks, where appropriate. Communication interface **1050** may include one or more communication interfaces **1050**, where appropriate.

[0239] Bus **1060** may include hardware, software, or both coupling components of computer system **1000** to each other. As an example, bus **1060** may include an Accelerated Graphics Port (AGP) or other graphics bus, an Enhanced Industry Standard Architecture (EISA) bus, a front-side bus (FSB), a HYPERTRANSPORT (HT) interconnect, an Industry Standard Architecture (ISA) bus, an INFINIBAND interconnect, a low-pin-count (LP C) bus, a memory bus, a Micro Channel Architecture (MCA) bus, a Peripheral Component Interconnect (PCI) bus, a PCI-Express (P Cle) bus, a serial advanced technology attachment (SATA) bus, a Video Electronics Standards Association local bus (VLB), or another suitable bus or a combination of two or more of these. Bus **1060** may include one or more buses **1060**, where appropriate.

[0240] Various modules, circuits, systems, methods, or algorithm steps described in connection with the implementations disclosed herein may be implemented as electronic hardware, computer software, or any suitable combination of hardware and software. Computer software (which may be referred to as software, computer-executable code, computer code, a computer program, computer instructions, or instructions) may be used to perform various functions described or illustrated herein, and computer software may be configured to be executed by or to control the operation of computer system **1000**. As an example, computer software may include instructions configured to be executed by processor **1010**. Owing to the interchangeability of hardware and software, the various illustrative logical blocks, modules, circuits, or algorithm steps have been described generally in terms of functionality. Whether such functionality is implemented in hardware, software, or a combination of hardware and software may depend upon the particular application or design constraints imposed on the overall system.

[0241] A computing device may be used to implement various modules, circuits, systems, methods, or algorithm steps disclosed herein. As an example, all or part of a module, circuit, system, method, or algorithm disclosed herein may be implemented or performed by a general-purpose single- or multi-chip processor, a digital signal processor (DSP), an ASIC, a FPGA, a GPU, any other suitable programmable-logic device, discrete gate or transistor logic, discrete hardware components, or any suitable combination thereof. A general-purpose processor may be a microprocessor, or any conventional processor, controller, microcontroller, GPU, or state machine. A processor may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0242] One or more implementations of the subject matter described herein may be implemented as one or more computer programs (e.g., one or more modules of computer-program instructions encoded or stored on a computer-readable non-transitory storage medium). As an example, the steps of a method or algorithm disclosed herein may be implemented in a processor-executable software module which may reside on a computer-readable non-transitory storage medium. A computer-readable non-transitory storage medium may include any suitable storage medium that may be used to store or transfer computer software and that may be accessed by a computer system.

Herein, a computer-readable non-transitory storage medium or media may include one or more semiconductor-based or other integrated circuits (ICs) (such, as for example, field-programmable gate arrays (FPGAs) or application-specific ICs (ASICs)), hard disk drives (HDDs), hybrid hard drives (HHDs), optical discs (e.g., compact discs (CDs), CD-ROM, digital versatile discs (DVDs), Blu-ray discs, or laser discs), optical disc drives (ODDs), magneto-optical discs, magneto-optical drives, floppy diskettes, floppy disk drives (FDDs), magnetic tapes, flash memories, solid-state drives (SSDs), RAM, RAM-drives, ROM, SECURE DIGITAL cards or drives, any other suitable computer-readable non-transitory storage media, or any suitable combination of two or more of these, where appropriate. A computer-readable non-transitory storage medium may be volatile, non-volatile, or a combination of volatile and non-volatile, where appropriate.

[0243] Certain features described herein in the context of separate implementations may also be combined and implemented in a single implementation. Conversely, various features that are described in the context of a single implementation may also be implemented in multiple implementations separately or in any suitable sub-combination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination may in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.

[0244] While operations may be depicted in the drawings as occurring in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all operations be performed. Further, the drawings may schematically depict one more example processes or methods in the form of a flow diagram or a sequence diagram. However, other operations that are not depicted may be incorporated in the example processes or methods that are schematically illustrated. For example, one or more additional operations may be performed before, after, simultaneously with, or between any of the illustrated operations. Moreover, one or more operations depicted in a diagram may be repeated, where appropriate. Additionally, operations depicted in a diagram may be performed in any suitable order. Furthermore, although particular components, devices, or systems are described herein as carrying out particular operations, any suitable combination of any suitable components, devices, or systems may be used to carry out any suitable operation or combination of operations. In certain circumstances, multitasking or parallel processing operations may be performed. Moreover, the separation of various system components in the implementations described herein should not be understood as requiring such separation in all implementations, and it should be understood that the described program components and systems may be integrated together in a single software product or packaged into multiple software products.

[0245] Various embodiments have been described in connection with the accompanying drawings. However, it should be understood that the figures may not necessarily be drawn to scale. As an example, distances or angles depicted in the figures are illustrative and may not necessarily bear an exact relationship to actual dimensions or layout of the devices illustrated.

[0246] One or more of the figures described herein may include example data that is prophetic. For example, the graphs illustrated in FIGS. **3-16**, **19**, and **45** may include or may be referred to as prophetic examples.

[0247] The scope of this disclosure encompasses all changes, substitutions, variations, alterations, and modifications to the example embodiments described or illustrated herein that a person having ordinary skill in the art would comprehend. The scope of this disclosure is not limited to the example embodiments described or illustrated herein. Moreover, although this disclosure describes or illustrates respective embodiments herein as including particular components, elements, functions, operations, or steps, any of these embodiments may include any combination or permutation of any of the components, elements, functions, operations, or steps described or illustrated anywhere herein that a person having ordinary skill in the art would comprehend.

[0248] The term “or” as used herein is to be interpreted as an inclusive or meaning any one or any combination, unless expressly indicated otherwise or indicated otherwise by context. Therefore, herein, the expression “A or B” means “A, B, or both A and B.” As another example, herein, “A, B, or C” means at least one of the following: A; B; C; A and B; A and C; B and C; A, B, and C. An exception to this definition will occur if a combination of elements, devices, steps, or operations is in some way inherently mutually exclusive.

[0249] As used herein, words of approximation such as, without limitation, “approximately,” “substantially,” or “about” refer to a condition that when so modified is understood to not necessarily be absolute or perfect but would be considered close enough to those of ordinary skill in the art to warrant designating the condition as being present. The extent to which the description may vary will depend on how great a change can be instituted and still have one of ordinary skill in the art recognize the modified feature as having the required characteristics or capabilities of the unmodified feature. In general, but subject to the preceding discussion, a numerical value herein that is modified by a word of approximation such as “approximately” may vary from the stated value by $\pm 0.5\%$, $\pm 1\%$, $\pm 2\%$, $\pm 3\%$, $\pm 4\%$, $\pm 5\%$, $\pm 10\%$, $\pm 12\%$, or $\pm 15\%$. The term “substantially constant” refers to a value that varies by less than a particular amount over any suitable time interval. For example, a value that is substantially constant may vary by less than or equal to 20%, 10%, 1%, 0.5%, or 0.1% over a time interval of approximately 10.s^{sup.4} s, 10.s^{sup.3} s, 10.s^{sup.2} s, 10 s, 1 s, 100 ms, 10 ms, 1 ms, 100 μ s, 10 μ s, or 1 μ s. The term “substantially constant” may be applied to any suitable value, such as for example, an optical power, an electrical current, a wavelength, an optical or electrical frequency, or an optical or electrical phase.

[0250] As used herein, the terms “first,” “second,” “third,” etc. may be used as labels for nouns that they precede, and these terms may not necessarily imply a particular ordering (e.g., a particular spatial, temporal, or logical ordering). As an example, a system may be described as determining a “first result” and a “second result,” and the terms “first” and “second” may not necessarily imply that the first result is determined before the second result.

[0251] As used herein, the terms “based on” and “based at least in part on” may be used to describe or present one or more factors that affect a determination, and these terms may not exclude additional factors that may affect a determination. A determination may be based solely on those factors which are presented or may be based at least in part on those factors. The phrase “determine A based on B” indicates that B is a factor that affects the determination of A. In some instances, other factors may also contribute to the determination of A. In other instances, A may be determined based solely on B.

[0252] Various example aspects included in this disclosure may be presented in a range format. It should be understood that a description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of a disclosed aspect. Accordingly, the description of a range should be considered to have specifically disclosed all suitable sub-ranges as well as individual numerical values within that range, unless expressly indicated otherwise or indicated otherwise by context. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed sub-ranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6, etc., as well as individual values within that range, for example, 1.1, 2, 2.3, 5, and 5.9.

[0253] Although the foregoing embodiments have been described in some detail for purposes of clarity of understanding, the invention is not limited to the details provided. There may be numerous alternative ways of implementing the invention. The disclosed embodiments are illustrative and not restrictive.

Claims

1. A system comprising: a pump light source configured to produce a pump beam of light at a pump frequency; a Stokes light source configured to produce a Stokes beam of light at a Stokes frequency, wherein the pump and Stokes frequencies are offset by a frequency offset Ω ; a photonic integrated circuit (PIC) comprising a plurality of optical waveguides configured to: direct the pump and Stokes beams of light to a sample; receive a Raman signal produced by coherent Raman scattering of the pump and Stokes beams of light at the sample; and direct the Raman signal to an optical receiver; and the optical receiver, wherein the optical receiver is configured to detect the Raman signal, the optical receiver comprising: a probe light source configured to produce a probe beam of light at a probe frequency; an optical detector configured to coherently mix a portion of the Raman signal with at least a portion of the probe beam of light to produce a corresponding photocurrent signal; and an electronic circuit configured to produce a digital output signal corresponding to the photocurrent signal.
2. The system of claim 1, wherein the PIC further comprises a waveguide combiner configured to combine the pump and Stokes beams of light to produce a combined pump-Stokes beam of light that is directed to the sample by one of the optical waveguides.
3. The system of claim 1, wherein the PIC further comprises a waveguide combiner configured to combine the Raman signal and the probe beam of light to produce a combined probe-Raman signal that is directed to the optical detector by one of the optical waveguides.
4. The system of claim 3, wherein the waveguide combiner is further configured to direct at least a portion of the pump beam of light to the sample, wherein the pump beam of light and the Raman signal propagate in opposite directions along one of the optical waveguides.
5. The system of claim 1, wherein the pump beam of light and the Stokes beam of light are directed to the sample separately by different optical waveguides.
6. The system of claim 1, further comprising an optical circulator comprising a first port, a second port, and a third port, wherein the optical circulator is configured to: receive the pump and Stokes beams at the first port, and direct the pump and Stokes beams to exit the circulator via the second port, wherein the second port is coupled to an optical waveguide that directs the pump and Stokes beams to the sample; and receive the Raman signal at the second port and direct the Raman signal to exit the circulator via the third port, wherein the third port is coupled to another optical waveguide that directs the Raman signal to the optical receiver.
7. The system of claim 1, wherein the PIC further comprises an evanescent waveguide, wherein the Raman signal is produced by evanescent-field portions of the pump and Stokes beams that extend into the sample while the pump and Stokes beams propagate along the evanescent waveguide.
8. The system of claim 1, further comprising a hollow waveguide, wherein the sample comprises a fluid or gas that flows into the hollow waveguide, and the Raman signal is produced while the pump and Stokes beams propagate through the hollow waveguide.
9. The system of claim 1, wherein one of the optical waveguides comprises a grating configured to direct the pump and Stokes beams of light to the sample along a direction that is at an angle with respect to a plane of the PIC.
10. The system of claim 9, wherein a second optical waveguide comprises a grating configured to receive the Raman signal and direct the Raman signal to propagate along the second waveguide to the optical receiver.
11. The system of claim 9, wherein the optical detector is one of a plurality of optical detectors, wherein each detector is configured to receive a portion of the Raman signal produced from a different part of the sample.
12. The system of claim 1, further comprising a mirror configured to direct the pump and Stokes beams of light to the sample along a direction that is at an angle with respect to a plane of the PIC.
13. The system of claim 1, further comprising an optical fiber coupled to one of the optical waveguides, wherein the optical fiber is configured to receive the pump and Stokes beams from the

optical waveguide and direct the pump and Stokes beams to the sample.

14. The system of claim 13, wherein the optical fiber is further configured to receive the Raman signal and direct the Raman signal to the optical waveguide.

15. The system of claim 13, wherein the optical fiber is a first optical fiber, and the system further comprises a second optical fiber configured to receive the Raman signal and direct the Raman signal to a second optical waveguide, wherein the second waveguide directs the Raman signal to the optical receiver.

16. The system of claim 13, wherein the system is configured to be affixed to a patient's body, wherein the optical fiber extends into the patient's body and directs the pump and Stokes beams into the patient's body.

17. The system of claim 13, wherein the optical fiber is coated with a biocompatible coating.

18. The system of claim 1, wherein the system is configured to be affixed to a patient's body.

19. The system of claim 18, wherein the sample comprises sweat, skin, interstitial fluid, blood, or saliva.

20. The system of claim 18, wherein the sample comprises a part of the patient's body, and the pump and Stokes beams interact with the sample via an evanescent waveguide or a hollow waveguide.

21. The system of claim 18, wherein the system is configured to direct the pump and Stokes beams into the patient's body.

22. The system of claim 18, further comprising one or more optical fibers configured to be inserted into the patient's body, wherein the pump and Stokes beams are directed into the patient's body and the Raman signal is collected and directed to the system by the one or more optical fibers.

23. The system of claim 1, wherein the PIC is made from a flexible material.

24. The system of claim 1, wherein the PIC is attached to a flexible substrate.

25. The system of claim 1, wherein the PIC comprises: silica-on-silicon, silicon-on-insulator (SOI), indium phosphide (InP), silicon nitride (SiN), lithium niobate (LiNbO₃), lithium tantalate (LiTaO₃), graphene, or a polymer.

26. The system of claim 1, wherein the PIC is a multi-chip PIC that comprises two or more secondary PICs, each of the secondary PICs comprising one or more of the optical waveguides.

27. The system of claim 1, further comprising a rechargeable battery configured to supply power to the system, wherein the battery is configured to be charged by wired charging or wireless charging.

28. The system of claim 1, further comprising a processor configured to determine a characteristic of the photocurrent signal based on the digital output signal.

29. The system of claim 1, further comprising a processor configured to communicate with an external computer using a wired connection or a wireless connection.

30. The system of claim 1, further comprising a display configured to display a value, image, or graph associated with the sample.
